

GATES LEARJET CORPORATION
maintenance manual

CHAPTER

30

**ICE & RAIN
PROTECTION**

Gates Learjet Corporation

RECORD OF TEMPORARY REVISIONS

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GENERAL - DESCRIPTION AND OPERATION

1. General

A. The aircraft is equipped with various systems to prevent formation of ice and to remove rain. Ice protection is provided for the following areas and components of the aircraft.

- (1) Wing leading edge.
- (2) Engine inlet duct and guide vanes.
- (3) Cockpit windshields.
- (4) Pitot tubes, static ports and angle of attack transducer vanes.
- (5) Radome.
- (6) Horizontal stabilizer leading edge.

2. Description

- A. Engine bleed air is used to prevent ice formation on the wing leading edge, nacelle inlet and the windshield.
- B. Electrical heaters are used to prevent ice formation on the pitot tubes, static ports, angle of attack transducer vanes and horizontal stabilizer leading edge.
- C. The alcohol system is used to prevent ice formation on the radome and is used on the pilot's windshield in case of windshield defog system malfunction.
- D. Indicating and warning lights and readout gages provide visual indication of the various ice protection system conditions.

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WING LEADING EDGE ANTI-ICE SYSTEM - DESCRIPTION AND OPERATION

1. Description

- A. Wing leading edge anti-icing is accomplished by directing engine bleed air through a diffuser tube installed in the wing leading edge. Bleed air is exhausted from the wing leading edge thru a scupper.
- B. The system consists of a wing root diffuser, a wing leading edge diffuser tube, a thermoswitch, a temperature sensor, a pressure regulator, a wing temperature indicator, a warning light, a system switch, and a circuit breaker.
- C. Component Description
 - (1) The temperature indicator is installed in the pilot subpanel.
 - (a) On temperature indicators with red, green, and yellow color ranges: the red zone - below 35°F (2° C), indicates moisture will freeze on surface or that system has failed; the green zone - above 35°F (2°C), indicates moisture will not freeze on surface or that system is functioning properly; and the yellow zone - overheating, indicates a possible system malfunction.
 - (b) On temperature indicators with blue, green, and red color ranges: the blue zone - below 35°F (2° C), indicates moisture will freeze on surface or that system has failed; the green zone - above 35°F (2°C), indicates moisture will not freeze on surface and system is functioning; and the red zone - overheating, indicates a possible system malfunction.
 - (2) On Aircraft 25-061, 25-070 thru 25-208 equipped with a Whittaker Control Valve (pressure regulator type), the control valve regulates bleed air pressure at 16 psig to the wing leading edge. The regulator is installed in a bleed air duct forward of the bleed air manifold and aft of frame 25.
On Aircraft 25-209 and Subsequent and prior Aircraft equipped with a Sterer Control Valve, the valve allows bleed air to flow to the wing leading edge. The shutoff valve is installed in a bleed air duct aft of frame 25 and forward of the bleed air manifold.
 - (3) The wing root diffuser tubes (RH and LH) have small holes so bleed air can be exhausted toward the wing root area.
 - (4) The wing diffuser tubes (RH and LH) exhaust bleed air to the wing leading edge and thru scuppers (RH and LH) located at WS 184.
 - (5) The wing temperature sensor is installed on the inside of the RH wing leading edge at WS 31 and provides a resistance circuit to the wing temperature indicator.
 - (6) The thermoswitch is installed at WS 33 in the RH wing adjacent to the wing-to-fuselage fairing and completes a ground circuit to the WING OV HT warning light (red) if temperature reaches 215°F (102°C).
 - (7) The WING OV HT warning light is located on the glareshield warning light panel.

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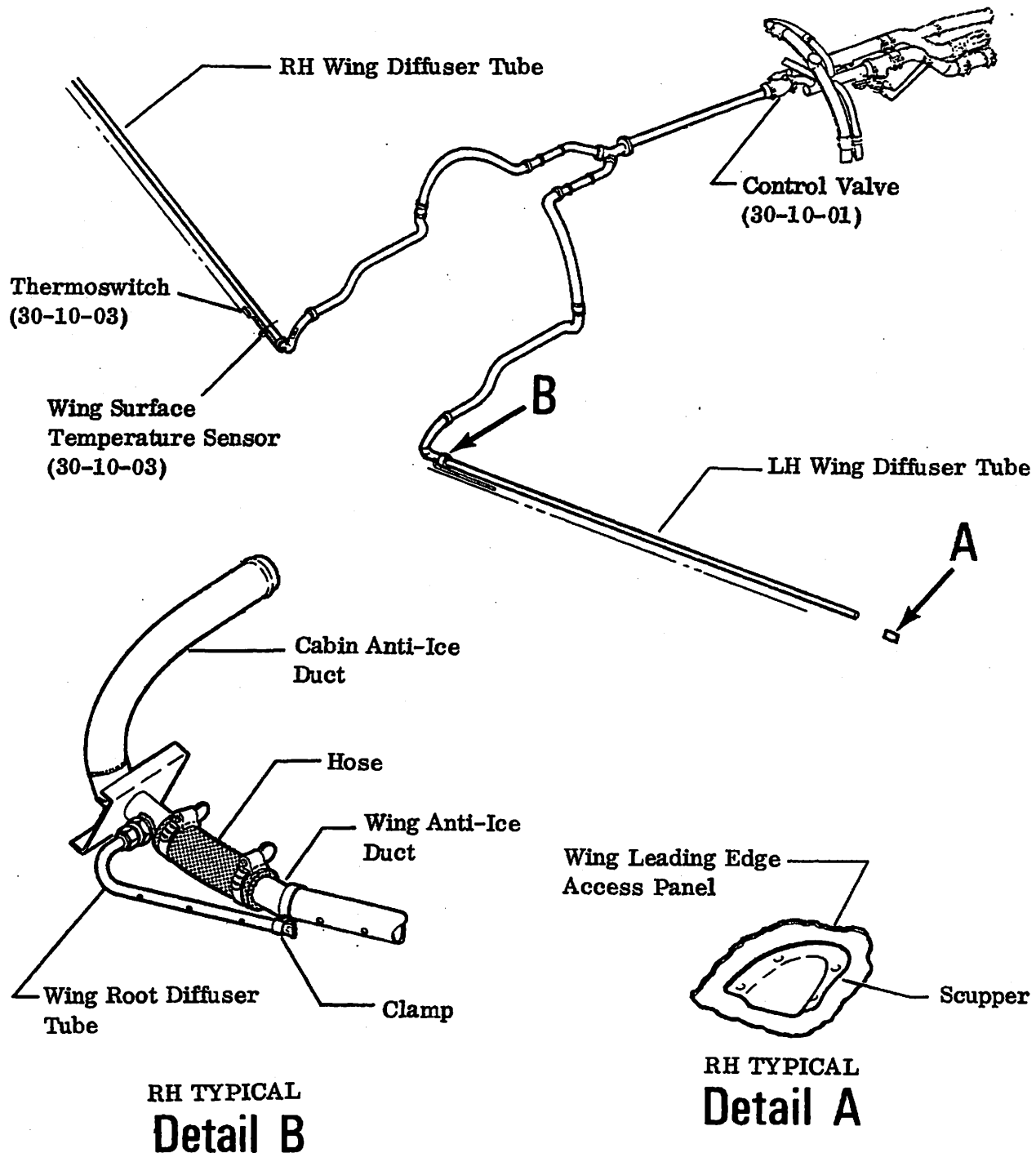
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Wing Leading Edge Anti-Ice Components Installation
Figure 1

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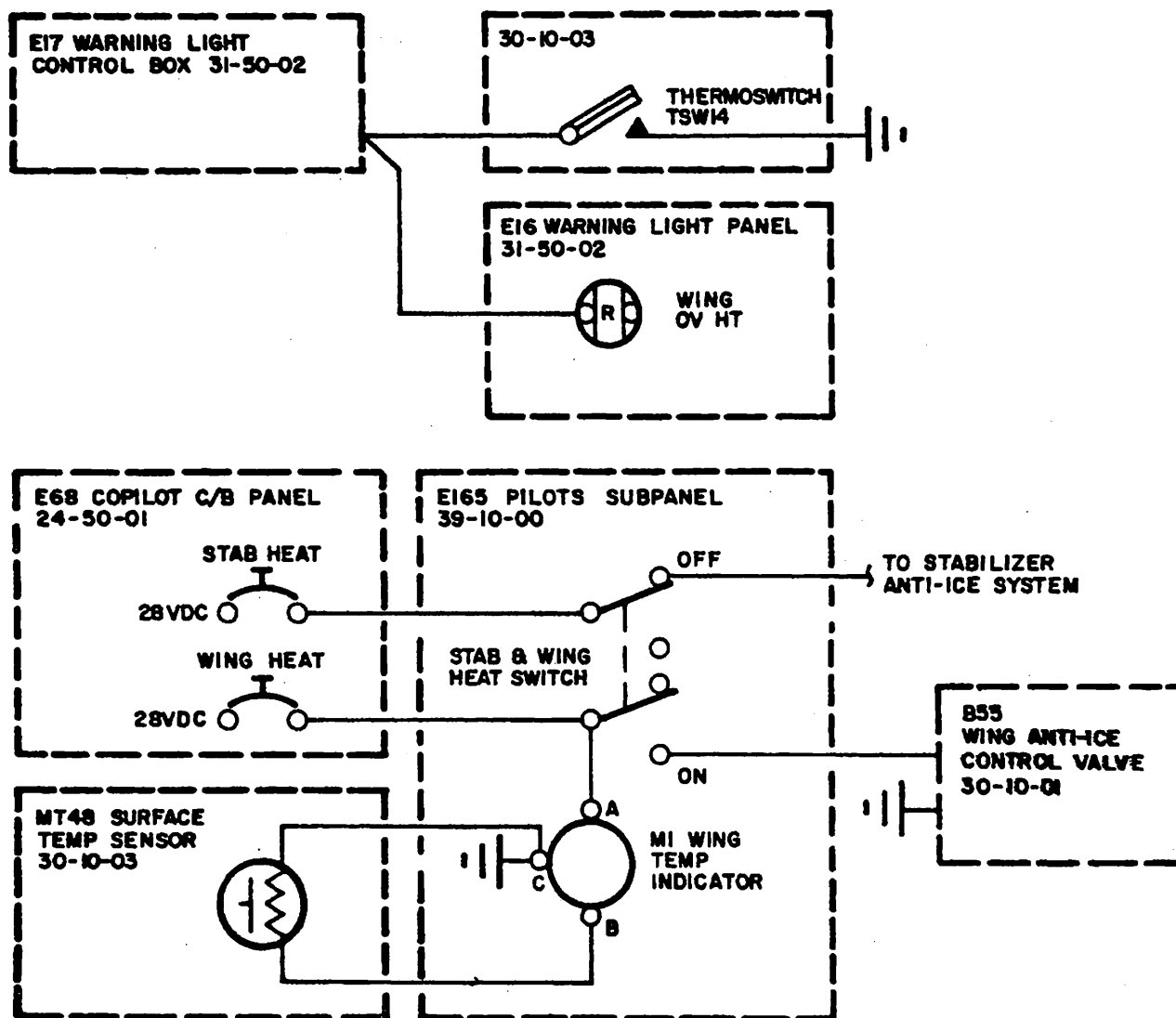
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2. Operation (See figures 2 and 3)

- A. With the Stab & Wing Heat Switch set to ON, 28 vdc is applied to the wing anti-ice control valve. The control valve opens and allows bleed air to flow thru the ducting to the wing leading edge. On Aircraft equipped with a Whittaker control valve, the valve regulates bleed air flow to 16 psig. Aircraft equipped with a Sterer control valve utilize a spring loaded, poppet type shutoff valve. A thermoswitch installed on the wing rib in the RH wing monitors the temperature of the wing leading edge structure and completes a ground circuit to the WING OV HT warning light if the temperature reaches 215°F. The wing temperature sensor, installed on the inner surface of the wing leading edge, monitors the leading edge surface temperature and provides the crew with a visual readout on the wing temperature indicator.



Wing Anti-Ice System Electrical Control Schematic

Figure 2

EFFECTIVITY: ALL

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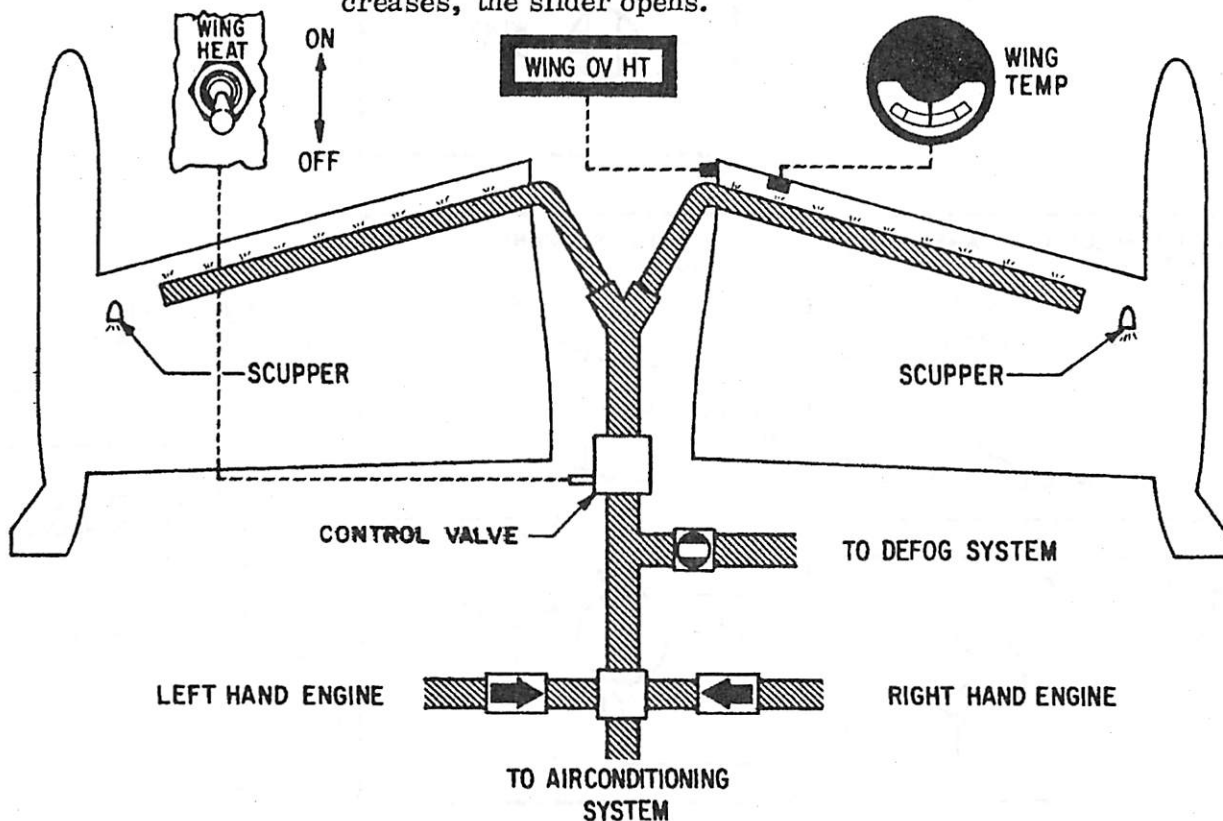
B. Component Operation - Pressure Regulator (25-061, 25-070 thru 25-208 equipped with Whittaker pressure regulator) (See figure 4)

(1) Nonregulating Position (Closed)

- (a) With the Wing and Stab Heat Switch set to OFF, power is removed from the solenoid, allowing it to close. With the solenoid closed, air pressure is trapped in pressure control chamber (PC₂). Air pressure, regulated to 16 psig is metered thru the pilot valve chamber to pressure control chamber (PC₁). Due to the construction of the valve, bleed air pressure (P₁) will bleed into pressure control chamber (PC₂) between the valve housing and slider. Pressure in chamber (PC₂) increases, overriding the pressure in chamber (PC₁) and moves the slider closed.

(2) Regulating Position (Open)

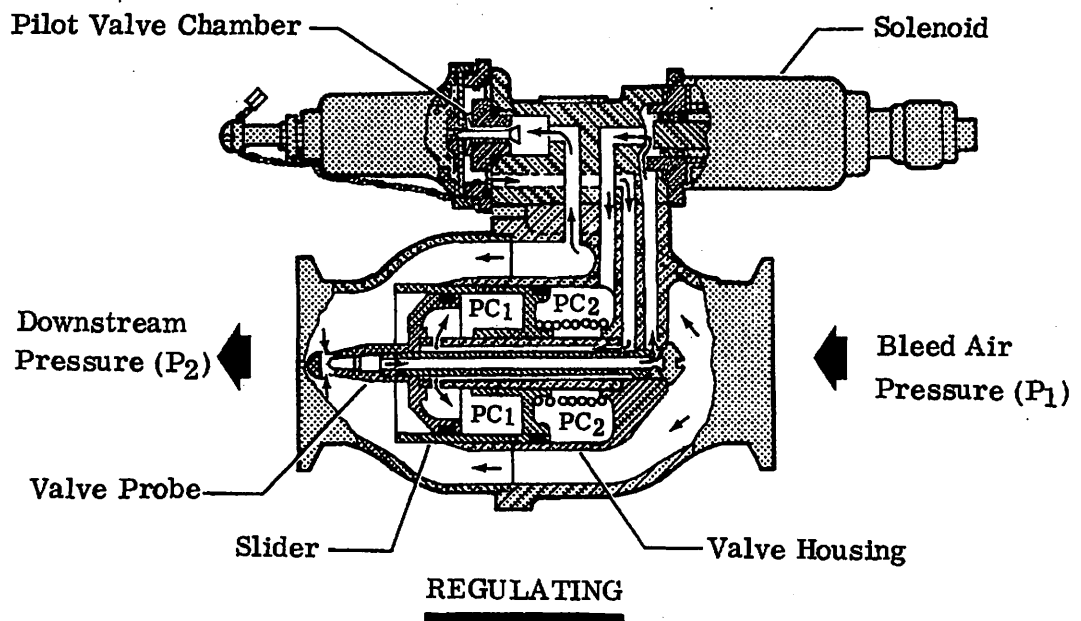
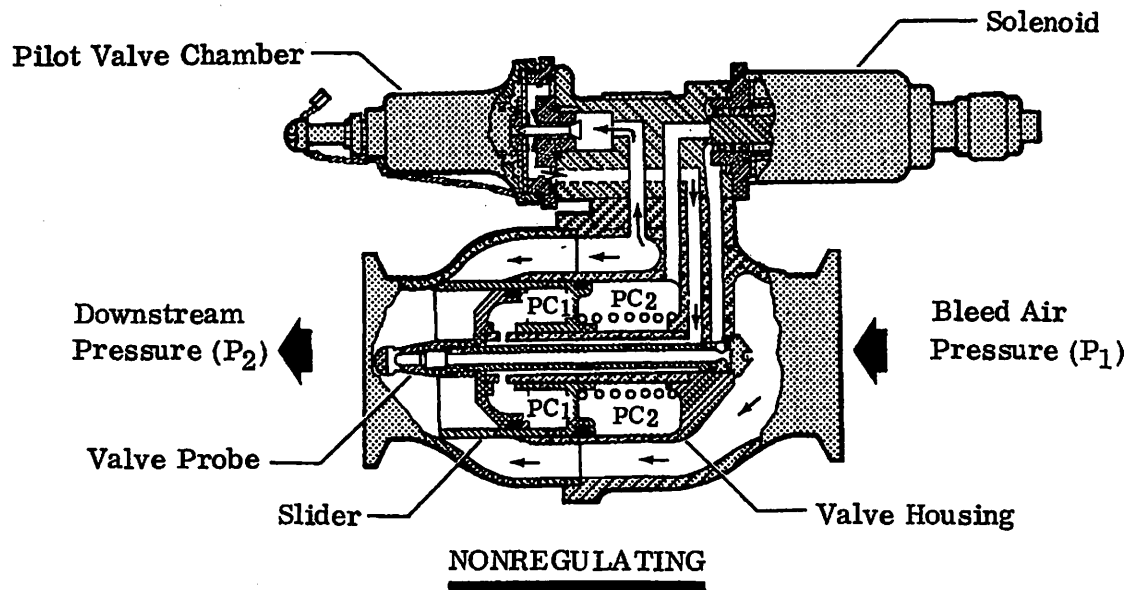
- (a) With the Wing and Stab Heat Switch set to ON, power is applied and the solenoid is opened. With the solenoid valve open, the trapped air in pressure control chamber (PC₂) decreases, pressure in chamber (PC₁) overrides the spring in chamber (PC₂) and moves the slider open, allowing bleed airflow downstream. As the downstream pressure (P₂) increases, it is sensed thru the valve probe into chamber (PC₂). As pressure in chamber (PC₂) increases, the slider moves closed and as pressure decreases, the slider opens.



Wing Leading Edge Anti-Ice System Schematic
Figure 3

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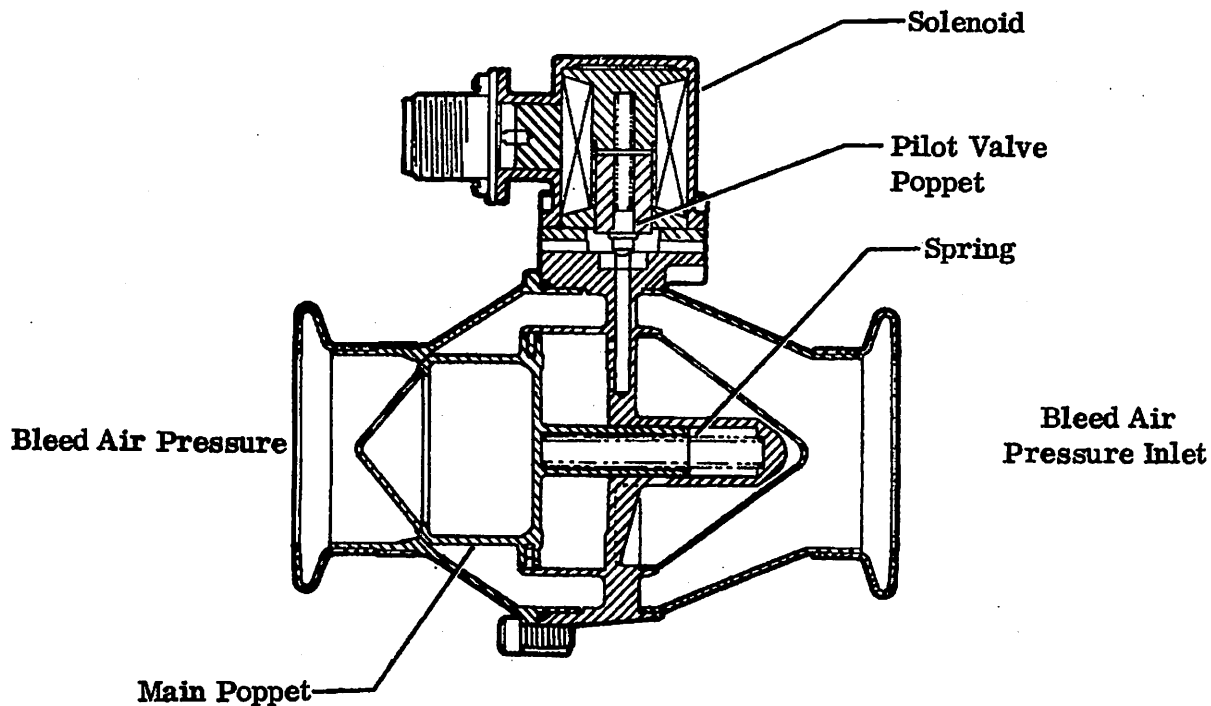
Pressure Regulator Schematic
Figure 4

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C. Component Operation - Shutoff Valve (25-209 and Subsequent and Prior Aircraft equipped with Sterer Shutoff Valve) (See figure 5)

- (1) When the Stabilizer and Wing Heat Switch is set to OFF, inlet air pressure enters the cavity behind the main poppet through two orifices and the entire valve housing up to the main poppet seat is pressurized. This, in addition to the spring force keeps the valves closed. When the valve is energized (Stabilizer and Wing Heat Switch set to ON), the pilot valve poppet retracts and vents the pressure in the cavity to ambient. Inlet pressure on the outside of the cavity overcomes the spring force and drives the poppet open. This allows bleed air to flow through the valve.



Shutoff Valve Schematic
Figure 5

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WING LEADING EDGE ANTI-ICE SYSTEM - TROUBLE SHOOTING

1. Tools and Equipment

NOTE: Equivalent substitutes may be used in lieu of the following.

NAME	PART NUMBER	MANUFACTURER	USE
Voltmeter	3430A	Hewlett Packard	General
Voltmeter	260	Simpson	General

2. Trouble Shooting

- A. Trouble shooting the wing leading edge anti-ice system can be accomplished by isolating the trouble to the control circuit, bleed air system, or the temperature indicating circuit. See figure 101 for trouble shooting procedure.

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ON AIRCRAFT WITH RED, GREEN, AND YELLOW COLOR RANGES, "STAB & WING HEAT" SWITCH SET TO OFF, TEMPERATURE INDICATOR POINTER IN RED ZONE.

ON AIRCRAFT WITH BLUE, GREEN, AND RED COLOR RANGES, "STAB & WING HEAT" SWITCH SET TO OFF. TEMPERATURE INDICATOR POINTER IN BLUE ZONE.

CHECK AIR FLOW FROM WING SCUPPERS, IF

OK. (NO AIR FLOW), REFER TO 30-10-02 FOR FUNCTIONAL TEST OF TEMPERATURE INDICATOR. IF

OK, CONTINUITY CHECK WIRING FROM INDICATOR TO TEMPERATURE SENSOR IN WING, IF

OK, REPLACE WING TEMPERATURE SENSOR.

NOT OK, REPLACE WIRING.

NOT OK, REPLACE INDICATOR.

NOT OK (AIRFLOW PRESENT), THE CONTROL VALVE HAS FAILED OPEN OR AN ELECTRICAL SHORT HAS OCCURRED. REMOVE ELECTRICAL PLUG FROM VALVE AND CHECK FOR 28 VDC, IF

OK (NO 28 VDC PRESENT), REPLACE CONTROL VALVE

NOT OK (28 VDC PRESENT), CHECK ELECTRICAL WIRING FOR SHORT.

ON AIRCRAFT WITH RED, GREEN, AND YELLOW COLOR RANGES, "STAB & WING HEAT" SWITCH SET TO ON, TEMPERATURE INDICATOR DOES NOT MOVE FROM RED ZONE.

ON AIRCRAFT WITH BLUE, GREEN, AND RED COLOR RANGES, "STAB & WING HEAT" SWITCH SET TO ON, TEMPERATURE INDICATOR DOES NOT MOVE FROM BLUE ZONE.

CHECK AIR FLOW FROM WING SCUPPERS, IF

OK (NO AIR FLOW), REFER TO 30-10-02 FOR FUNCTIONAL TEST OF TEMPERATURE INDICATOR. IF

OK, CONTINUITY CHECK WIRING FROM INDICATOR TO TEMPERATURE SENSOR IN WING. IF

OK, REPLACE TEMPERATURE SENSOR.

NOT OK, REPLACE WIRING.

NOT OK, REPLACE INDICATOR.

NOT OK (NO AIRFLOW), THE CONTROL VALVE HAS FAILED CLOSED OR THE POWER CIRCUIT FROM THE SYSTEM SWITCH TO THE CONTROL VALVE IS OPEN. REMOVE ELECTRICAL PLUG FROM VALVE AND CHECK FOR 28 VDC. IF

OK (28 VDC PRESENT), REMOVE AND REPLACE CONTROL VALVE.

NOT OK (28 VDC ABSENT), PERFORM CONTINUITY CHECK OF WIRING FROM POWER SOURCE THRU SWITCH TO VALVE.

Wing Leading Edge Anti-Ice Trouble Shooting
Figure 101

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WING LEADING EDGE ANTI-ICE - MAINTENANCE PRACTICES

1. General

- A. The wing leading edge is normally not removed unless damage has occurred. The thermoswitch and temperature sensor are accessible by removing the wing-to-fuselage fairing.
- B. Refer to 30-10-03 for maintenance practices on the pressure regulator.

2. Tools and Equipment

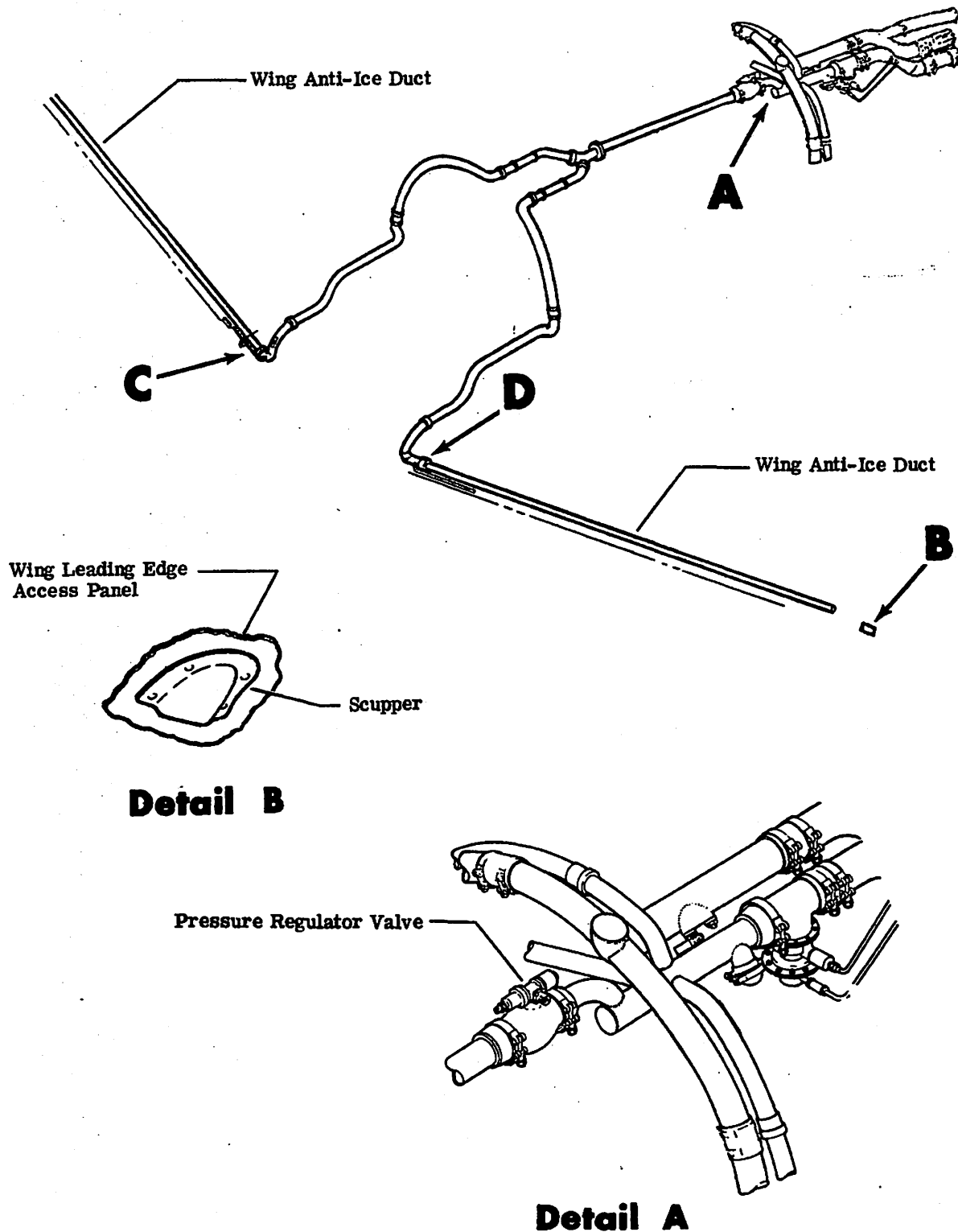
NOTE: Equivalent substitutes may be used in lieu of the following list.

NAME	PART NUMBER	MANUFACTURER	USE
Sandpaper	#380	Local	For cleaning surface.
Heat Sink Grease	#340	Dow Corning	For thermoswitch installation.
Bonding Material	#281	Dow Corning	Sensor Installation

3. Removal/Installation

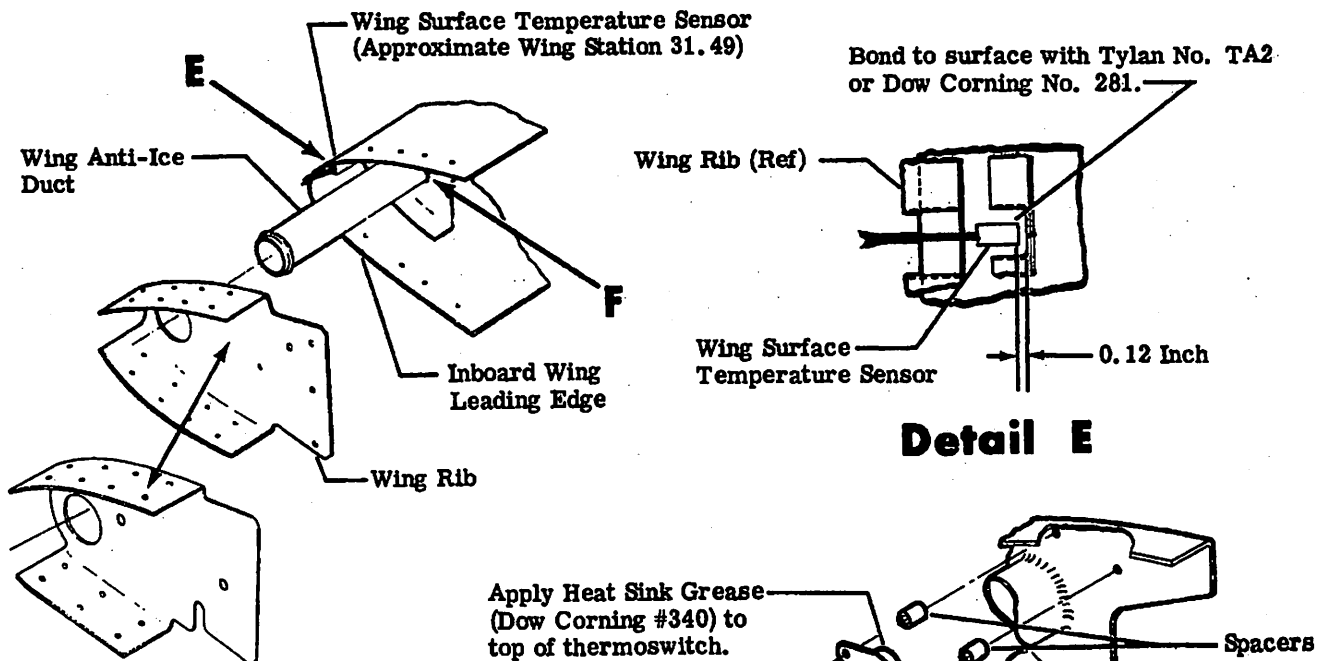
- A. Remove Thermoswitch or Temperature Sensor (See figure 201)
 - (1) Remove attaching screws and right wing-to-fuselage fairing from aircraft.
 - (2) Remove clamp securing wing root diffuser tube to wing anti-ice duct. Loosen and remove wing root diffuser tube from aircraft.
 - (3) Loosen clamps and disconnect hose from cabin anti-ice duct and wing anti-ice duct. Disconnect thermoswitch and sensor wiring at splices.
 - (4) Remove screws and wing rib from aircraft.
 - (5) If thermoswitch is to be replaced, remove attaching screws, spacers, and thermoswitch from aircraft.
 - (6) If temperature sensor is to be replaced, break bond and remove sensor from aircraft.
- B. Install Thermoswitch or Temperature Sensor (See figure 201)
 - (1) Clean all surfaces of aerodynamic filler and sensor bonding material with #380 sandpaper. Wipe with a clean cloth dampened with methyl ethyl ketone. Scrape sealant from wing rib and immediate area of wing rib installation. Wipe with a clean cloth dampened with methyl ethyl ketone.
 - (2) When installing thermoswitch, apply heat sink grease (Dow Corning #340, manufactured by Dow Corning Corp., Midland, Mich.) and position sensor as shown in detail F.

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Wing Leading Edge Anti-Ice Components Installation
Figure 201 (Sheet 1 of 2)

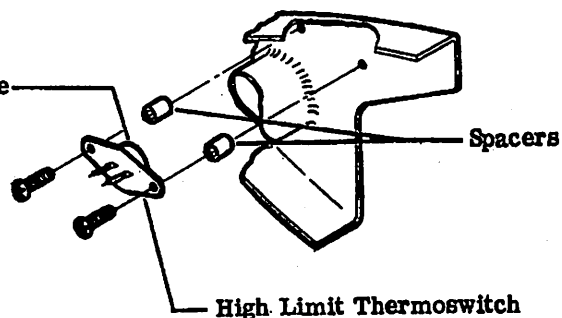
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Detail E

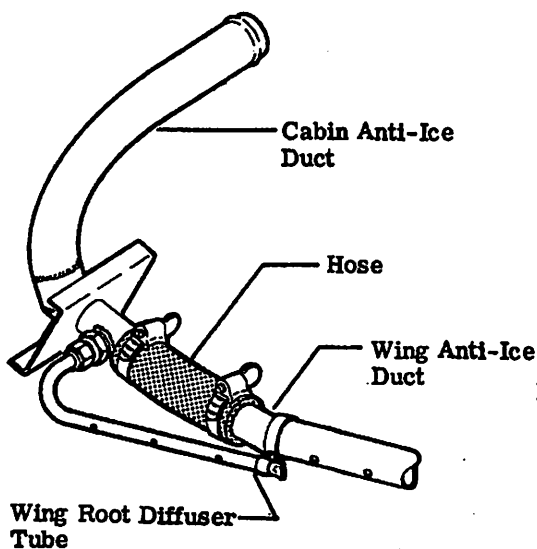
Aircraft 25-206 and Subsequent and Earlier Aircraft equipped with Reduced Approach Speed System.

Apply Heat Sink Grease (Dow Corning #340) to top of thermoswitch.



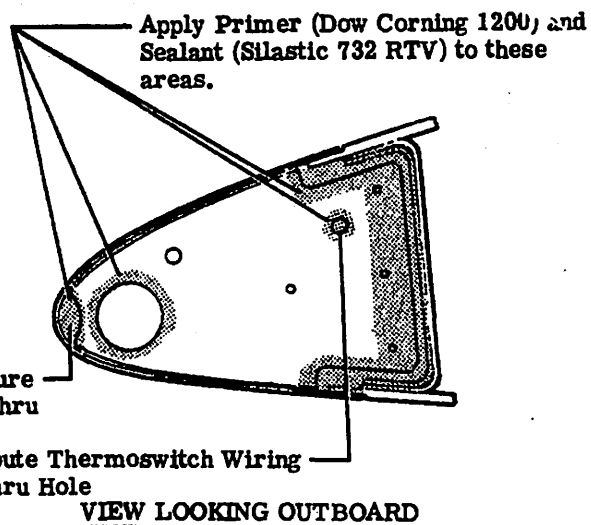
Detail F

Detail C



**TYPICAL
Detail D**

Wing Leading Edge Anti-Ice Components Installation
Figure 201 (Sheet 2 of 2)





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- (3) Install thermoswitch and secure with spacers and screws.
- (4) Route wiring through wing rib as shown and splice.
- (5) When installing sensor, apply bonding material (Tylan No. TA2, manufactured by Tylan Corp., Torrance Calif., or Dow Corning #281, manufactured by Dow Corning Corp., Midland, Mich.) and position sensor as shown in detail E.
- (6) Route sensor wiring as shown and splice.
- (7) Install wing rib and secure with screws.
- (8) Install hose and secure with clamps. Install wing root diffuser tube and secure with clamp.
- (9) Install wing-to-fuselage fairing.



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PRESSURE REGULATOR - DESCRIPTION AND OPERATION

1. General

- A. The pressure regulator regulates bleed air pressure to the wing leading edge at 16 psig. The regulator is installed between the bleed air manifold and a duct assembly which leads to both anti-ice systems.

2. Operation (See figure 1)

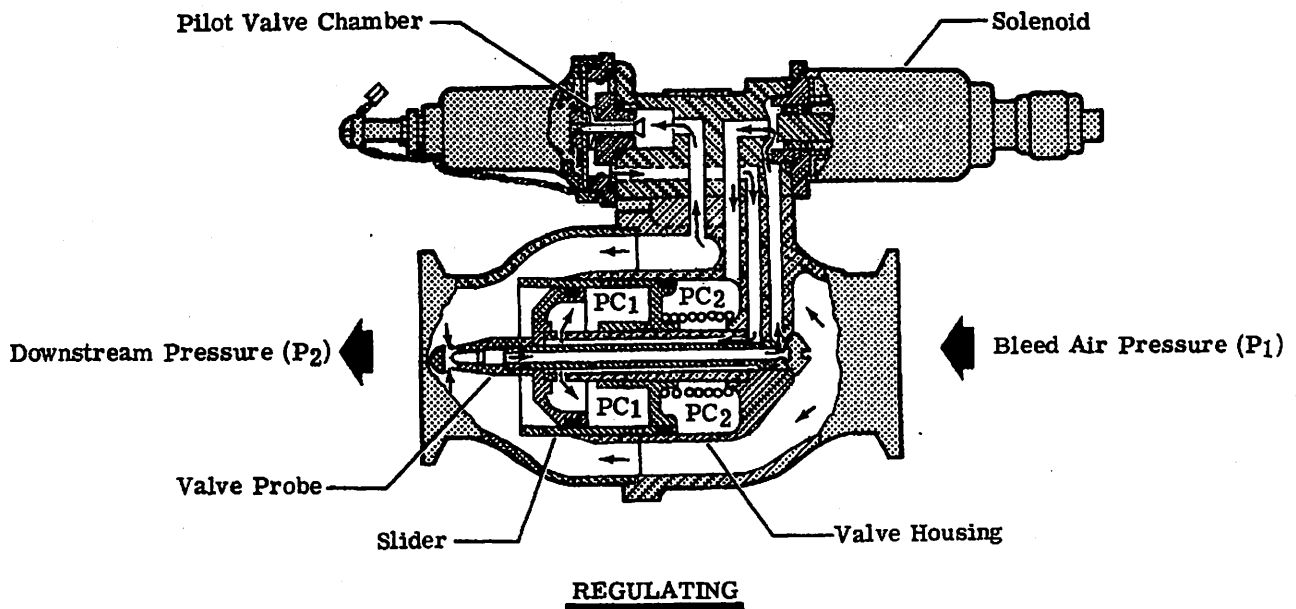
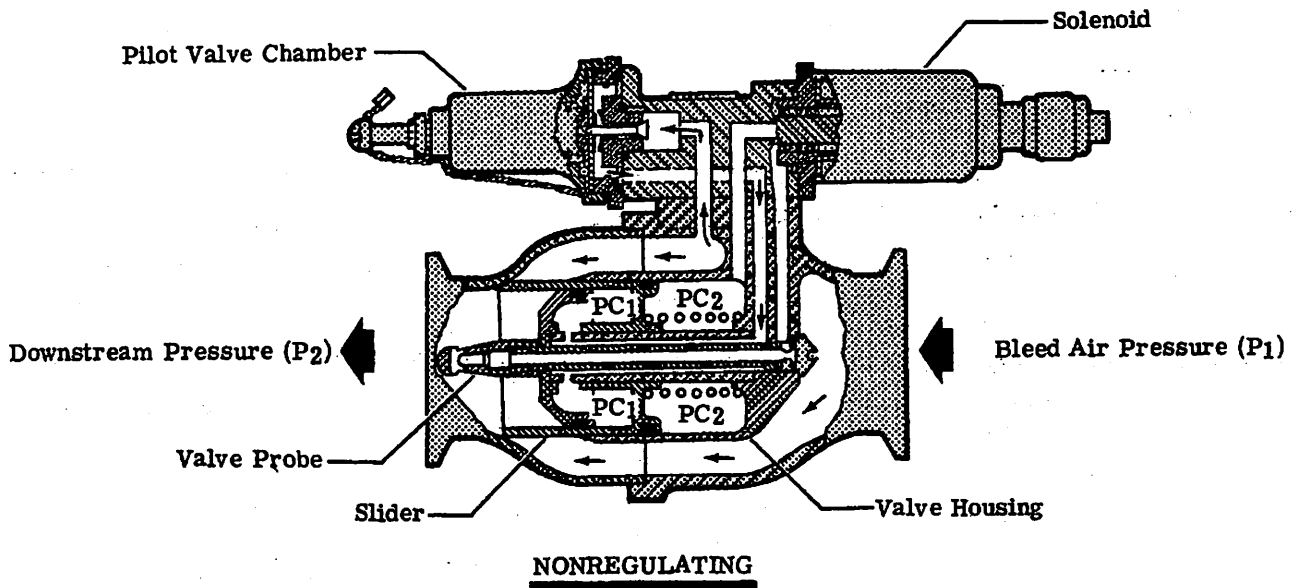
A. Nonregulating Position (closed).

- (1) With the Wing and Stab Heat Switch set to OFF, power is removed from the solenoid, allowing it to close. With the solenoid closed, air pressure is trapped in pressure control chamber (PC₂). Air pressure, regulated to 16 psig is metered through the pilot valve chamber to pressure control chamber (PC₁). Due to the construction of the valve, bleed air pressure (P₁) will bleed into pressure control chamber (PC₂) between the valve housing and slider. Pressure in chamber (PC₂) increases, overriding the pressure in chamber (PC₁) and moves the slider closed.

B. Regulating Position (open).

- (1) With the Wing and Stab Heat Switch set to ON, power is applied and the solenoid is opened. With the solenoid valve open, the trapped air in pressure control chamber (PC₂) is bled off through the valve probe. As the pressure in chamber (PC₂) decreases, pressure in chamber (PC₁) overrides the spring in chamber (PC₂) and moves the slider open, allowing bleed airflow downstream. As the downstream pressure (P₂) increases, it is sensed through the valve probe into chamber (PC₂). As pressure in chamber (PC₂) increases, the slider moves closed and as pressure decreases, the slider opens.

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Pressure Regulator Schematic
Figure 1

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CONTROL VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Aircraft 25-061, 25-070 thru 25-208 are equipped with Whittaker pressure regulators or Sterer shutoff valves. Aircraft 25-209 and Subsequent are equipped with Sterer shutoff valves.

Removal and installation procedures for both Whittaker and Sterer Control Valves are identical.

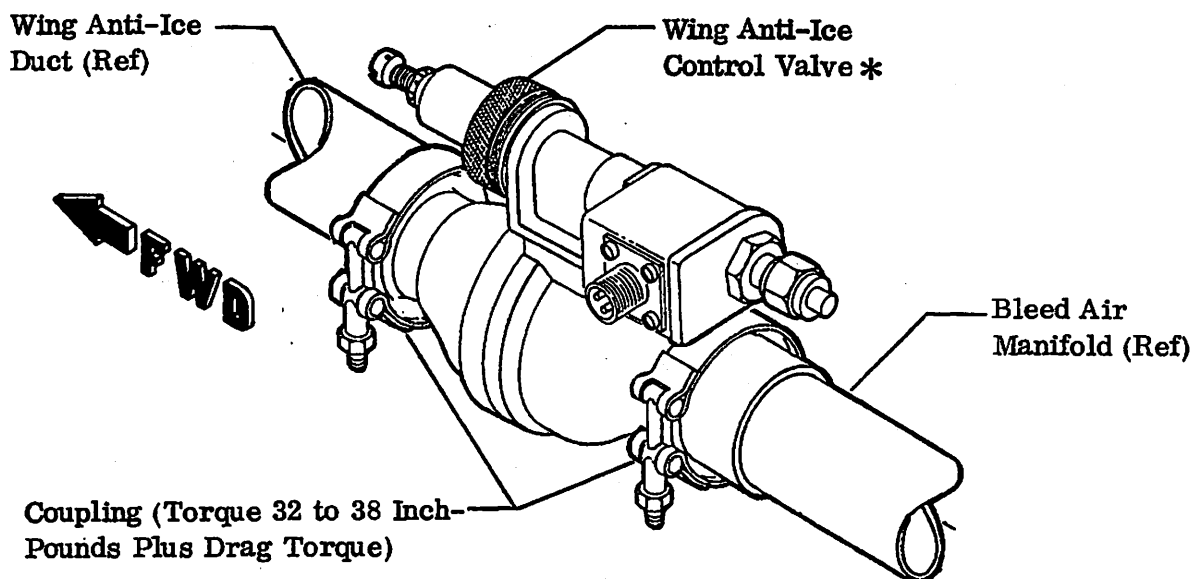
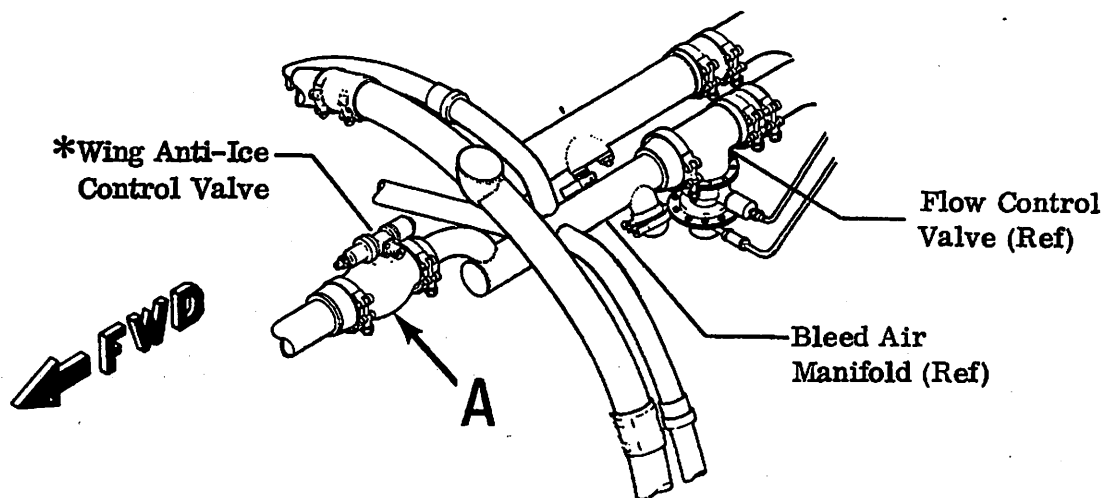
- A. Remove Control Valve (See figure 201).
 - (1) Lower tailcone access door and remove electrical power from aircraft.
 - (2) Remove insulation jackets and disconnect electrical connector from control valve solenoid.
 - (3) Loosen coupling and remove control valve from aircraft.
- B. Install Control Valve (See figure 201)
 - (1) Install control valve and secure with coupling. Torque couplings to 32 to 38 inch-pounds plus drag torque.

NOTE: Drag torque is the amount of torque required to overcome the friction of any self-locking nut. This nut friction (drag torque) must be added to the torque callout to assure proper torquing. The tailcone bleed air ducting incorporates stainless steel clamps, nuts and bolts, which require higher nut-friction than the standard self-locking nuts. In some instances nut friction may exceed the required torque values.

- (2) Connect electrical connector to control valve solenoid and install insulation jackets.
- (3) Restore electrical power to aircraft and secure tailcone access door.

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Detail A

* Aircraft 25-061, 25-070 thru 25-208 are equipped with Whittaker pressure regulators or Sterer shutoff valves. Aircraft 25-209 and Subsequent are equipped with Sterer shutoff valves.

Wing Anti-Ice Control Valve Installation
Figure 201

13-4D-8

EFFECTIVITY: ALL

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WING TEMPERATURE INDICATOR - MAINTENANCE PRACTICES

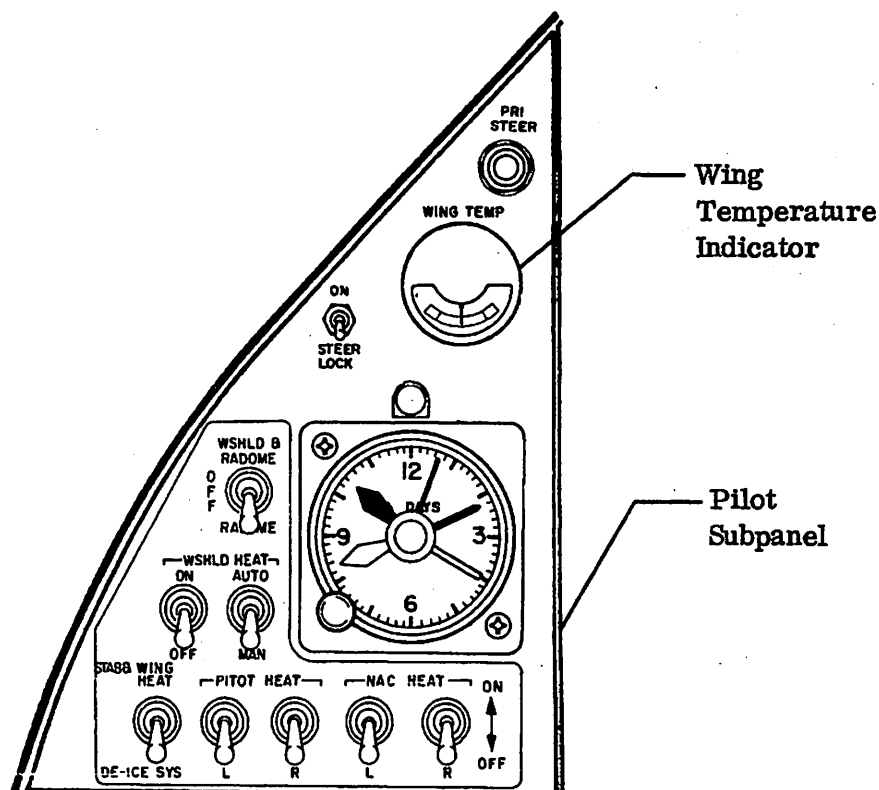
1. Removal/Installation

A. Remove Wing Temperature Indicator (See figure 201)

- (1) Remove electrical power from aircraft.
- (2) Remove attaching parts and instrument panel overlay.
- (3) Remove attaching parts and instrument panel from its supports sufficiently to gain access to indicator electrical wiring. (Refer to Chapter 39.)
- (4) Disconnect electrical wiring from temperature indicator.
- (5) Remove attaching parts and temperature indicator from aircraft.

B. Install Wing Temperature Indicator (See figure 201)

- (1) Install temperature indicator and attaching parts.
- (2) Connect electrical wiring to temperature indicator.
- (3) Install instrument panel and secure with attaching parts. (Refer to Chapter 39.)
- (4) Install instrument panel overlay and secure with attaching parts.
- (5) Restore electrical power to aircraft.



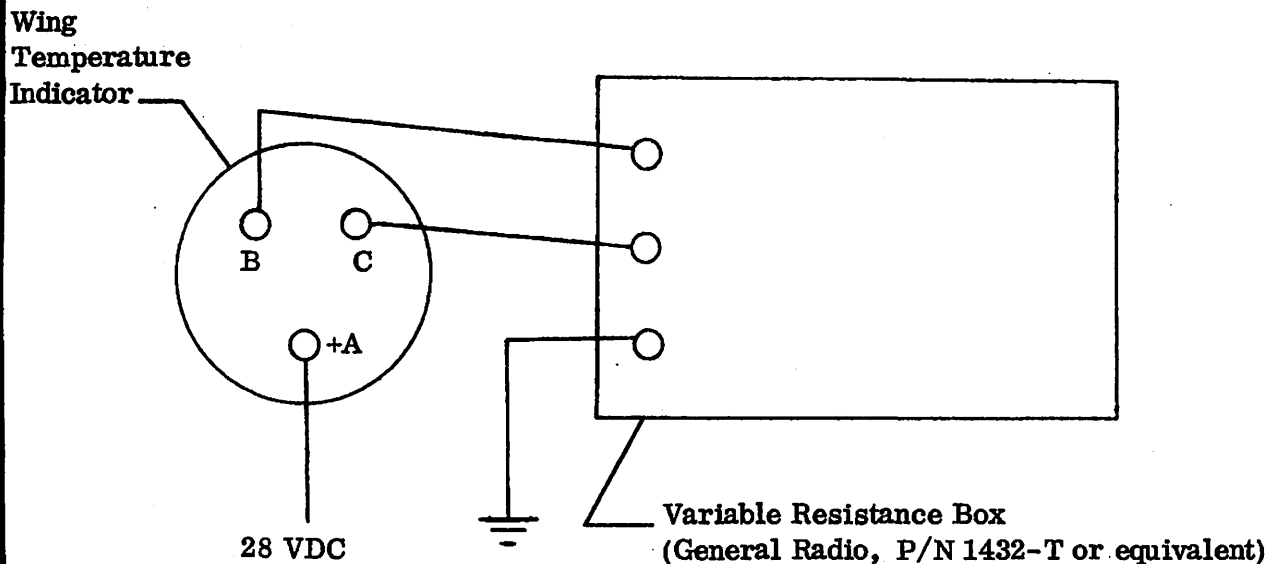
Wing Temperature Indicator Installation
Figure 201

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2. Adjustment/Test

A. Perform functional test as follows: (See figure 202)

- (1) Connect variable resistance box (General Radio P/N 1432-T or equivalent) to indicator as shown in figure 202.
- (2) On aircraft with red, green, and yellow color ranges:
 - (a) Adjust resistance until temperature indicator pointer is at intersection of red and green arcs. Resistance shall be $99.6 (+2, -0)$ ohms.
 - (b) Adjust resistance until temperature indicator pointer is at intersection of green and yellow arcs. Resistance shall be $131 (\pm 5)$ ohms.
- (3) On aircraft with blue, green, and red color ranges:
 - (a) Adjust resistance until temperature indicator pointer is at intersection of blue and green arcs. Resistance shall be $99.6 (+2, -0)$ ohms.
 - (b) Adjust resistance until temperature indicator pointer is at intersection of green and red arcs. Resistance shall be $131 (\pm 5)$ ohms.



Wing Temperature Indicator Functional Test Schematic
Figure 202

EFFECTIVITY: ALL

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THERMOSWITCH AND TEMPERATURE SENSOR - MAINTENANCE PRACTICES

1. Tools and Equipment

NOTE: Equivalent substitutes may be used in lieu of the following.

NAME	PART NUMBER	MANUFACTURER	USE
Sandpaper	380 grit	Local	For cleaning surface.
Heat Sink Grease	340	Dow Corning	For thermoswitch installation.
Bonding Material	281	Dow Corning	Sensor installation.

2. Removal/Installation

A. Remove Thermoswitch or Temperature Sensor (See figure 201)

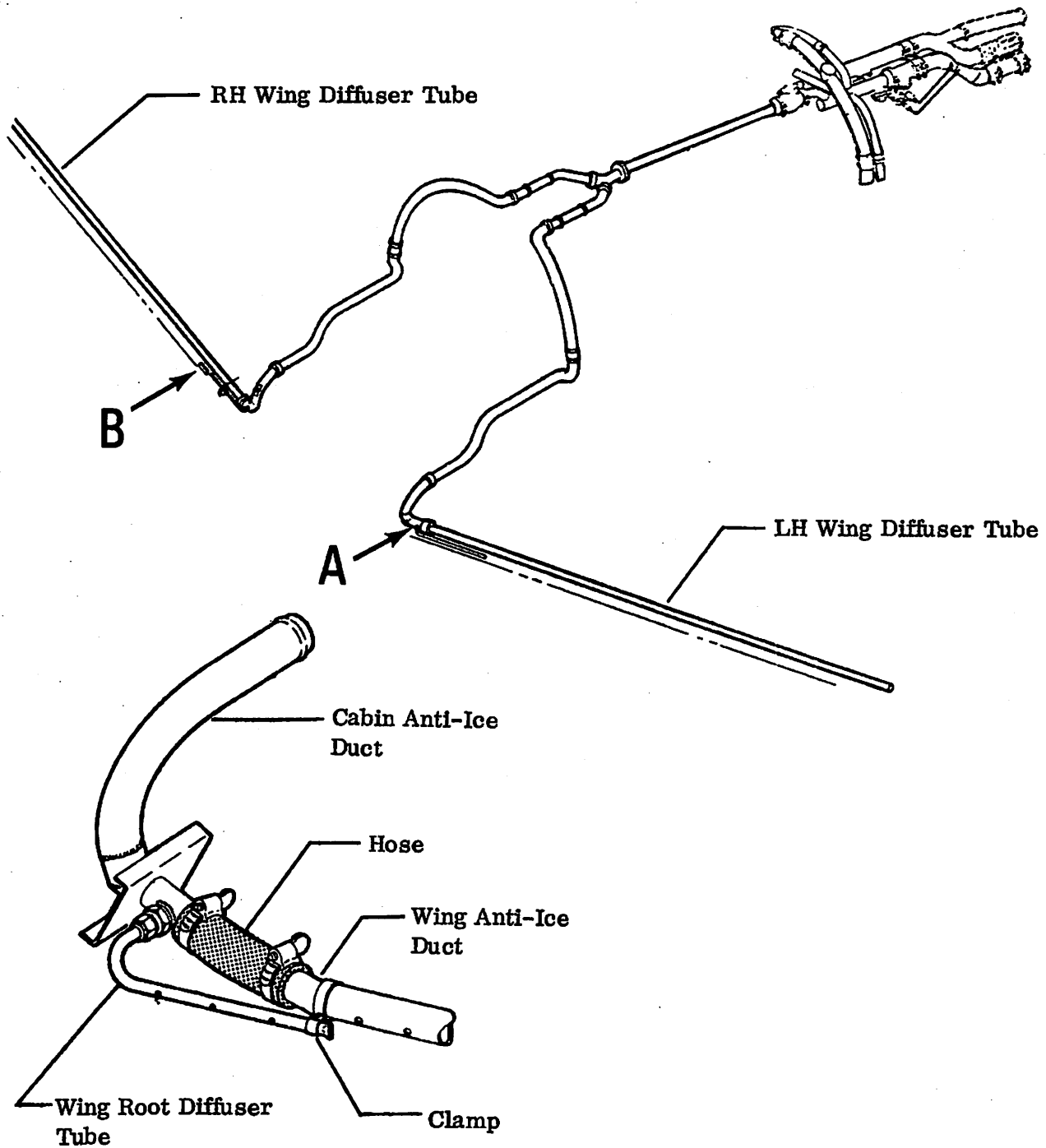
- (1) Remove attaching parts and RH wing-to-fuselage fairing from aircraft.
- (2) Remove clamp securing wing root diffuser tube to wing anti-ice duct. Loosen and remove wing root diffuser tube from aircraft.
- (3) Loosen clamps and disconnect hose from cabin anti-ice duct and wing anti-ice duct. Disconnect thermoswitch and sensor wiring at splices.
- (4) Remove attaching parts and wing rib from aircraft.
- (5) If thermoswitch is to be replaced, remove attaching parts and thermoswitch from aircraft.
- (6) If temperature sensor is to be replaced, break bond and remove sensor from aircraft.

B. Install Thermoswitch or Temperature Sensor (See figure 201)

- (1) Clean all surfaces of aerodynamic filler and sensor bonding material with 380 grit sandpaper. Wipe with a clean cloth dampened with methyl ethyl ketone. Scrape sealant from wing rib and immediate area of wing rib installation. Wipe with a clean cloth dampened with methyl ethyl ketone.
- (2) When installing thermoswitch, apply heat sink grease (Dow Corning No. 340, manufactured by Dow Corning Corp., Midland, Mich.) and position sensor as shown in detail D.
- (3) Install thermoswitch and secure with attaching parts.
- (4) Route wiring through wing rib as shown and splice.
- (5) When installing sensor, apply bonding material (Tylan No. TA2, manufactured by Tylan Corp., Torrance, Calif., or Dow Corning No. 281, manufactured by Dow Corning Corp., Midland, Mich.) and position sensor as shown in detail C.

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RH TYPICAL

Detail A

Wing Temperature Sensor and Thermoswitch Installation
Figure 201 (Sheet 1 of 2)

4-15C
B12-13

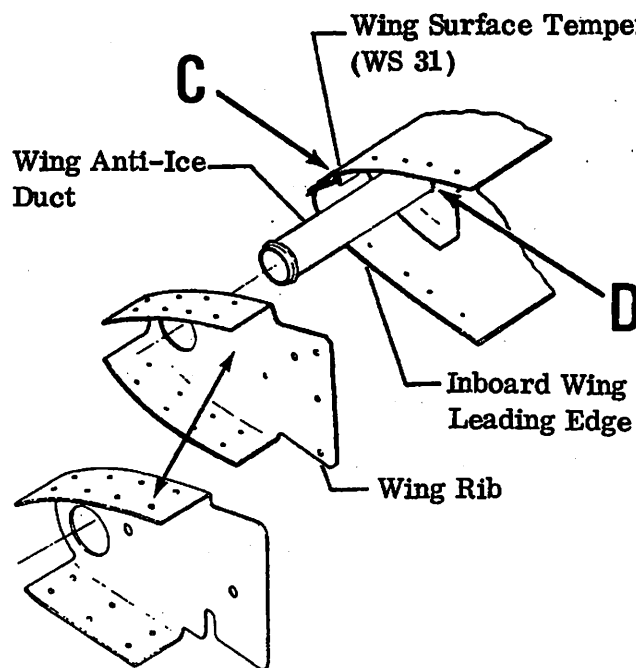
EFFECTIVITY: ALL

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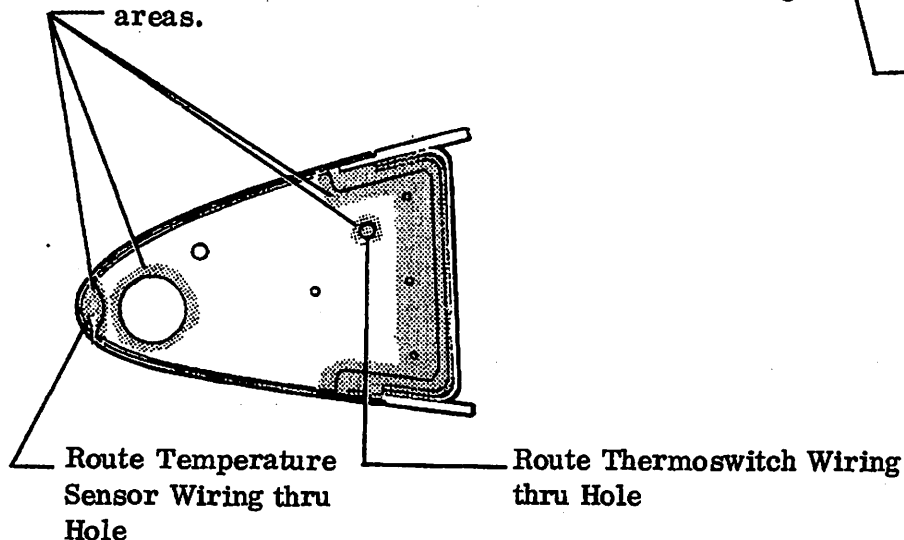
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Aircraft 25-206 and Subsequent and Earlier Aircraft equipped with Reduced Approach Speed System.

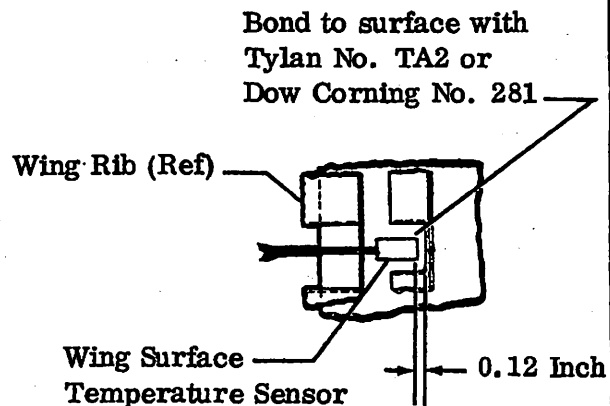
Detail B

Apply primer (Dow Corning 1200) and sealant (Silastic 732 RTV) to these areas.

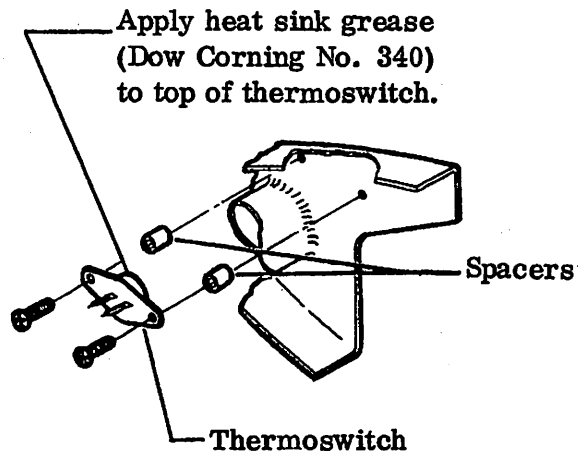


VIEW LOOKING OUTBOARD

Wing Temperature Sensor and Thermoswitch Installation
Figure 201 (Sheet 2 of 2)



Detail C



Detail D

B4-16C
D14-83B

EFFECTIVITY: ALL

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- (6) Route sensor wiring as shown and splice.
- (7) Install wing rib and secure with attaching parts.
- (8) Install hose and secure with clamps. Install wing root diffuser tube and secure with clamp.
- (9) Install wing-to-fuselage fairing.

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**HORIZONTAL STABILIZER LEADING EDGE ANTI-ICE
DESCRIPTION AND OPERATION**

1. **General**

A. Horizontal stabilizer leading edge anti-icing is accomplished by an electrically operated heat blanket bonded to the stabilizer leading edge.

2. **Description**

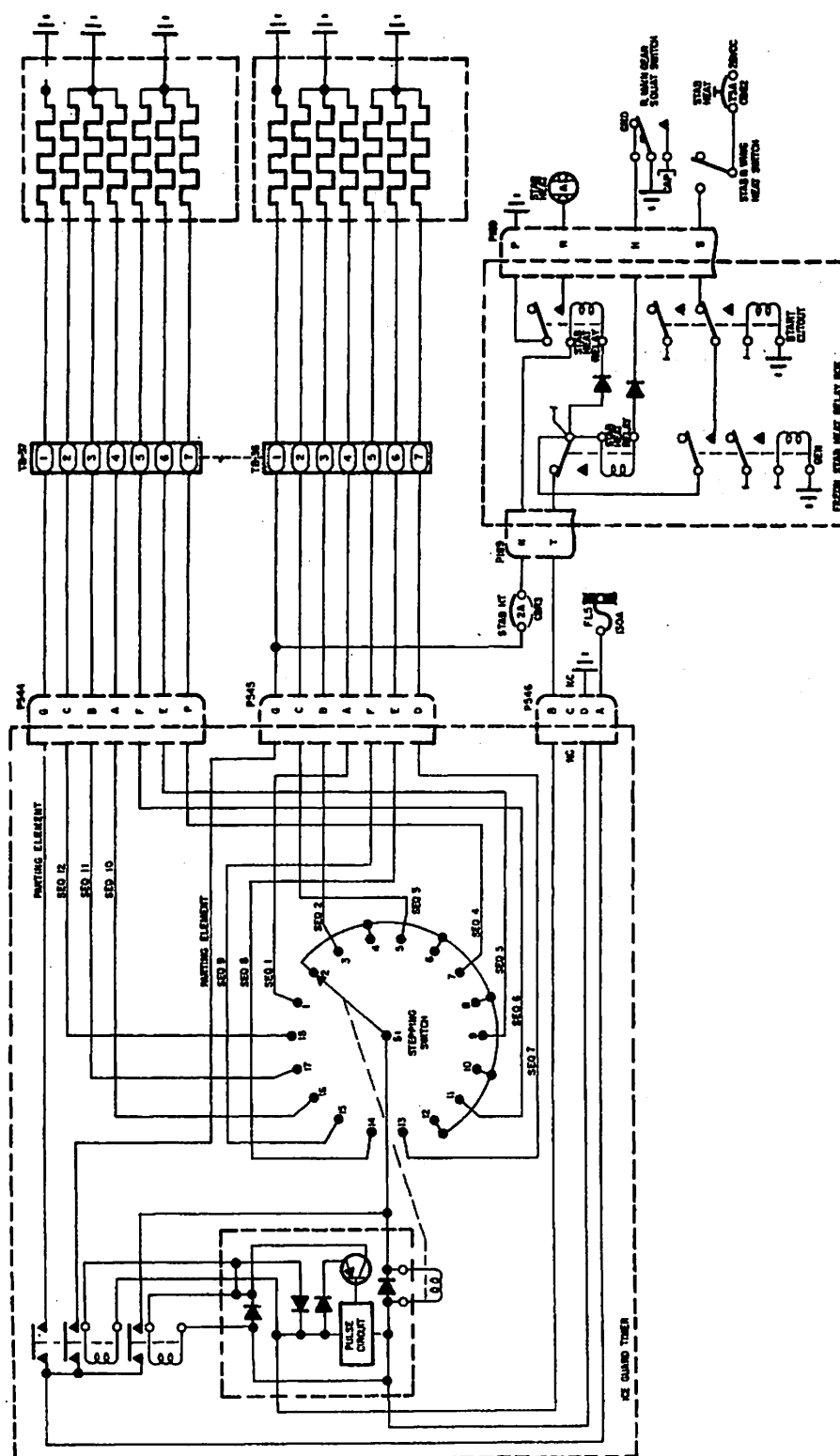
A. The stabilizer anti-ice system consists of an electrical blanket bonded to each half of the leading edge, a sequence timer, a caution light, a system switch, a circuit breaker, a current limiter and utilizes two relays within the freon and stabilizer heat relay box.

3. **Operation** (See figure 1)

CAUTION: DO NOT ATTEMPT TO OPERATIONALLY TEST THE SYSTEM WHILE THE AIRCRAFT IS ON THE GROUND. ALTHOUGH THE SYSTEM WILL OPERATE WITH THE SQUAT SWITCH DISABLED (OR WITH THE AIRCRAFT ON JACKS), OVERHEATING OF THE BLANKETS WILL OCCUR.

A. The Stab & Wing Heat Switch located on the instrument panel, controls electrical power to the sequence timer from the battery charging bus. The switch is wired through the stabilizer heat control relay to prevent use of the system during ground operation. With aircraft in flight and the Stab & Wing Heat Switch in the ON position, intermittent electrical power is distributed by the sequence timer to the individual heating elements in the blankets in a forward-to-aft sequence of 15 seconds duration each. The center or parting elements are supplied with continuous electrical power. A caution light (STAB HEAT) is located on the glareshield of the instrument panel and will illuminate when the parting elements encounter a loss of electrical power. The aircraft ammeters will reflect a pulsating current drain of approximately 60 amperes in 15-second cycles indicating proper operation of the system.

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Horizontal Stabilizer Anti-Ice System Control Schematic
Figure 1

EFFECTIVITY: ALL

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HORIZONTAL STABILIZER LEADING EDGE ANTI-ICE SYSTEM - MAINTENANCE PRACTICES

1. Trouble Shooting

- A. Since the system can only be operated during flight, the aircraft ammeters become the trouble shooting instruments. With the system operating properly, an even, pulsating current drain of approximately 60 amperes in 15-second cycles is shown on the ammeters. If the blankets check out properly, check continuity of wiring. A known good sequence timer may be substituted to eliminate a probable cause of system malfunction.

2. Removal/Installation

- A. Tools and Equipment

CAUTION: DO NOT INSTALL ANTI-ICE BLANKET IF TEMPERATURE IS BELOW 60°F OR RELATIVE HUMIDITY IS ABOVE 60%.

- DO NOT LIFT BLANKETS BY THE LEAD WIRES. THE BLANKETS SHOULD BE SUPPORTED TO PREVENT SAGGING WHILE BEING CARRIED. AVOID ANY TWISTING OR BENDING.
- DO NOT TRIM BLANKET.
- DO NOT PUNCTURE BLANKET TO RELEASE AIR BUBBLES TRAPPED DURING INSTALLATION.

NAME	PART NUMBER	MANUFACTURER	USE
Anti-Ice Blanket	4065-3481	Goodyear	
Cement	EC-1300L	3M	Bonding Blanket
Sealing Compound	No. 890B-1/2	Coast Pro-Seal	
Methyl Ethyl Ketone	(MEK)	Commercially Available	
Methyl Isobutyl Ketone	(MIBK)	Commercially Available	
2-Inch Rubber Roller		Commercially Available	
1/4-Inch Metal Hand Stitcher Roller		Commercially Available	

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NAME	PART NUMBER	MANUFACTURER	USE
2-Inch Paint Brush		Commercially Available	
Masking Tape		Commercially Available	
6-Inch Swab Stick		Commercially Available	
Cheesecloth		Commercially Available	
#160 Emery Cloth		Commercially Available	
Large Table		Commercially Available	

B. Remove Heat Blanket (See figure 201)

- (1) Remove vertical fin fairing, stabilizer fairing (on side that the blanket is being removed), and upper left access door.
- (2) Identify electrical wires corresponding to faulty blanket, then tag terminals on strip with wire number to facilitate installation.
- (3) Remove terminal block covers, cut wire bundle tie-wraps, then disconnect wires from terminal blocks.
- (4) Disconnect ground wires in upper left access hole.

NOTE: Mask off a "bonding area" 1/2 inch from periphery of blanket.

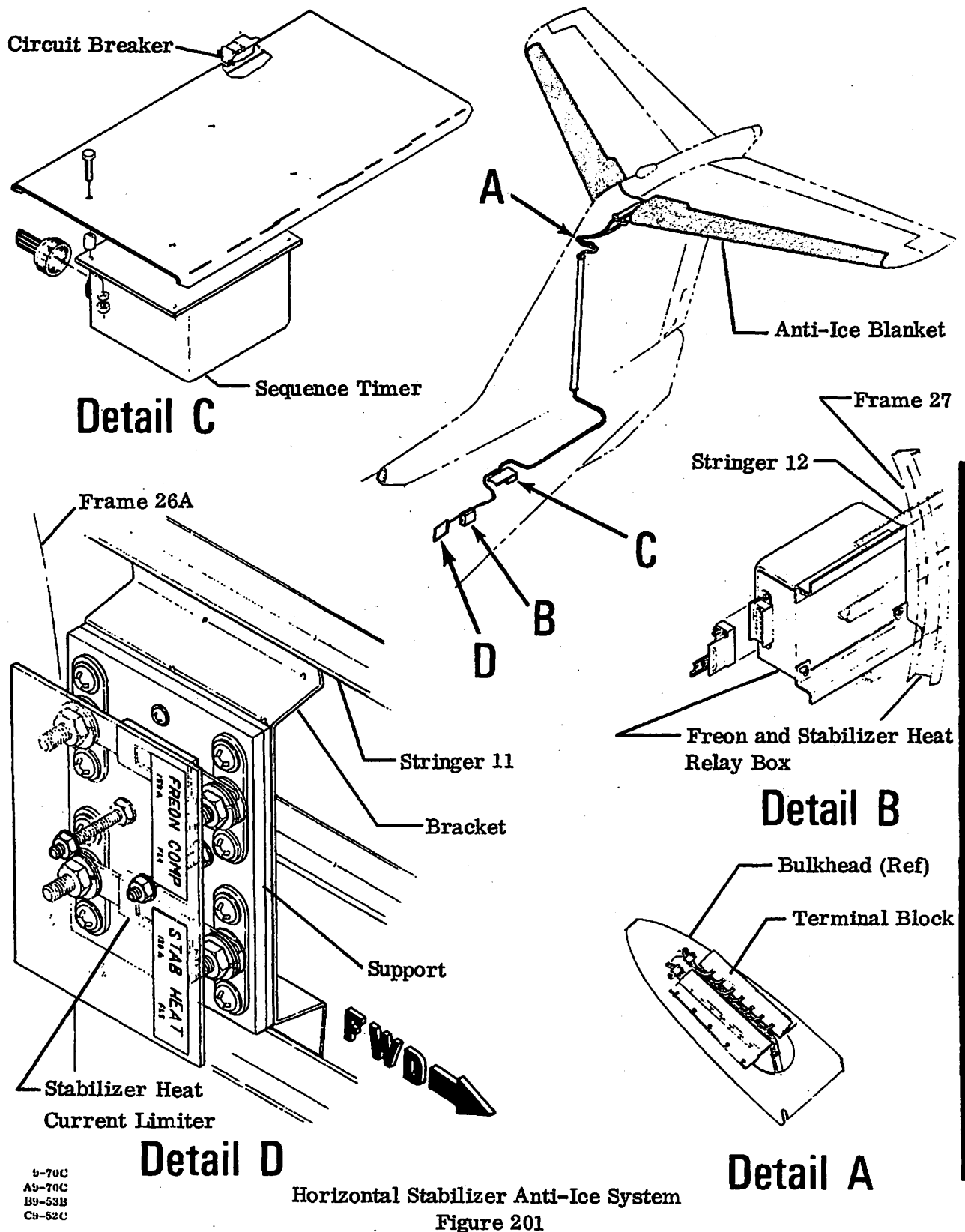
- (5) Beginning at one corner on the top side of the blanket, apply methyl ethyl ketone (MEK) at the juncture of the blanket and leading edge. As the MEK softens the cement, begin peeling the blanket forward and down around the leading edge until free of the stabilizer. Pull disconnected wires through the tube and remove blanket from aircraft.
- (6) Clean cement residue from leading edge with MEK.

C. Install Heat Blanket (See figure 201)

- (1) Install and secure sleeves over wiring.
- (2) Remove paint from bonding area (inside masking tape).
- (3) Thoroughly clean metal surface with MEK, wiping MEK off with clean cloth before it dries. Repeat this operation a minimum of 2 times.
- (4) Remove protective cloth and all traces of starch from blanket by soaking with warm water, then allow to dry thoroughly.

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- (5) Using a Wheatstone bridge or other suitable equipment capable of measuring milliohms, measure each element circuit resistance. Measure and record element circuit resistance as shown in figure 202.

CAUTION: BE CERTAIN THAT GOOD CONTACT IS OBTAINED BETWEEN THE METER LEADS AND THE CIRCUITS TO BE MEASURED. POOR CONTACT MIGHT INCREASE THE RESISTANCE VALUES OF A GOOD BOOT TO VALUES ABOVE ALLOWABLE LIMITS.

- (6) Do not install blanket if resistance is not 0.59 to 0.703 ohm.
- (7) Wash boot surfaces with MEK. Apply with clean, oil and lint-free, absorbent cloth. Wipe with fresh clean cloth before MEK dries. Allow to dry (minimum of 15 minutes). Hand buff bonding surface of boot with 180 grit abrasive paper.
- (8) Reclean bonding surface with clean cloth dampened with MEK. Change cloth frequently to avoid contamination of cleaned areas. Repeat clean operation minimum of 2 times.
- (9) Combine two parts of cement to one part of MEK, then mix thoroughly.
- (10) Apply one brush coat of cement to the heating blanket and corresponding masked area of the stabilizer. Apply cement quickly and avoid excessive rebrushing that can cause piling.
- (11) Allow cement to dry for 1 hour, then apply a second coat to both surfaces and allow to dry for a minimum of 20 minutes.
- (12) Thread heating element wires through holes in stabilizer.
- (13) Align the center parting line on blanket with the center of the leading edge and the wire connectors with the holes in the stabilizer. With blanket in this position, and using a swab moistened with MIBK, begin at the inboard end and moisten a 1-inch wide strip down the centerline. Press blanket against the stabilizer as the swab progresses toward the stabilizer tip. Immediately roll the adhered portion down chordwise to prevent air from being trapped.
- (14) Repeat the procedure progressing aft in approximately 2-inch strips until the blanket is fully adhered to the stabilizer.
- (15) Roll the taper edges of the blanket with the metal hand stitcher roller.

NOTE: The presence of any air bubbles will necessitate complete removal and installation of blanket.

- (16) Remove masking tape and remove excess cement with cloth dampened with MEK. Use care to avoid loosening edge of blanket with MEK.

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- (17) Recheck circuit resistances for possible installation damage.
- (18) Route electrical leads through the tube to the terminal block, cut to length, then install terminals. Do not connect wires at this time.
- (19) After a drying period of 72 hours, interconnect all leads and with a megger read the resistance between the leading edge skin and the leads. A 1-megohm reading or greater is acceptable. If this requirement is not met, allow 24 hours additional drying time, then recheck.
- (20) Seal the periphery of the blanket with a small fillet of 890B-1/2 sealing compound.
- (21) Connect terminals at terminal block and ground, install tie-wraps and terminal block cover, then reinstall fin fairing and access hole cover.
- (22) After sealer has dried, mask blanket and repaint exposed bonding area to match aircraft exterior.
- (23) Install stabilizer fairing.

3. Inspection/Check

A. Heat Blanket Functional Check

When any component of the stabilizer anti-ice system is replaced, a system functional check should be performed to verify proper operation.

- (1) Fabricate a functional test box in accordance with figure 203.
- (2) Lower tailcone access door and disconnect both aircraft batteries.
- (3) Disconnect electrical connectors P544 and P545 from ice guard timer and connect the test box between the ice guard timer and aircraft wiring.
- (4) Remove cover from Freon and Stabilizer heat relay box and install jumper wire between pins T and S of PJ189 (on inside of relay box). This will simulate the circuit as if one engine were operating.

NOTE: In the following steps involving positioning of the squat switches, the switches will be either in the air mode or in the ground mode.

Ground Mode - Aircraft is resting on gear with strut compressed.

Air Mode - Aircraft 25-061, 25-070 thru 25-205 not modified by AMK 75-12, the squat switch can be blocked to the air mode. On Aircraft 25-206 and Subsequent and aircraft modified by AMK 75-12, the aircraft must be placed on jacks so that the strut will extend.

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Note

To correct for temperature variation take the resistance value and subtract 0.0005 ohm for every 1° F above 68° or add 0.0005 ohm for every 1° below 68° F.

Circuit *	Required Value (Ohms)	Recorded Value (Ohms)	Corrected Value (Ohms)
1-6	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
2-3	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
4-3	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
5-3	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
7-8	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
9-8	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	
10-8	0.59 to 0.703 at 68° F	Resistance Read _____ - Meter Leads _____ Actual Resistance _____	

* Boot leads are numbered to identify circuits.

Anti-Ice Blanket Resistance Check
Figure 202

EFFECTIVITY: ALL

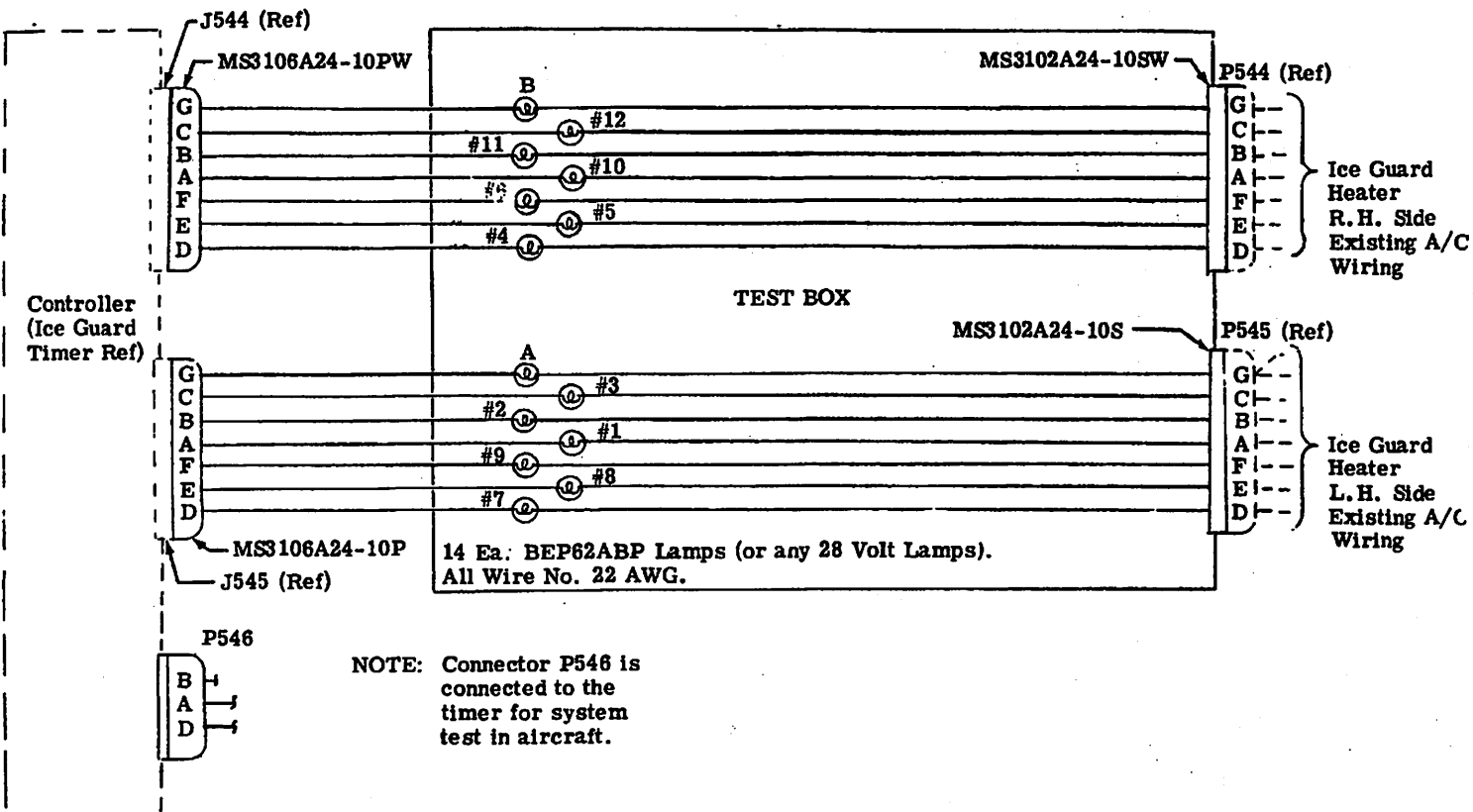
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- (5) Set the squat switches to the air mode.
- (6) Assure that the STAB HEAT circuit breaker is pulled and connect aircraft batteries.
- (7) Connect an external power cart to the aircraft. On Aircraft 25-090 and 25-103 and Subsequent, the Battery Switches will have to be set ON to connect the external power to the battery bus.

CAUTION: OBSERVE EXTERNAL POWER CART AMMETER WHEN SETTING STAB & WING HEAT SWITCH TO ON. AN AMMETER INCREASE OF MORE THAN 5 AMPERES INDICATES THAT THE TEST BOX IS IMPROPERLY CONNECTED.

- (8) Depress the STAB HEAT circuit breaker and set STAB & WING HEAT switch to ON. Check external power cart ammeter.
- (9) Lamps A and B will be illuminated throughout the test. These lamps correspond to the parting element of each ice guard.
- (10) Lamps numbering 1 thru 12 will sequence in order, but can start at any one of the 12 lamps, with only one lamp on at a time.
- (11) Each lamp will be illuminated approximately 15 (± 2) seconds.
- (12) The amber STAB HEAT caution light will illuminate.
- (13) Set squat switches to the ground mode. The sequencing lamps, parting element lamps (A and B) and the amber STAB HEAT caution light will go out.
- (14) Set STAB HEAT and Battery Switches to OFF. Disconnect external power cart and aircraft batteries.
- (15) Disconnect electrical plugs from test box and connect to ice guard timer.
- (16) Remove jumper wire from Freon and Stabilizer heat relay box and install relay box.
- (17) Connect aircraft batteries and secure tailcone access door.



Horizontal Stabilizer Functional Test Box
Figure 203

EFFECTIVITY: ALL

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NACELLE ANTI-ICE SYSTEM - DESCRIPTION AND OPERATION

1. Description

- A. Each nacelle anti-ice system consists of the INLET HTR warning light, a system switch, a current limiter, a temperature switch, an anti-ice relay and heating elements bonded to the nacelle nose cap assembly. On Aircraft certified for operation in France, a reed relay assembly is included in the nacelle anti-ice system.
- (1) Component Description
- (a) The INLET HTR (amber) warning light is located on the glare-shield warning light panel.
 - (b) The LH and RH Nacelle Heat Switches are located on the pilot's subpanel.
 - (c) The LH and RH current limiters are located on the current limiter panel.
 - (d) The temperature overheat switches are an integral part of the nacelle heaters.
 - (e) The anti-ice relays are located on each nacelle bulkhead.
 - (f) On Aircraft certified for operation in France, a reed relay assembly consists of three reed relays. Each reed relay senses the current flow thru the leads of the heater elements and energizes the relay, opening the ground circuit to the INLET HTR warning light. If one or more heating elements fail, the relays will deenergize and complete the ground to illuminate the INLET HTR warning light.

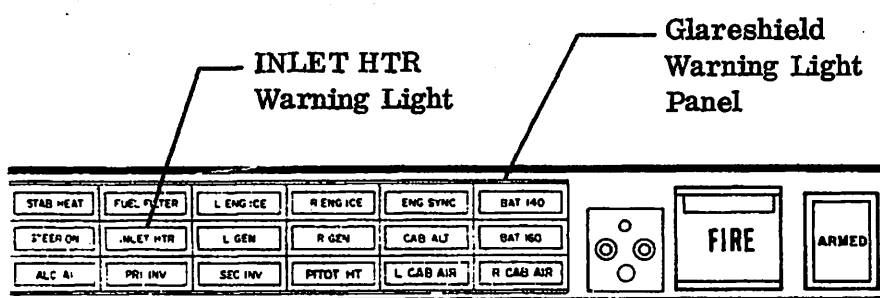
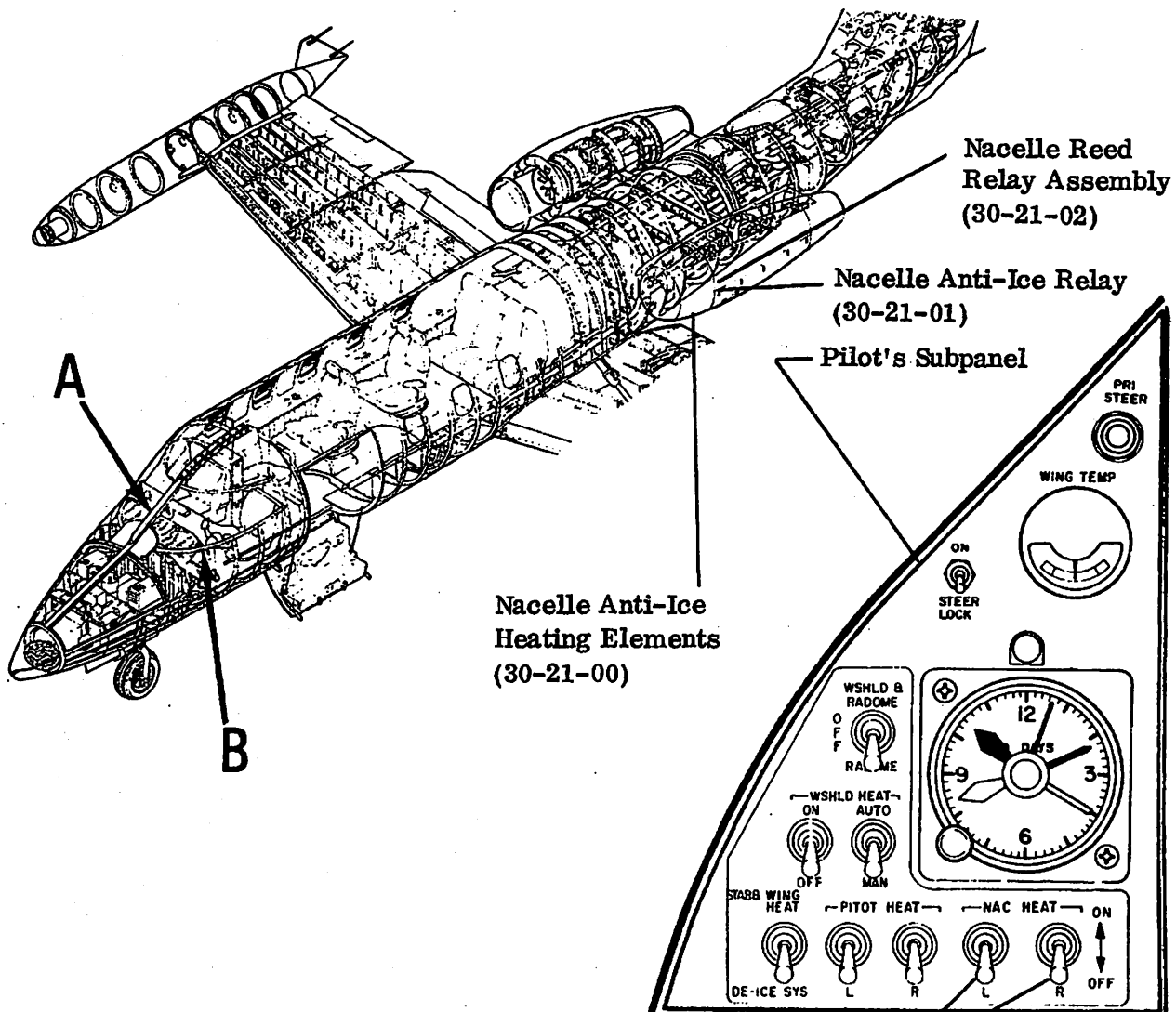
2. Operation (See figure 1)

CAUTION: AVOID UNNECESSARY OPERATION OF THE NACELLE ANTI-ICE SYSTEM. SET THE NACELLE HEAT SWITCHES LOCATED ON THE PILOT'S SUBPANEL TO OFF AND PULL THE NAC HT CIRCUIT BREAKERS LOCATED ON THE COPILOT'S CIRCUIT BREAKER PANEL. UNNECESSARY OPERATION OF NACELLE ANTI-ICE SYSTEM WHILE THE AIRCRAFT IS ON THE GROUND WILL REDUCE THE SERVICE LIFE OF THE HEATING ELEMENTS.

- A. Setting the applicable Nacelle Heat Switch to ON allows power to energize the applicable inlet heater via the nacelle anti-ice relay. If inlet temperature rises above 190° (±3°)F while aircraft is on the ground, the temperature overheat switch will energize the INLET HTR warning light. The Nacelle Heat Switch should then be set to OFF. The system is protected by two current limiters. The INLET HTR warning light is inoperative when the aircraft is in flight.

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Detail A

Nacelle Anti-Ice System Locator
Figure 1

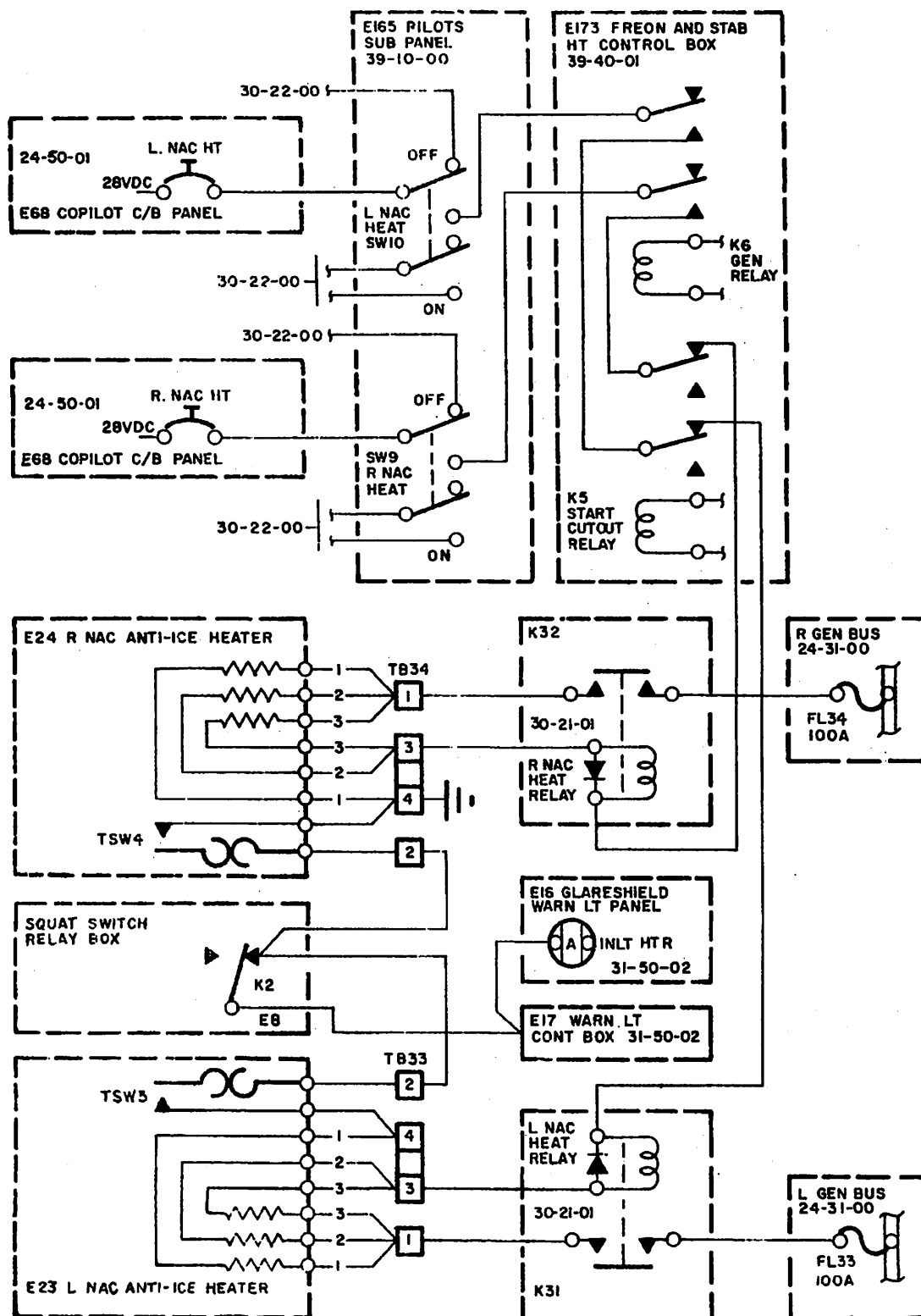
LH and RH Nacelle Heat Switches

Detail B

EFFECTIVITY: ALL

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Nacelle Anti-Ice Electrical Control Schematic
Figure 2 (Sheet 1 of 2)

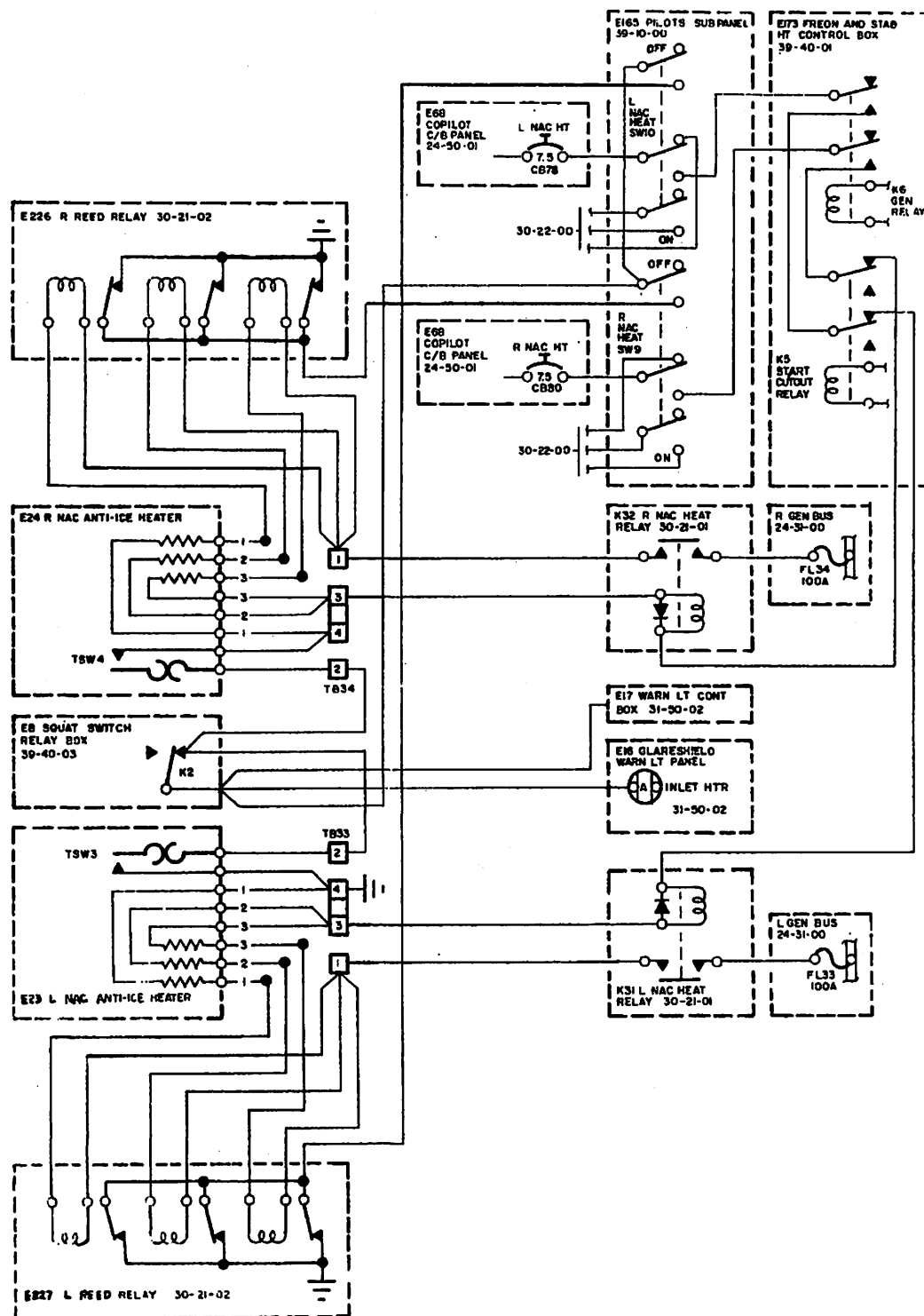
EFFECTIVITY: ALL, except Aircraft Certified for
Operation in France

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Nacelle Anti-Ice Electrical Control Schematic
Figure 2 (Sheet 2 of 2)

EFFECTIVITY: Aircraft Certified for Operation in France

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NACELLE ANTI-ICE RELAY - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Removal and installation procedures for LH and RH nacelle anti-ice relays are identical.

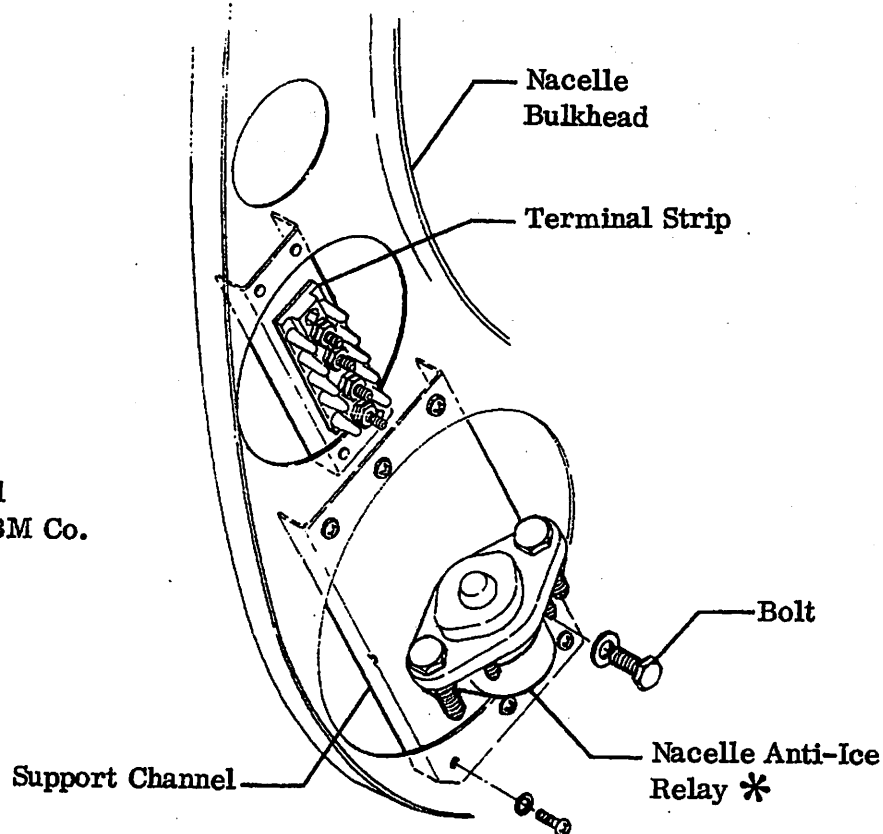
A. Remove Anti-Ice Relay (See figure 201)

- (1) Remove electrical power from aircraft.
- (2) Remove engine lower nacelle.
- (3) Disconnect electrical wiring from relay. Tag all wiring to aid re-installation.
- (4) Remove attaching parts and relay from support channel on nacelle bulkhead.

B. Install Anti-Ice Relay (See figure 201)

- (1) Bond relay to support channel using EC 776, 3M Co. Install relay and secure with attaching parts.
- (2) Connect electrical wiring to relay.
- (3) Install engine lower nacelle.
- (4) Restore electrical power to aircraft.

*Bond relay to support channel using EC 776, 3M Co.



Nacelle Anti-Ice Relay Installation
Figure 201

9-300B

EFFECTIVITY: ALL

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NACELLE ANTI-ICE REED RELAY - MAINTENANCE PRACTICES

1. Removal/Installation

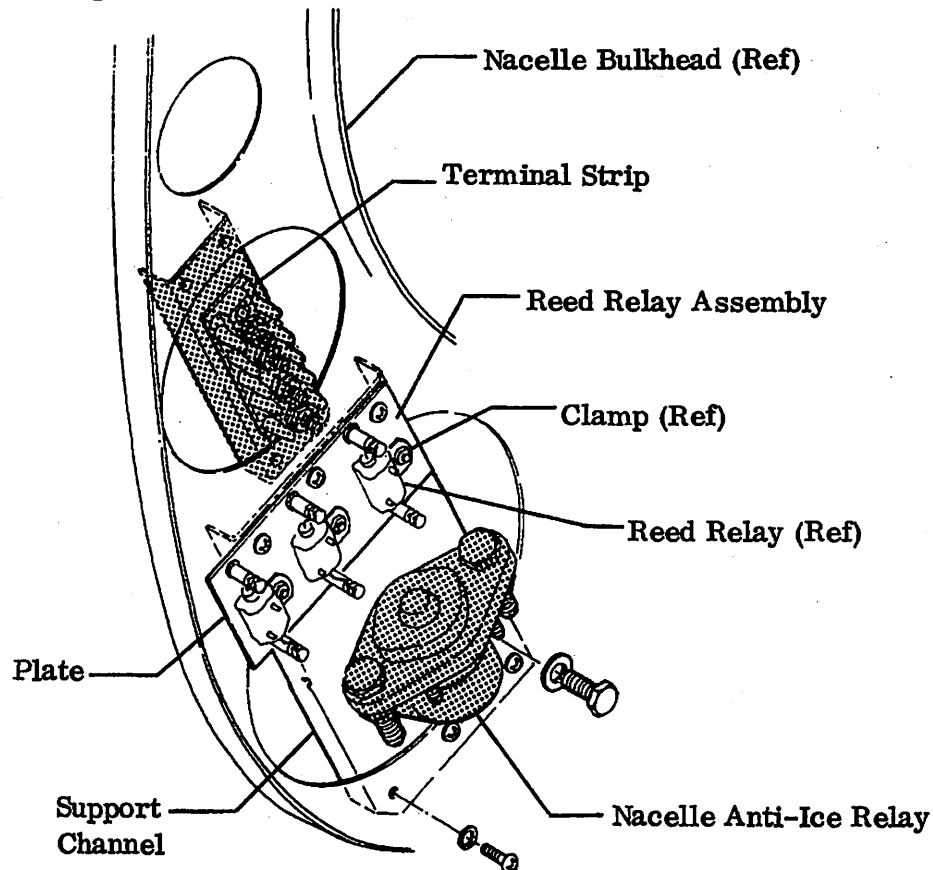
NOTE: Removal and installation procedures for LH and RH nacelle anti-ice reed relays are identical.

A. Remove Anti-Ice Reed Relay Assembly (See figure 201)

- (1) Remove electrical power from aircraft.
- (2) Remove engine lower nacelle.
- (3) Disconnect electrical wiring from relays. Tag wiring to aid reinstallation.
- (4) Remove attaching parts and reed relay assembly from support on nacelle bulkhead.

B. Install Anti-Ice Reed Relay Assembly (See figure 201)

- (1) Install reed relay assembly and secure with attaching parts.
- (2) Connect electrical wiring to reed relays.
- (3) Restore electrical power to aircraft.
- (4) Install engine lower nacelle.



Nacelle Anti-Ice Reed Relay Installation
Figure 202

9-300B

EFFECTIVITY: Aircraft Certified for Operation in France

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MEMORANDUM FOR THE DIRECTOR

RE: [Illegible]

DATE: [Illegible]

1. [Illegible]

2. [Illegible]

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NACELLE ANTI-ICE HEATING ELEMENTS - MAINTENANCE PRACTICES

1. Removal/Installation

A. Nacelle Heating Elements

NOTE: The heating elements are an integral part of the nacelle nose cap assembly. Refer to SSK 904 and SSK 922 for specific installation instructions.

2. Inspection/Check

A. Nacelle Anti-Ice Heating Elements Resistance Check

NOTE: The nacelle anti-ice heating elements should be checked for serviceability in accordance with current inspection requirements, Chapter 5. Perform resistance check using a certified Wheatstone Bridge tester, low resistance test leads, and a thermometer with an accuracy of $\pm 1^{\circ}\text{F}$. The nacelle anti-ice heating elements are checked for resistance as follows:

- (1) Remove electrical power from aircraft.
- (2) Remove LH engine lower nacelle.
- (3) Disconnect six wire terminals numbered 1, 2, and 3 from terminal posts.
- (4) Zero Wheatstone Bridge Tester by following instructions shown on lid of tester.
- (5) Connect test leads to tester posts, measure resistance, and record on a form similar to figure 201.
- (6) Measure resistance of each of the three heating element circuits and the three heating element circuits connected in parallel. Record the resistance values in figure 201.

CAUTION: GOOD CONTACT MUST BE OBTAINED BETWEEN TEST LEADS AND THE CIRCUITS TESTED. POOR CONTACT MAY RESULT IN AN INCREASE OF RESISTANCE OF A SERVICEABLE HEATING ELEMENT ABOVE THE ALLOWABLE LIMITS.

- (7) Determine ambient temperature in degrees Fahrenheit by using a thermometer.
- (8) Correct for temperature variation by taking the actual resistance value and subtract 1% for every 5°F above 75°F or add 1% for every 5°F below 75°F . If ambient temperature is above 80°F or below 70°F , corrections for temperature variation must be made. Record corrected values in figure 201.

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EXAMPLE: Circuit No. 1
 Temperature 80° F
 Resistance of Meter Leads 0.135 ohms
 Resistance Read 0.950 ohms
 Compute for corrected value.

Circuit	Required Value (Ohms)	Recorded Value (Ohms)	Corrected Value (Ohms)
Circuit No. 1	0.836 ±10% (0.7942 to 0.8778 at 75° F)	Resistance Read 0.950 - Meter Test Leads -0.135 Actual Resistance 0.815	0.807

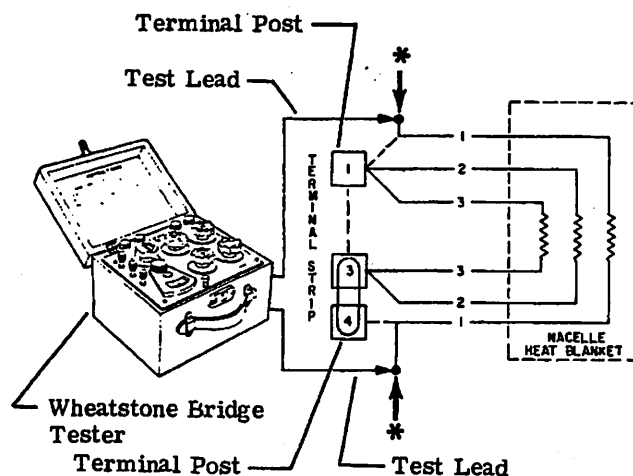
OPERATION:

Subtract meter test leads (0.135 from resistance reading (0.950). The result is the actual resistance (0.815).

Correct for temperature (80° F by subtracting 1% of actual resistance (0.815). The result is the corrected value (0.807).

CONCLUSION:

The corrected value (0.807) ohms is within the required value. Circuit No. 1 is O.K.



* Disconnect wire terminals from terminal posts before performing resistance check.

TABLE I

Circuit	Required Value (Ohms)	Recorded Value (Ohms)	Corrected Value (Ohms)
Circuit No. 1	0.836 ±10% (0.7524 to 0.9196 at 75° F)	Resistance Read _____ - Meter Test Leads _____ Actual Resistance _____	
Circuit No. 2	0.795 ±10% (0.7155 to 0.8745 at 75° F)	Resistance Read _____ - Meter Test Leads _____ Actual Resistance _____	
Circuit No. 3	0.891 ±10% (0.8019 to 0.9801 at 75° F)	Resistance Read _____ - Meter Test Leads _____ Actual Resistance _____	
Total (Circuits 1, 2 & 3 in Parallel)	0.27961 ±10% (0.251649 to 0.307571 at 75° F)	Resistance Read _____ - Meter Test Leads _____ Actual Resistance _____	

Nacelle Anti-Ice Heating Elements Resistance Check
 Figure 201

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- (9) Compare actual resistance values with corrected values for circuits 1, 2, and 3 and total circuits 1, 2, and 3 in parallel. The nacelle anti-ice heating elements are serviceable if: the corrected resistance values for all three circuits fall within the specified tolerances; or the corrected resistance value for the three circuits checked in parallel fall within the specified tolerances; or the four corrected values fall within the specified tolerances.
- (10) Connect the six terminal wires to the terminal posts.
- (11) Install LH engine lower nacelle.
- (12) Repeat steps (1) thru (12) for RH engine.
- (13) Restore electrical power to aircraft.

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ENGINE ANTI-ICE SYSTEM - DESCRIPTION AND OPERATION

1. Description (See figure 1)

- A. Each engine anti-ice system consists of an anti-ice control valve, an anti-ice pressure switch, a warning light on the glareshield warning light panel, and a system switch.
- B. Component Description
 - (1) The anti-ice control valve is located on the engine. For further information, refer to the Engine Maintenance Manual SEI-186.
 - (2) The anti-ice pressure switch is located on the engine.
 - (3) The L(R) ENG ICE (amber) warning light is located on the glareshield warning light panel and indicates system malfunction.
 - (4) The Nacelle Heat Switches (L and R) are located on the pilot's subpanel.

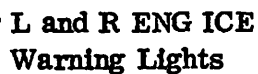
2. Operation (See figure 2)

- A. When the Nacelle Heat Switch is set to ON, the applicable anti-ice control valve is deenergized and 28 vdc is applied to the anti-ice pressure switch.

NOTE: Under normal no-ice conditions, the anti-ice control valves are electrically energized and held to the closed position. When anti-icing is required, the valves are deenergized (opened); therefore, with a complete electrical power failure, engine anti-ice is still available.

- B. With the anti-ice control valve open (deenergized), bleed air from the compressor section is directed to the front frame manifold and radially inward to the engine forward frame struts, inlet guide vanes, and inlet bullet nose cone. A drop in anti-ice duct pressure is sensed by the pressure switch which energizes the L(R) ENG ICE warning lights to indicate a malfunction in the anti-ice system.

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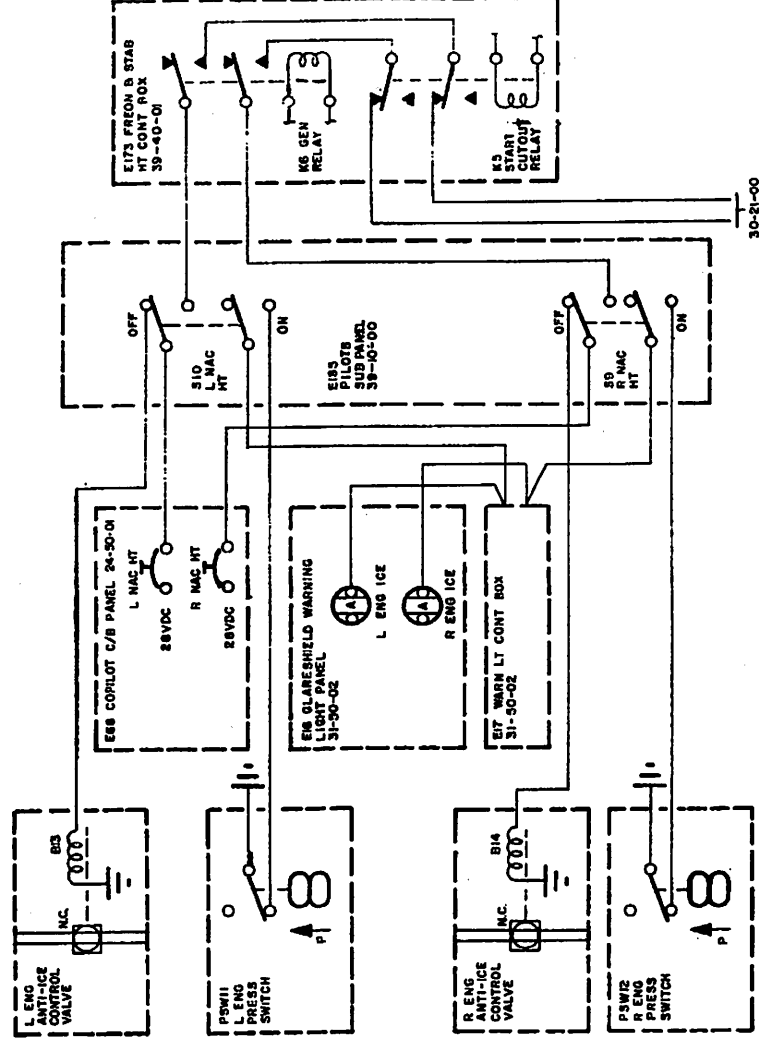


Detail B

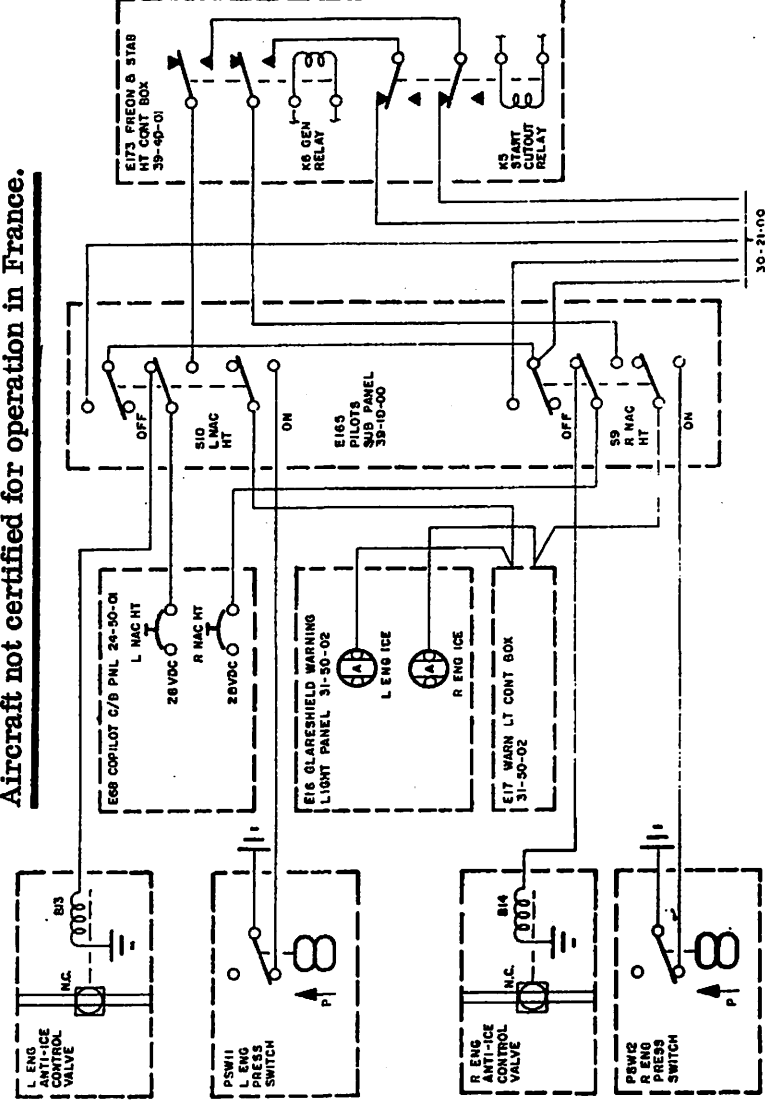
LH and RH Nacelle Heat Switches

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Aircraft not certified for operation in France.



Aircraft certified for operation in France.

Engine Anti-Ice Electrical Control Schematic

Figure 2

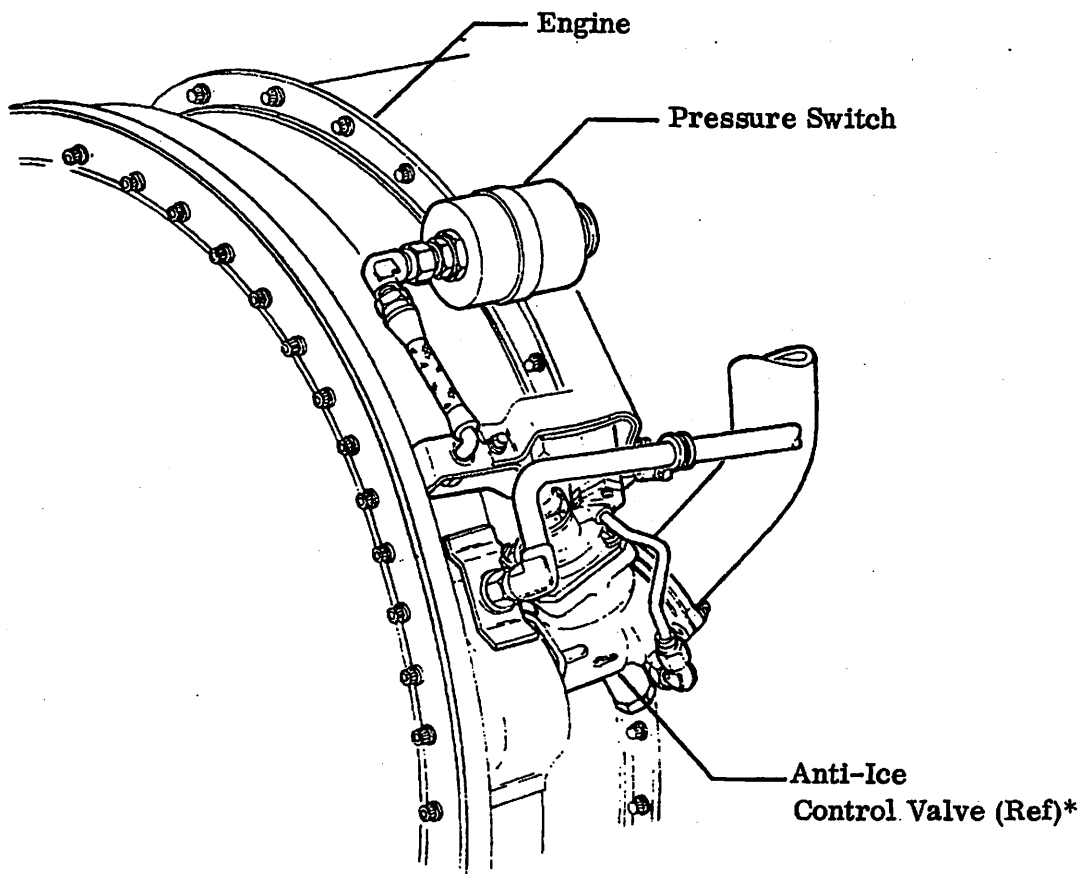
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ENGINE ANTI-ICE PRESSURE SWITCH - MAINTENANCE PRACTICES

1. Removal/Installation

- A. Remove Engine Anti-Ice Pressure Switch (See figure 201)
- (1) Remove engine nacelle.
 - (2) Disconnect electrical connector from pressure switch.
 - (3) Disconnect plumbing from pressure switch.
 - (4) Remove attaching parts and pressure switch from engine.
- B. Install Engine Anti-Ice Pressure Switch (See figure 201)
- (1) Install pressure switch and secure with attaching parts.
 - (2) Connect plumbing to pressure switch.
 - (3) Connect electrical connector to pressure switch.
 - (4) Install engine nacelle.



*Refer to Engine Maintenance Manual SEI-186
for further information.

Engine Anti-Ice Pressure Switch Installation
Figure 201

7-9A-1

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2. Adjustment/Test

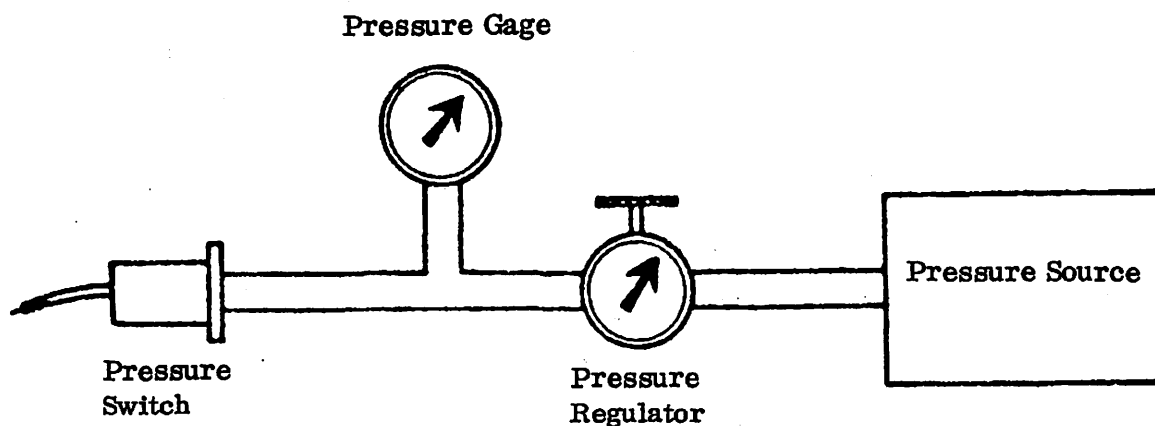
A. Perform Functional Test of Engine Anti-Ice Pressure Switch (See figure 202)

- (1) Connect variable pressure source to pressure switch.
- (2) Connect an ohmmeter across pins B and C of electrical connector. Ohmmeter shall indicate open circuit with no pressure applied.
- (3) Slowly increase pressure until switch actuates (continuity between pins B and C). Switch actuation shall occur at 6 psi as indicated on test pressure gage.
- (4) While holding pressure on switch, connect ohmmeter across pins A and B of electrical connector. Ohmmeter shall indicate an open circuit.
- (5) Slowly decrease pressure until switch resets (continuity between pins A and B). Switch shall reset at 4 psi minimum.
- (6) Disconnect pressure switch from test setup and install in aircraft.

3. Inspection/Check

A. Engine Anti-Ice Pressure Switch Inspection

- (1) Inspect electrical connector for damage or corrosion.
- (2) Inspect pressure port for foreign matter.



Engine Anti-Ice Pressure Switch Functional Test Setup
Figure 202

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PITOT AND STATIC - DESCRIPTION AND OPERATION

1. Description

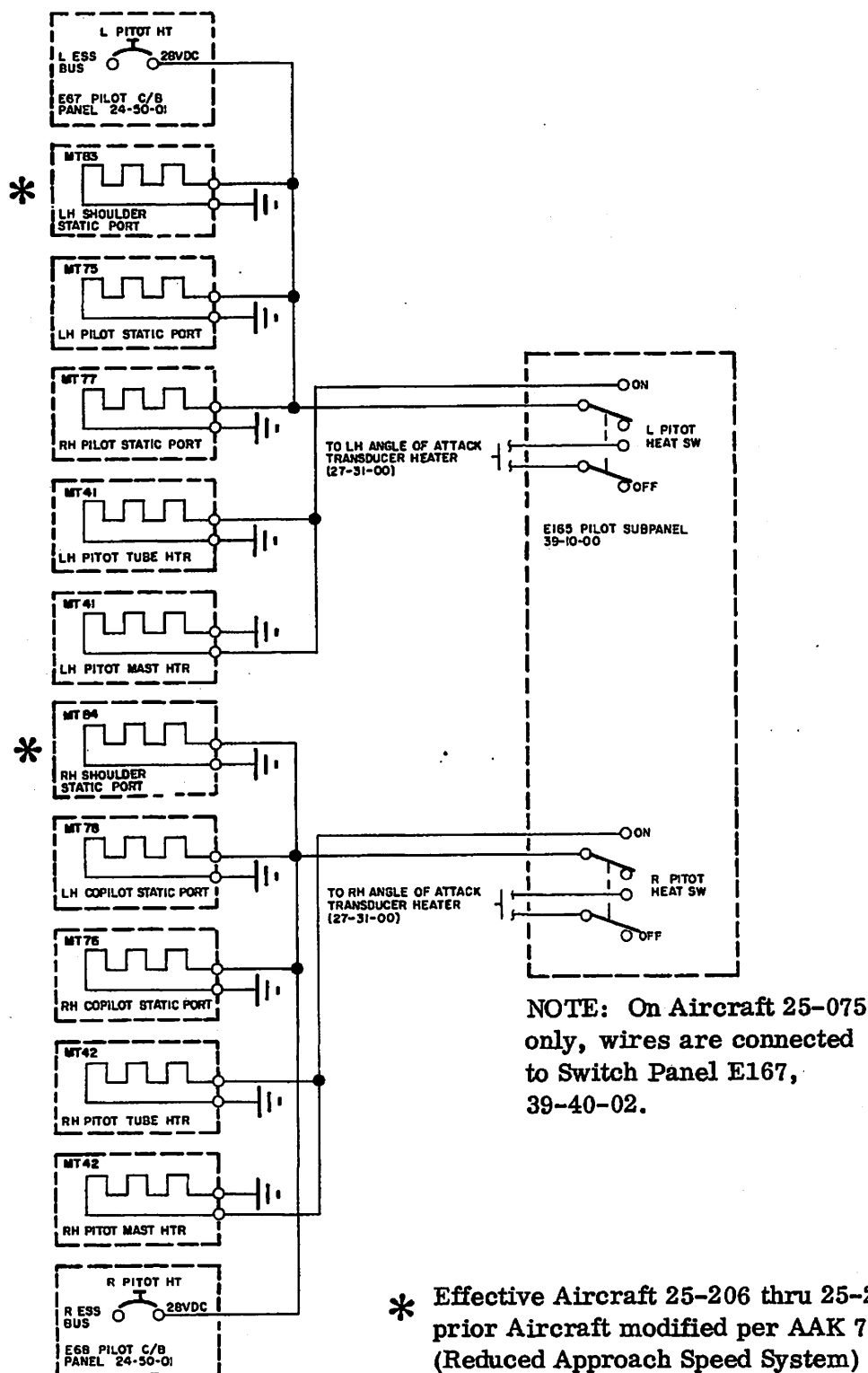
- A. Integral heating elements are used in six static ports, the two pitot tubes and masts, and the two angle of attack transducers.
- B. Each pitot tube and pitot mast is equipped with an electrical heating element to prevent moisture from freezing on the tube and mast. The pitot tubes and masts heating elements are powered by 28 vdc and controlled by the Pitot Heat Switches. The pitot tubes and masts are located on each side of the nose compartment at FS 150. Refer to Chapter 34 for pitot tube maintenance practices.
- C. Each of the four heated static ports is equipped with an electrical heating element to prevent moisture from freezing and obstructing the port openings. Four static ports are located at FS 155 and FS 159 on the LH side and FS 155 and 159 on the RH side.
- D. Two shoulder static ports are located in front of the windshield at RBL and LBL 18. On Aircraft 25-206 and Subsequent and prior Aircraft modified per AAK 76-4 (Reduced Approach Speed System), heating elements are installed on each shoulder static port to prevent moisture from freezing and obstructing the port openings.
- E. Each angle of attack transducer is equipped with an electrical heating element to prevent moisture from freezing on the transducer vane. The heater is powered by 28 vdc and is controlled by the Pitot Heat Switches.
- F. On Aircraft 25-290 and Subsequent and prior Aircraft modified per AAK 79-2 (Installation of Pitot Heat Indicating System), a pitot current sensor system is installed to monitor current flow to both pitot tubes and masts. For pitot current sensor system operation, refer to 30-31-00.

2. Operation (See figure 1)

- A. The static ports are powered by 28 vdc and are heated when the Battery Switches are set to ON.
- B. The pitot tubes and masts and angle of attack transducer vanes will heat when the Pitot Heat Switches are set to ON.

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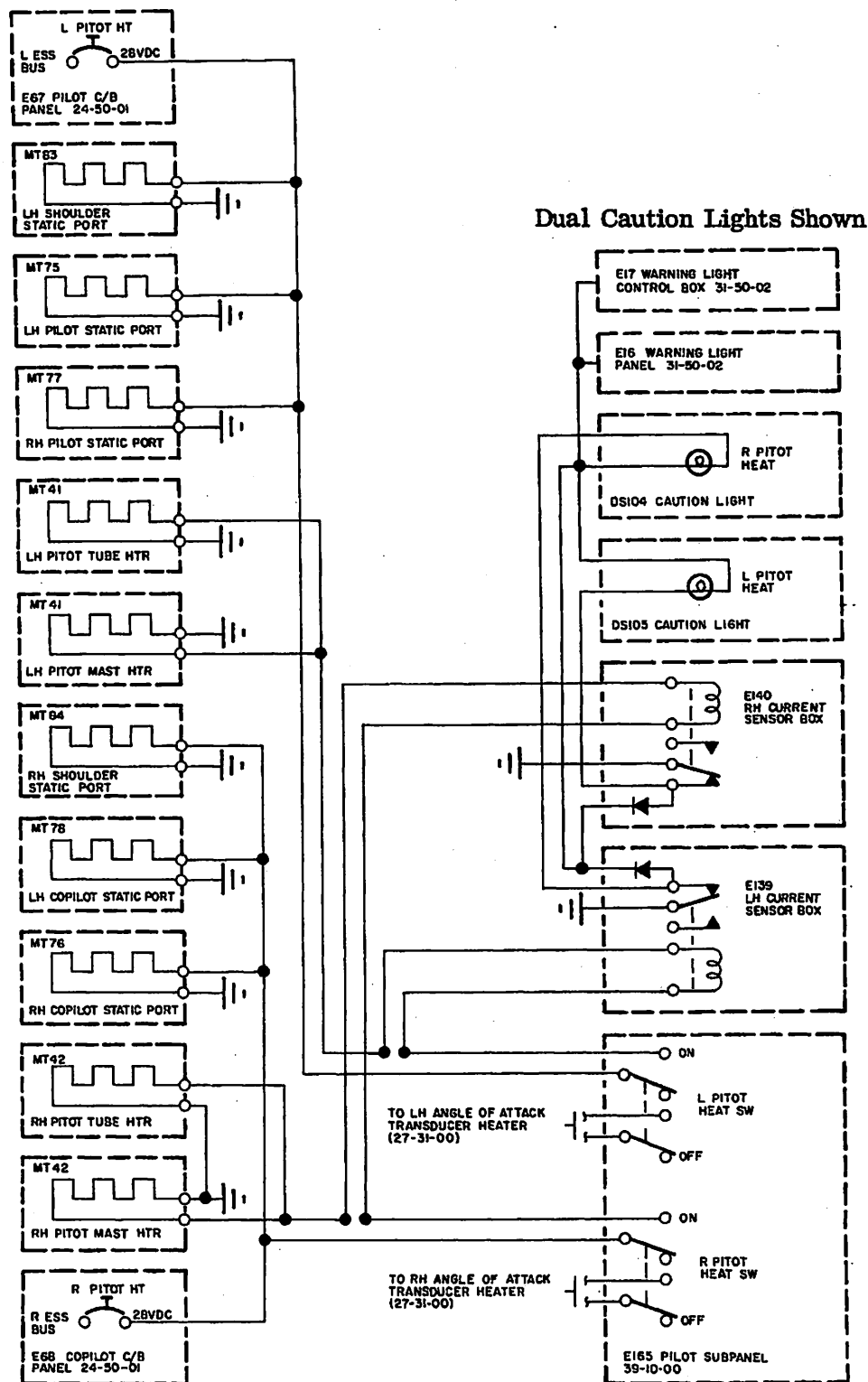
Pitot and Static Systems Electrical Control Schematic
Figure 1 (Sheet 1 of 4)

EFFECTIVITY: 25-061, 25-070 thru 25-289 Not Modified per AAK 79-2 (Installation of Pitot Heat Indicating System)

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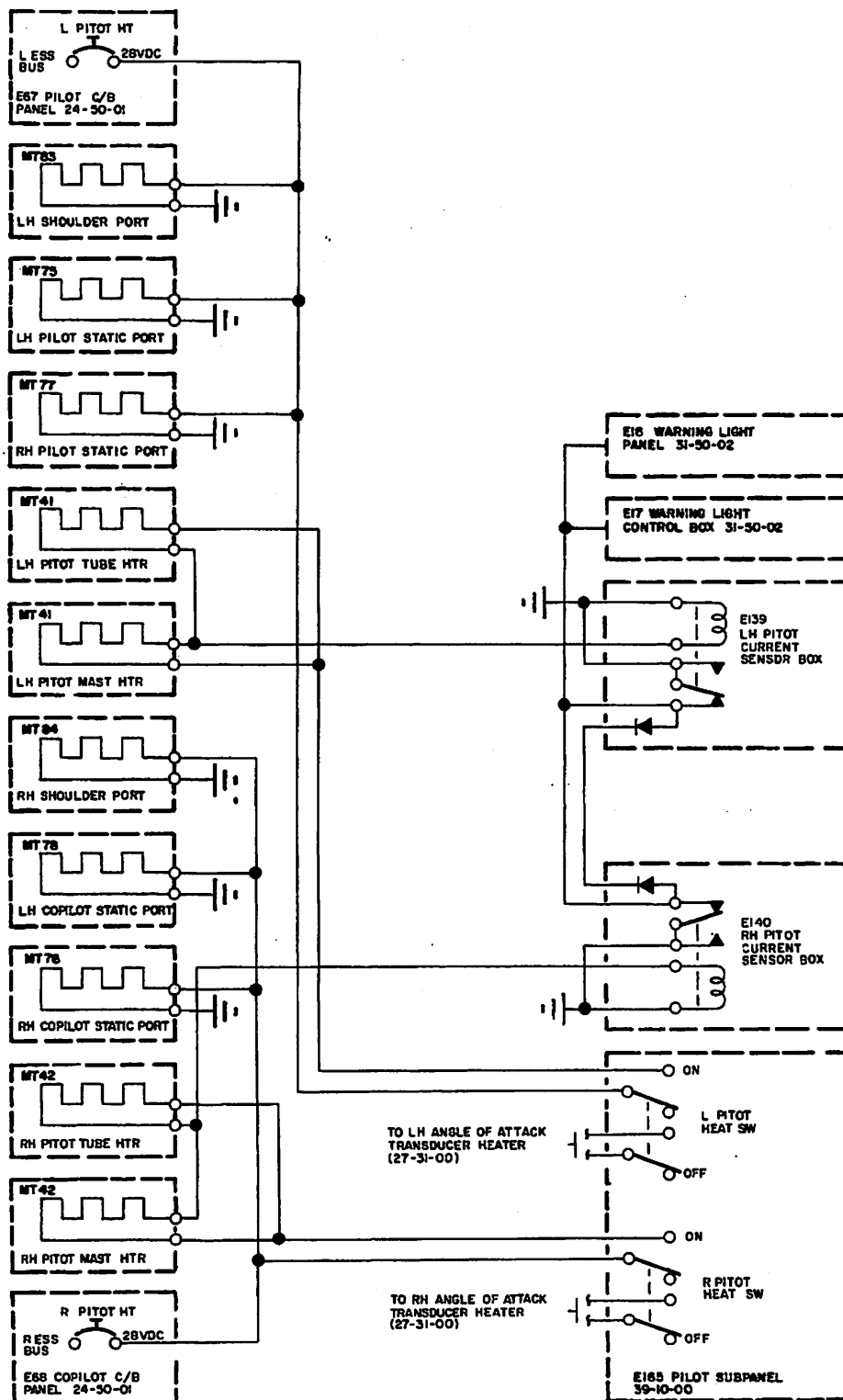
Pitot and Static Systems Electrical Control Schematic
Figure 1 (Sheet 2 of 4)

EFFECTIVITY: 25-061, 25-070 thru 25-289 Modified
per AAK 79-2 (Installation of Pitot Heat
Indicating System)

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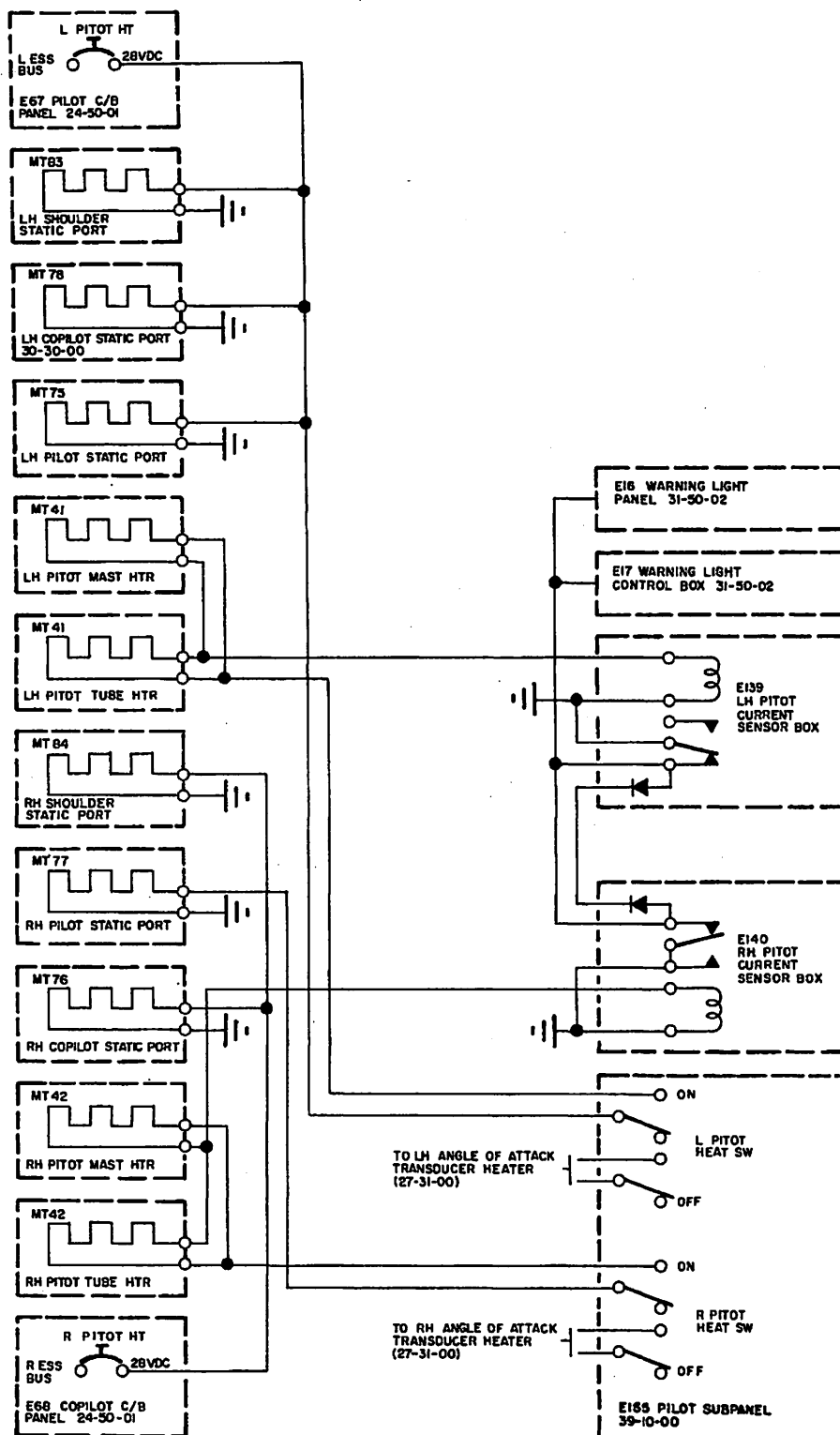
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Pitot and Static Systems Electrical Control Schematic
Figure 1 (Sheet 3 of 4)

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Pitot and Static Systems Electrical Control Schematic
Figure 1 (Sheet 4 of 4)

EFFECTIVITY: 25-339 AND SUBSEQUENT

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PITOT AND STATIC - TROUBLE SHOOTING

1. General

- A. Trouble shooting the pitot and static anti-ice systems can be isolated to one of four areas: (1) circuit breaker failure, (2) switch failure, (3) component failure or (4) loose or damaged wiring.
- B. To verify operation, hold hand near the static ports, pitot heads and angle of attack transducer vane.

2. Tools and Equipment

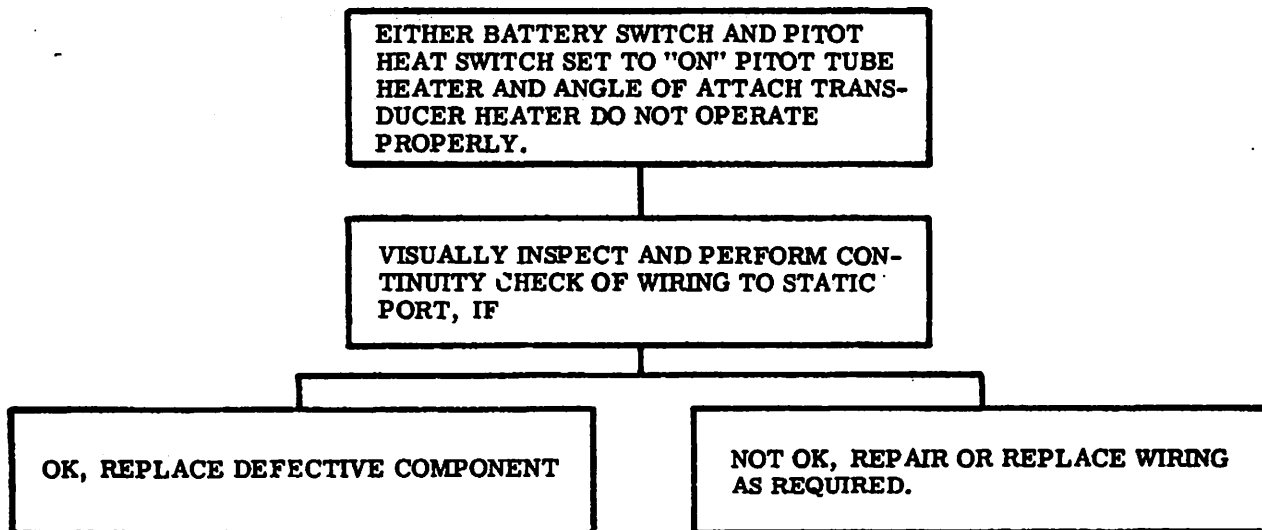
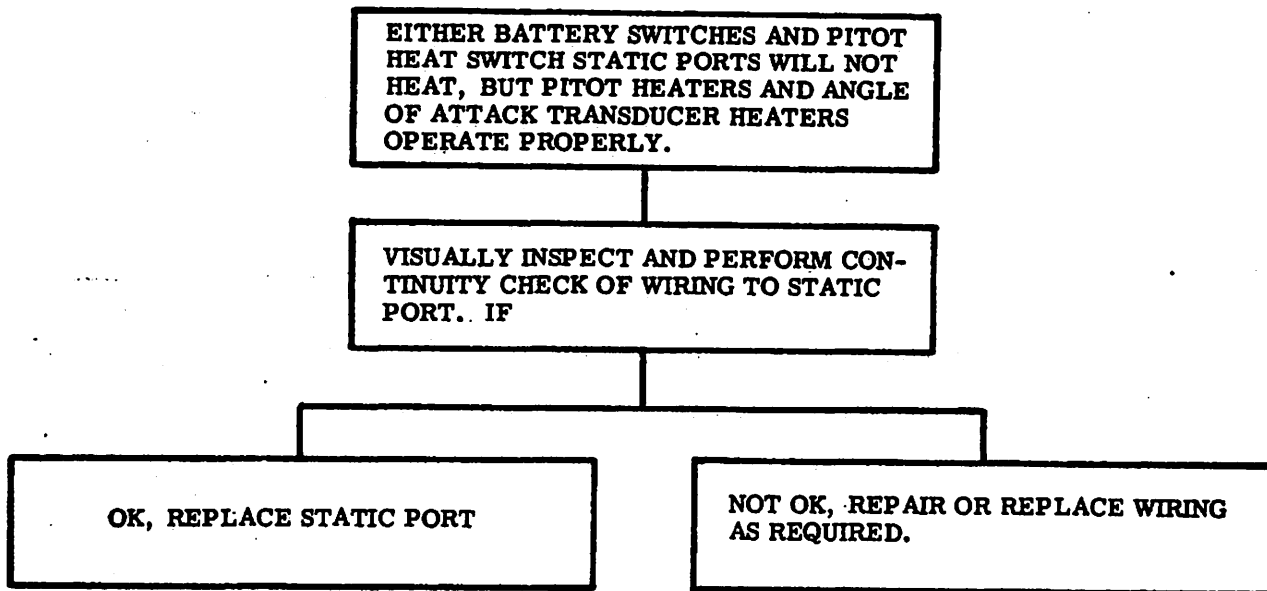
NOTE: Equivalent substitutes may be used in lieu of the following items.

NAME	PART NUMBER	MANUFACTURER	USE
Voltmeter	3430A	Hewlett Packard	For checking heater circuits.
Voltmeter	260	Simpson	For checking heater circuits.

3. Trouble Shooting

- A. See figure 101 for trouble shooting procedure.

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Pitot and Static Electrical System Trouble Shooting
Figure 101

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STATIC PORT - MAINTENANCE PRACTICES

1. Tools and Equipment

NAME	PART NUMBER	MANUFACTURER	USE
Rivet	MS20426AD3	Local	To install static ports.
Shim	2484077-3	Gates Learjet	To install static ports.
Adhesive Tape	Y-9460	3M Company	To install static ports.

2. Removal/Installation

A. Remove Static Port (See figure 201)

- (1) Remove nose compartment access doors.
- (2) Disconnect electrical wiring from static port at splice.
- (3) Disconnect tube assembly from static port. Cap tube assembly.
- (4) Locate and drill out rivets using a No. 40 (0.098) drill.
- (5) Remove static port from aircraft.

B. Install Static Port (See figure 201)

- (1) Install and secure static port with rivets. Apply sealant over rivet beads as shown in detail A.
- (2) On Aircraft 25-308 and Subsequent and prior Aircraft modified per AAK 79-10 (Installation of Wing Fences, Stall Strips and Boundary Layer Energizers), measure the distance the static port protrudes away from the aircraft skin. The distance of protrusion shall be 0.003 to 0.009 inch. If the protrusion distance is less than 0.003 inch, install 2484077-3 shims (as necessary to meet the tolerance) using Y-9460 adhesive transfer tape (product of Minnesota Mining and Manufacturing Co.). Assure that the holes of the shim align with the holes of the static port. Using a No. 52 (0.63 inch) bit as a punch, clear the adhesive from the shim and static port holes. Do not drill static port holes. Allow adhesive to cure at room temperature for 72 hours. Adhesive may be force-cured at 150°F for 1 hour.
- (3) Remove cap from tube assembly and connect tube assembly to static port.
- (4) Connect electrical wiring to static port.
- (5) Perform static system leak check. (Refer to Chapter 34.)

CAUTION: AFTER TOUCH-UP PAINTING HAS BEEN ACCOMPLISHED, ASSURE THAT STATIC PORT OPENINGS ARE NOT PLUGGED.

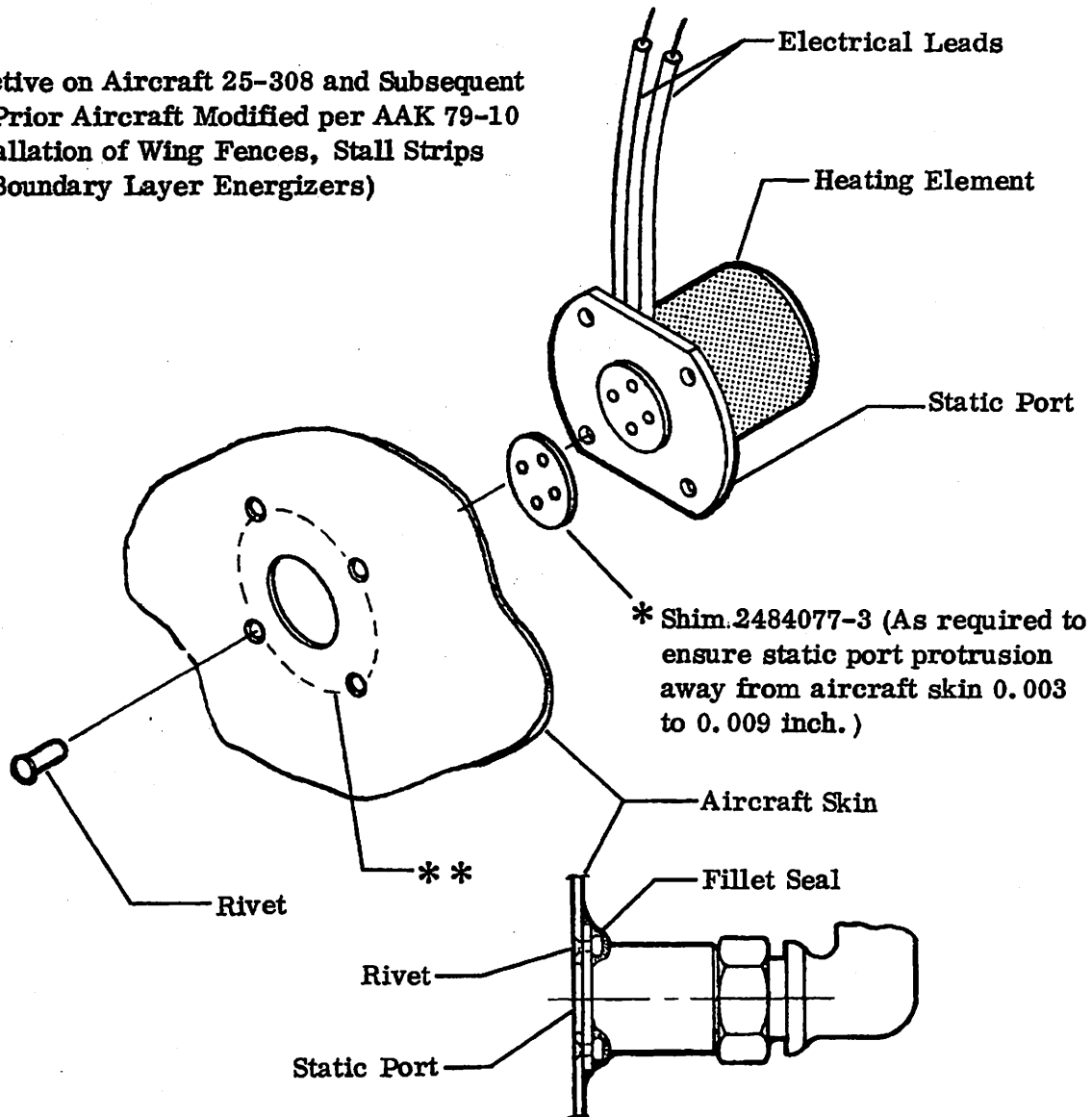
- (6) Touch-up aircraft paint as required. (Refer to Chapter 20.)

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- (7) On Aircraft 25-337, 25-342 and Subsequent and Prior Aircraft Modified for Horizontal Stabilizer Trim and Autopilot Improvement, assure one inch radius of skin around static port is free of paint, dirt, and foreign material.

* Effective on Aircraft 25-308 and Subsequent and Prior Aircraft Modified per AAK 79-10 (Installation of Wing Fences, Stall Strips and Boundary Layer Energizers)



** On Aircraft 25-337, 25-342 and Subsequent and Prior Aircraft Modified for Horizontal Stabilizer Trim and Autopilot Improvement, assure one inch radius of skin around static port is free of paint, dirt, and foreign material.

Static Port Installation
Figure 201

EFFECTIVITY: ALL

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C. Remove Shoulder Static Port (See figure 202)

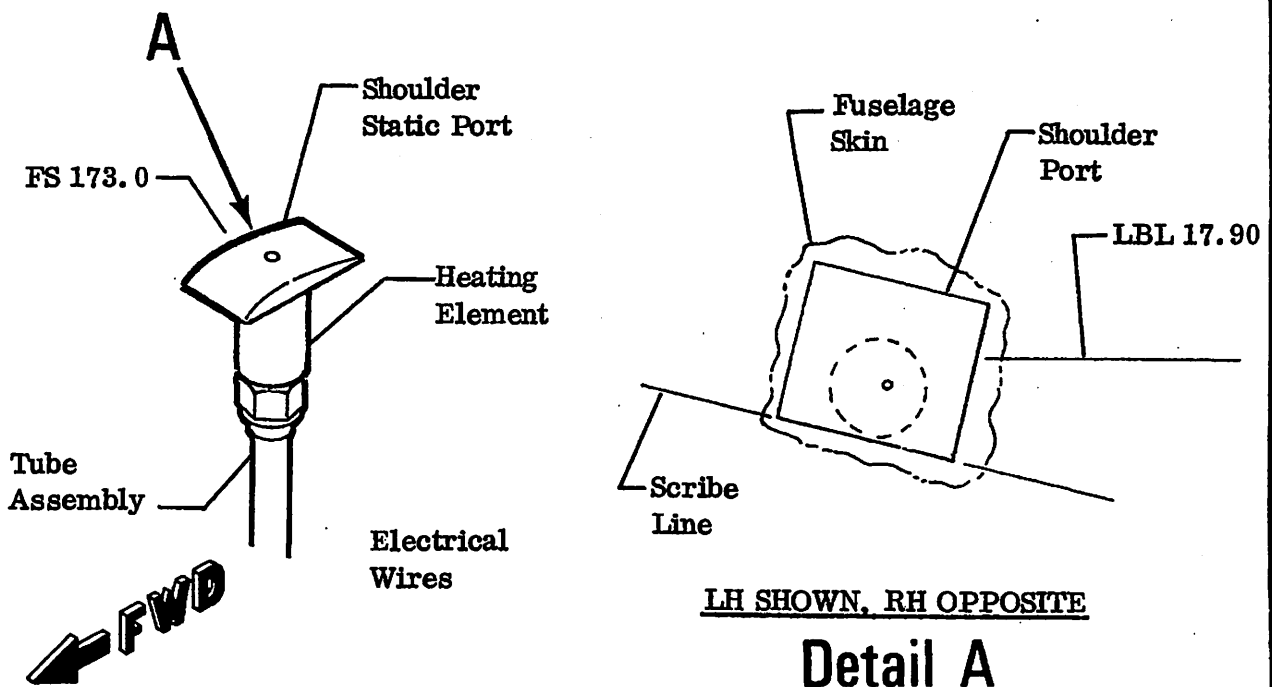
- (1) Remove nose compartment access doors.
- (2) Disconnect electrical wiring from shoulder static port at splice.
- (3) Disconnect tube assembly from shoulder static port and cap tube assembly.

NOTE: Note the position of the shoulder static port and scribe a line on the fuselage skin to aid installation of port.

- (4) Remove shoulder static port and adhesive from aircraft.

D. Install Shoulder Static Port (See figure 202)

- (1) Position shoulder static port along scribe line on fuselage skin. Install shoulder static port using RTV156 or RTV159 adhesive, General Electric, Waterford, N. Y.
- (2) Remove cap from tube assembly and connect tube assembly to shoulder static port.
- (3) Connect electrical wiring to shoulder static port.
- (4) Set Pitot Heat Switches to ON. LH and RH shoulder static ports shall become warm to touch.
- (5) Perform static system leak check.
- (6) Install nose compartment access doors.



Shoulder Static Port Installation
Figure 202

9-24B-3

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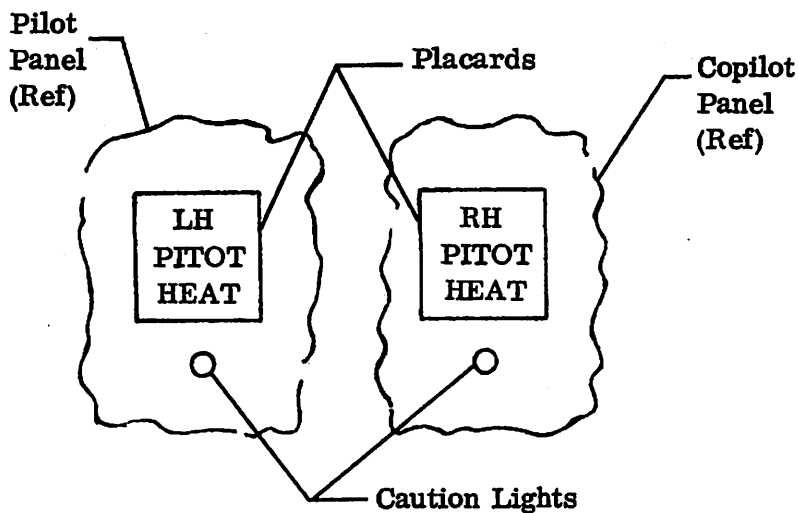
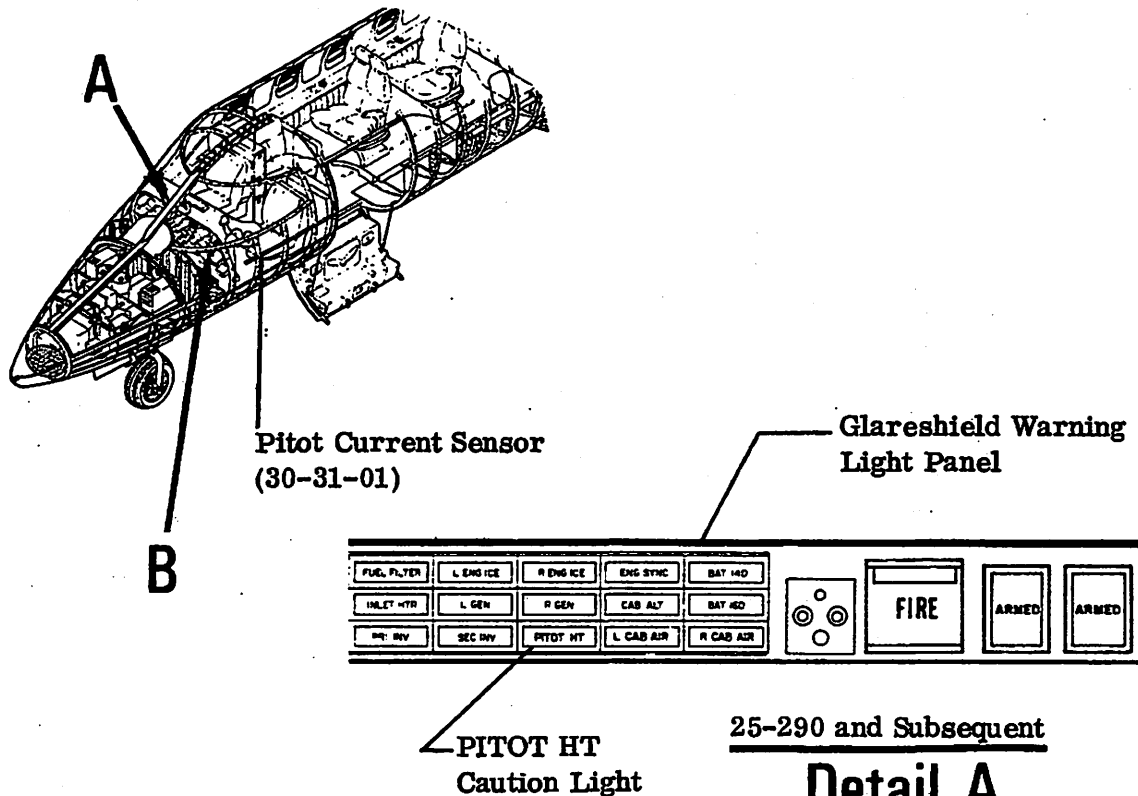
PITOT CURRENT SENSOR SYSTEM - DESCRIPTION AND OPERATION

1. Description (See figure 1)
 - A. The pitot current sensor system is installed to sense current flow to both pitot tubes and masts. On Aircraft 25-290 and Subsequent, the pitot current sensor boxes are located aft of frame 8 on stringers 11 and 12, one on each side of the fuselage. On prior Aircraft modified per AAK 79-2 (Installation of Pitot Heat Indicating System), pitot current sensors are located aft of frame 8 on stringer 12 on each side of fuselage. The sensor box consists of a relay which is wired in series with the pitot tube heater and ground.
 - B. PITOT HEAT caution lights monitor system operation. On Aircraft 25-290 and Subsequent, the PITOT HEAT caution light is located on the glareshield warning light panel. On prior Aircraft modified per AAK 79-2, (Installation of Pitot Heat Indicating System), caution light(s) and placard(s) are located on pilot's panel or copilot's panel.
2. Operation (Refer to 30-30-00, figure 1, for Electrical Control Schematic)
 - A. When the Pitot Heat Switch is set to ON, 28 vdc is applied through the pitot tube and mast heater elements and the pitot current sensor relay coil to ground. This ground removes ground from the PITOT HEAT caution light. If pitot heater elements fail or if wiring opens, power is removed from the pitot current sensor relay coil and a ground is applied to illuminate the PITOT HEAT caution light. When Pitot Heat Switch is set to OFF, no current flows through the pitot current sensor (which is normally closed) and a ground is applied to illuminate the PITOT HEAT caution light.
 - B. The PITOT HEAT caution light illuminates when:
 - (1) One or both Pitot Heat Switches are set to OFF.
 - (2) One or both pitot heat systems fail.

NOTE: Dual caution lights indicate which pitot heat system has failed.

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TYPICAL DUAL INSTALLATION
Aircraft modified per AAK 79-2
(Installation of Pitot Heat Indicating System)

Detail B

Pitot Current Sensor System Locator
Figure 1

EFFECTIVITY: 25-290 & Subsequent and Prior Aircraft
Modified per AAK 79-2 (Installation of
Pitot Heat Indicating System)

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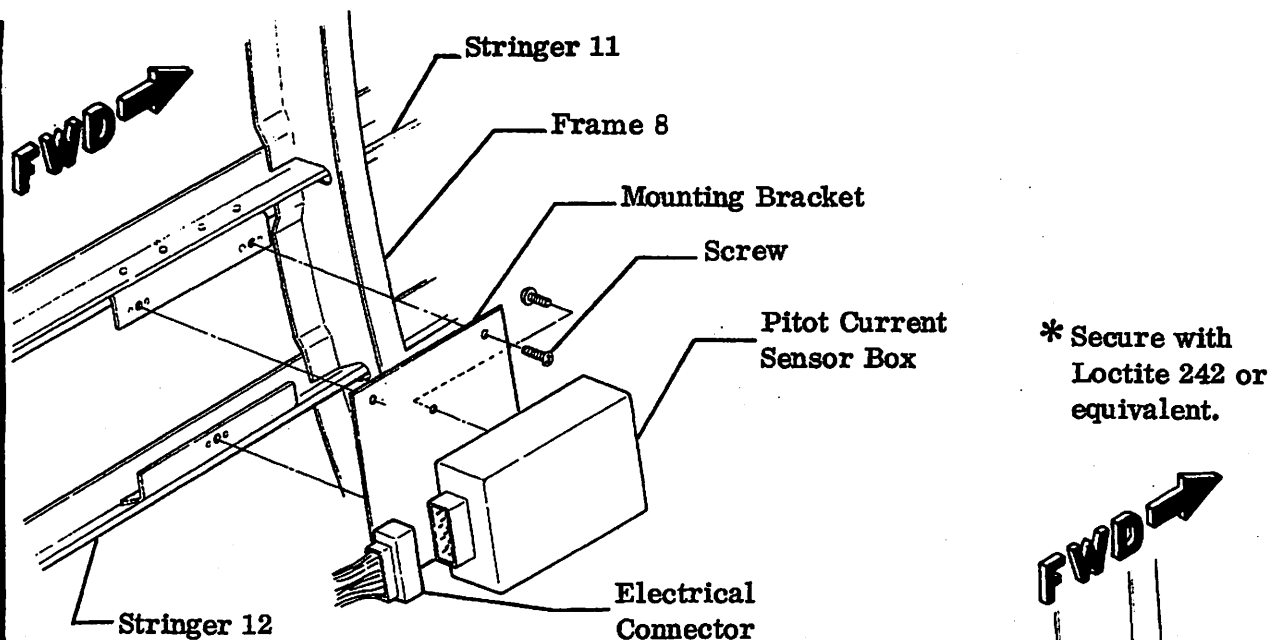
PITOT CURRENT SENSOR BOX - MAINTENANCE PRACTICES

1. Removal/Installation

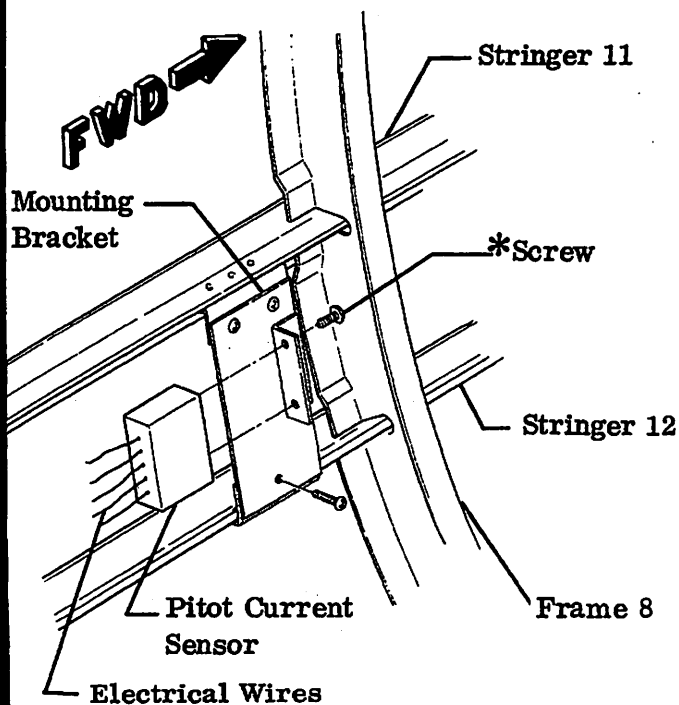
NOTE: Removal and installation for right and left pitot current sensor boxes are identical.

- A. Remove Pitot Current Sensor Box (See figure 201)**
(Aircraft 25-290 thru 25-330)
- (1) Remove electrical power from aircraft.
 - (2) Remove cockpit liner as required to gain access to frame 8 and stringers 11 and 12.
 - (3) Disconnect electrical connector from pitot current sensor box.
 - (4) Remove attaching part and mounting bracket with pitot current sensor box attached from aircraft.
- B. Install Pitot Current Sensor Box (See figure 201)**
(Aircraft 25-290 thru 25-330)
- (1) Position mounting bracket with pitot current sensor box attached (receptacle aft) and secure with attaching parts.
 - (2) Connect electrical connector to sensor box.
 - (3) Restore electrical power to aircraft.
 - (4) Perform operational check as follows:
 - (a) Set Pitot Heat Switches to ON.
 - (b) PITOT HEAT caution light will not illuminate.
 - (c) Set Pitot Heat Switches to OFF.
 - (d) Set Battery Switches to OFF.
 - (5) Install previously removed cockpit interior liner.
- C. Remove Pitot Current Sensor (See figure 201)**
(Aircraft 25-331 and Subsequent and prior Aircraft modified per AAK 79-2, Installation of Pitot Heat Indicating System)
- (1) Remove electrical power from aircraft.
 - (2) Remove cockpit interior liner as required to gain access to frame 8 and stringer 11 and 12.
 - (3) Disconnect electrical wiring from pitot current sensor.
 - (4) Remove attaching parts and pitot current sensor from aircraft.

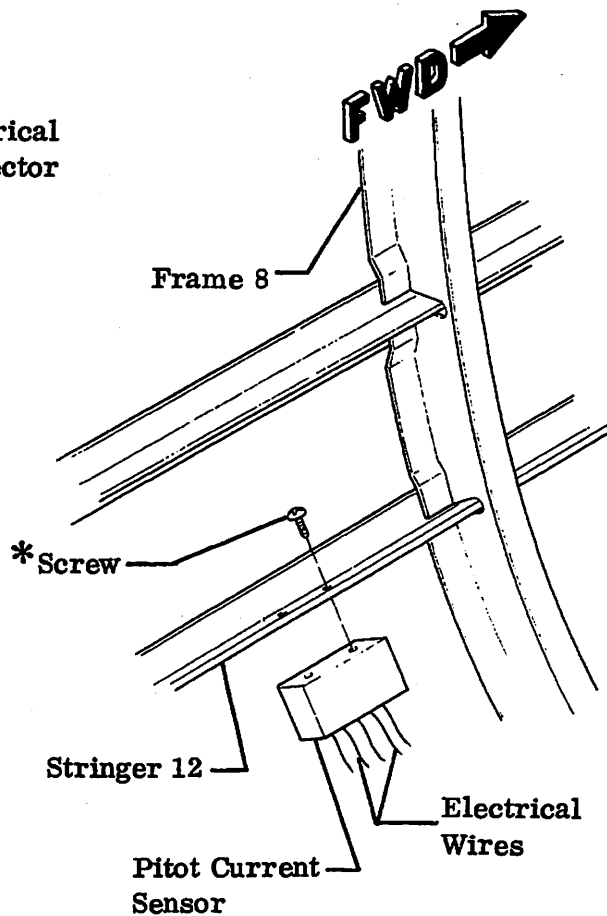
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LH INSTALLATION - RH TYPICAL
 25-290 thru 25-330



LH INSTALLATION - RH TYPICAL
 25-331 and Subsequent



LH INSTALLATION - RH TYPICAL
 Aircraft modified per AAK 79-2
 (Installation of Pitot Heat Indicating System)

Pitot Current Sensor Installation
 Figure 201

14-28C-1h

EFFECTIVITY: 25-290 and Subsequent and Prior
 Aircraft Modified per AAK 79-2
 (Installation of Pitot Heat Indicating System)

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- D. Install Pitot Current Sensor (See figure 201)
(Aircraft 25-331 and Subsequent and prior Aircraft modified per AAK 79-2,
Installation of Pitot Heat Indicating System)
- (1) Install pitot current sensor and secure with attaching parts. Secure screws with Loctite 242 or equivalent.
 - (2) Connect electrical wiring to pitot current sensor box.
 - (3) Restore electrical power to aircraft.
 - (4) Perform operational check as follows:
 - (a) Set Pitot Heat Switches to ON.
 - (b) PITOT HEAT caution light will not illuminate.
 - (c) Set Pitot Heat Switches to OFF.
 - (d) Set Battery Switches to OFF.
 - (5) Install previously removed cockpit interior liner.

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WINDSHIELD ANTI-ICE SYSTEM - DESCRIPTION AND OPERATION

1. Description (See figure 1)

- A. Windshield anti-icing is accomplished by directing engine bleed air through the defog ducting to the external defog outlets. Windshield anti-icing can be operated in either the automatic or manual mode.
- B. The system consists of two low limit thermostats, two high limit thermostats, two warning lights, an indicator light, a defog pressure regulator valve, a defog shutoff valve, two system control relays, a circuit breaker, and a manual control valve.
 - (1) Two low limit thermostats and two high limit thermostats are installed in the external defog nozzle outlets. One high and one low limit thermostat are installed in the LH and RH nozzles.
 - (2) Two WSHLD OV HT warning lights (red) are located on the glareshield warning light panel.
 - (3) The WSHLD HT indicator light (green) is located on the glareshield warning light panel.
 - (4) The control valve (B65) regulates bleed air to the external defog nozzles and is located in the bleed air ducting downstream of the shutoff valve (B30). Refer to 30-10-00 for further information on the control valve (B65).
 - (5) The shutoff valve (B30) is located in the bleed air ducting in the tailcone.
 - (6) Two system control relays are located in the pilot's subpanel.
 - (7) A manual defog control valve is installed in the bleed air duct upstream of the defog outlets. The manual control valve can be manually positioned, utilizing the IN-Normal, OUT-Defog control knob, to direct engine bleed air to the external outlets or to the footwarmers. The manual defog control valve incorporates a flapper-type check valve which checks the airflow out of the cabin in the event of pressurization system failure.

2. Operation (See figures 2 and 3)

- A. During an automatic mode of operation (Windshield Heat Switch to AUTO), the defog shutoff valve (B30) is opened through the deenergized contacts of the defog control relay and the green WSHLD HEAT light will illuminate. If bleed airflow reaches 215° F (±5° F) in either defog nozzle when the aircraft is on the ground, the contacts of the low limit thermostat complete the ground circuit to energize the defog control relay. This relay removes power from the open side of the defog shutoff valve and applies power to close the defog shutoff valve (B30). The low limit thermostat also completes a ground circuit through the squat switch relay panel to illuminate the applicable WSHLD OV HT warning light. When airflow temperature falls below 210° F (±5°), the contacts of the low limit thermostat opens, breaking the ground circuit to the defog control relay and deenergizing the relay. With the relay deenergized, power is removed from the closed side of the

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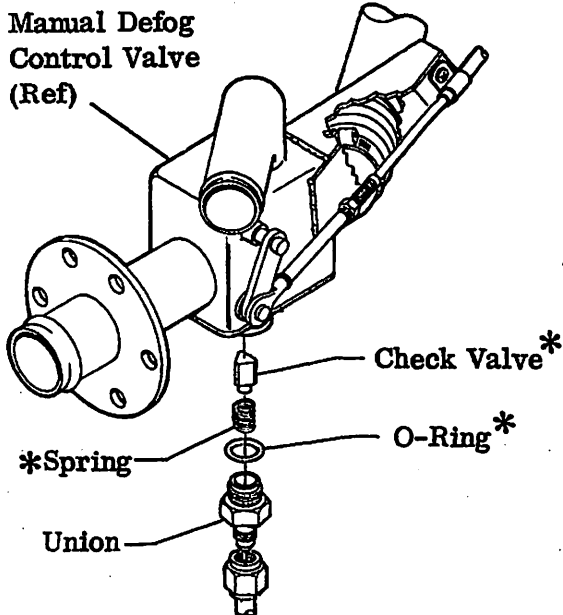
maintenance manual

defog shutoff valve and applied to the open side of the defog shutoff valve. When in flight, the automatic operation is the same, except the squat switch relay panel circuits to the WSHLD OV HT warning lights are opened. The high limit thermostatic switch then completes a ground circuit to light the applicable WSHLD OV HT warning light and close the defog control valve (B65) if bleed air temperature in the defog outlets reaches 250°F (±5°). When airflow temperature falls below 240°F (±5) the contacts of the high limit thermostatic switch open and the defog shutoff valve (B30) opens.

- B. The defog system may be manually operated by setting the Windshield Heat Switch to MAN and utilizing the Windshield Heat ON-OFF Switch to obtain the desired amount of airflow. If bleed airflow temperature reaches 250°F (in flight), the WSHLD OV HT warning light will illuminate and energize the defog control valve (B65), allowing it to close. If bleed air temperature reaches 215°F (±5°) when the aircraft is on the ground, the WSHLD OV HT will illuminate but the defog control valve (B65) will not close. In the manual mode the Windshield Heat Switch must be set to OFF to stop bleed airflow.
- C. For a description and operation of the windshield internal defog system, refer to Chapter 21.

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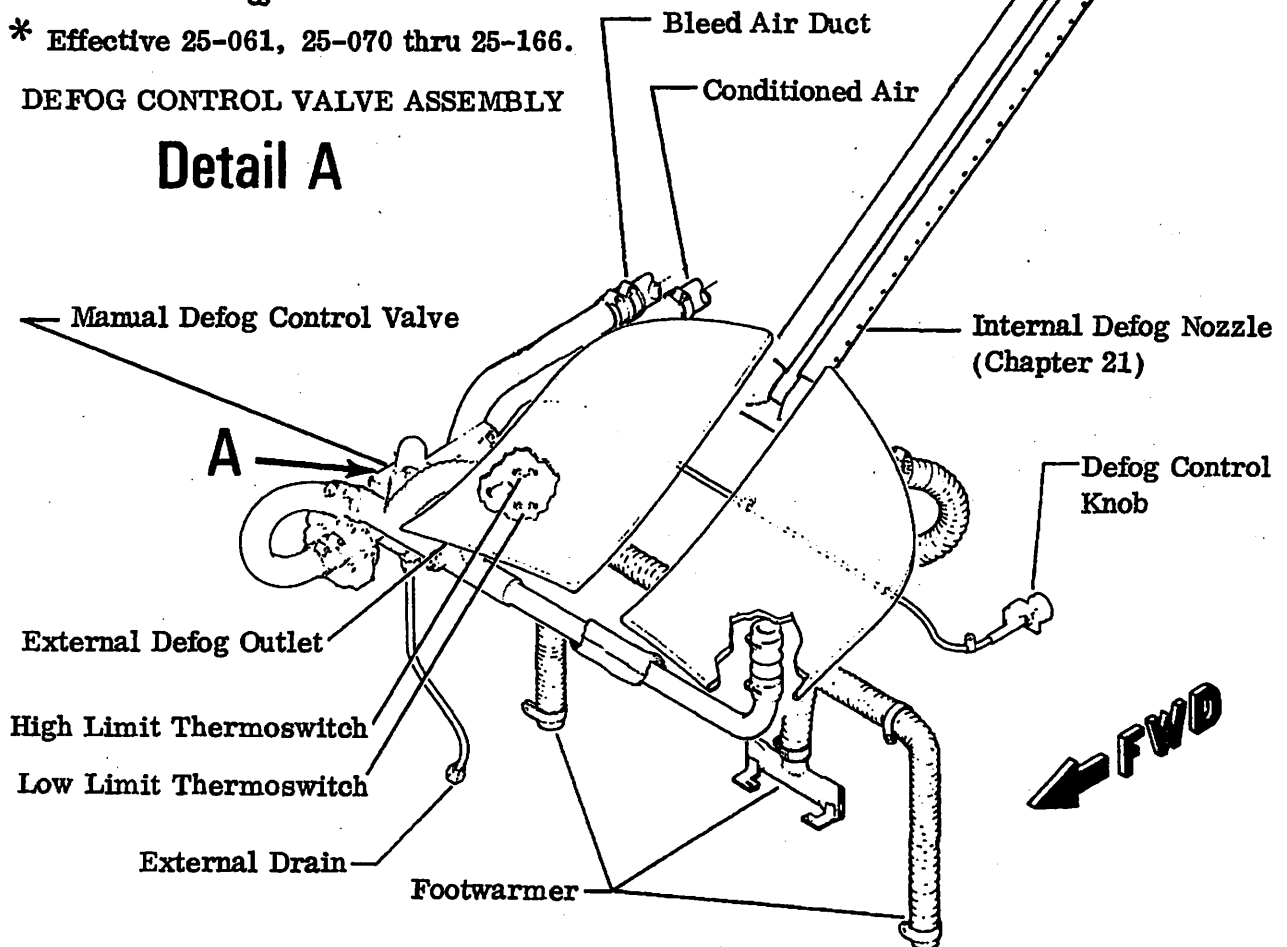
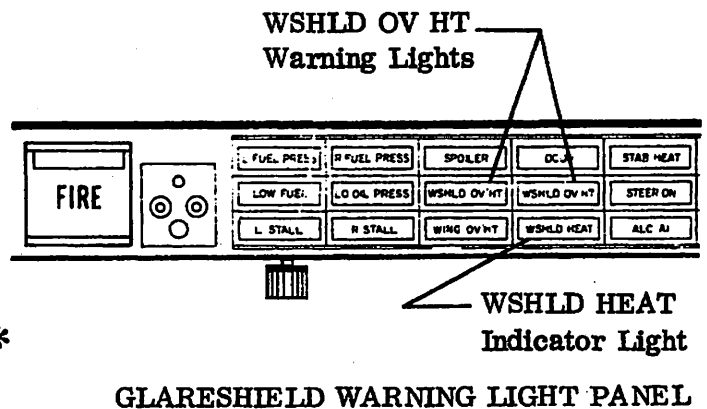
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* Effective 25-061, 25-070 thru 25-166.

DEFOG CONTROL VALVE ASSEMBLY

Detail A



Windshield Anti-Ice System Installation
Figure 1

13-34C-3
13-34C-6

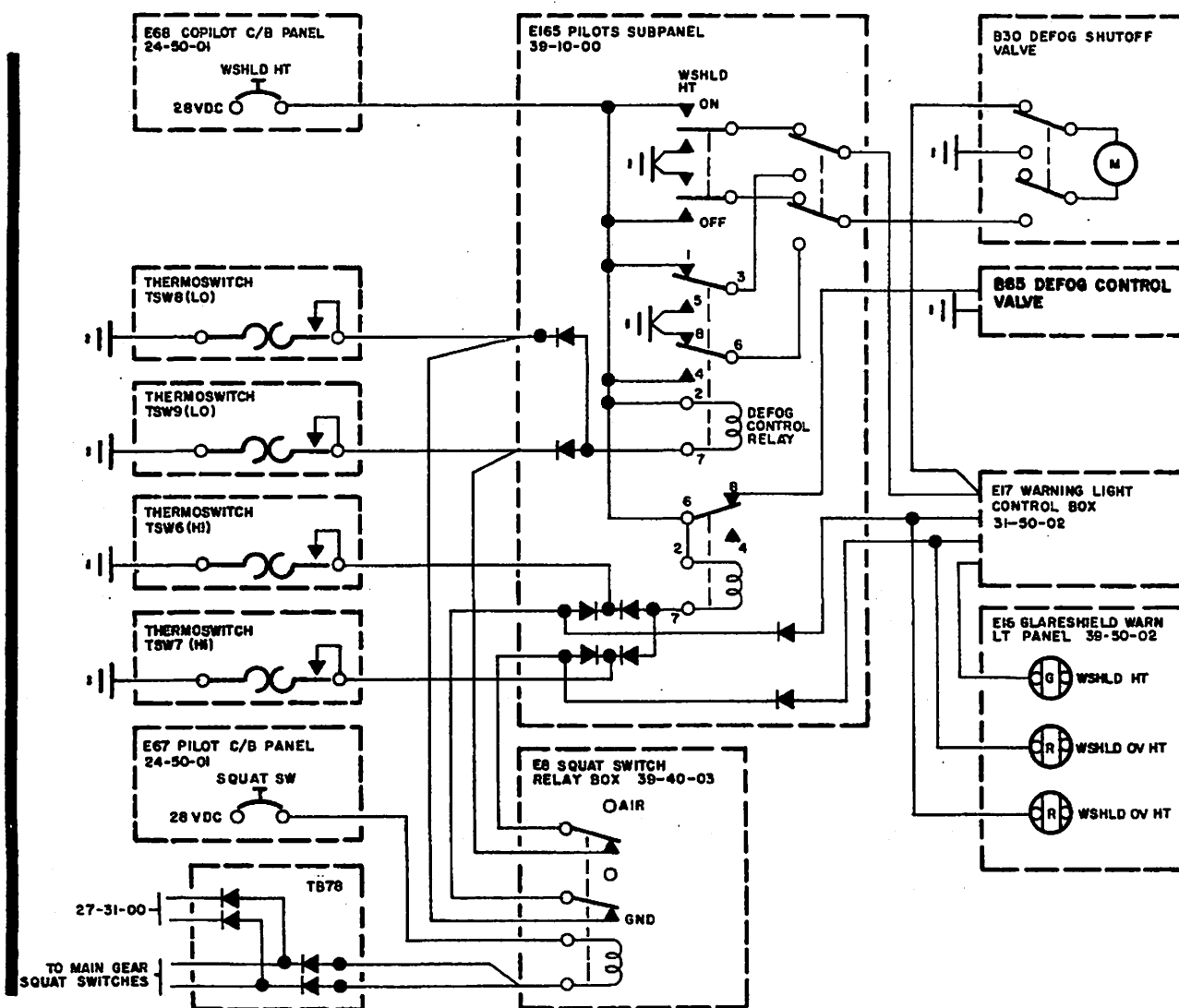
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Windshield Anti-Ice Control System Schematic
Figure 2

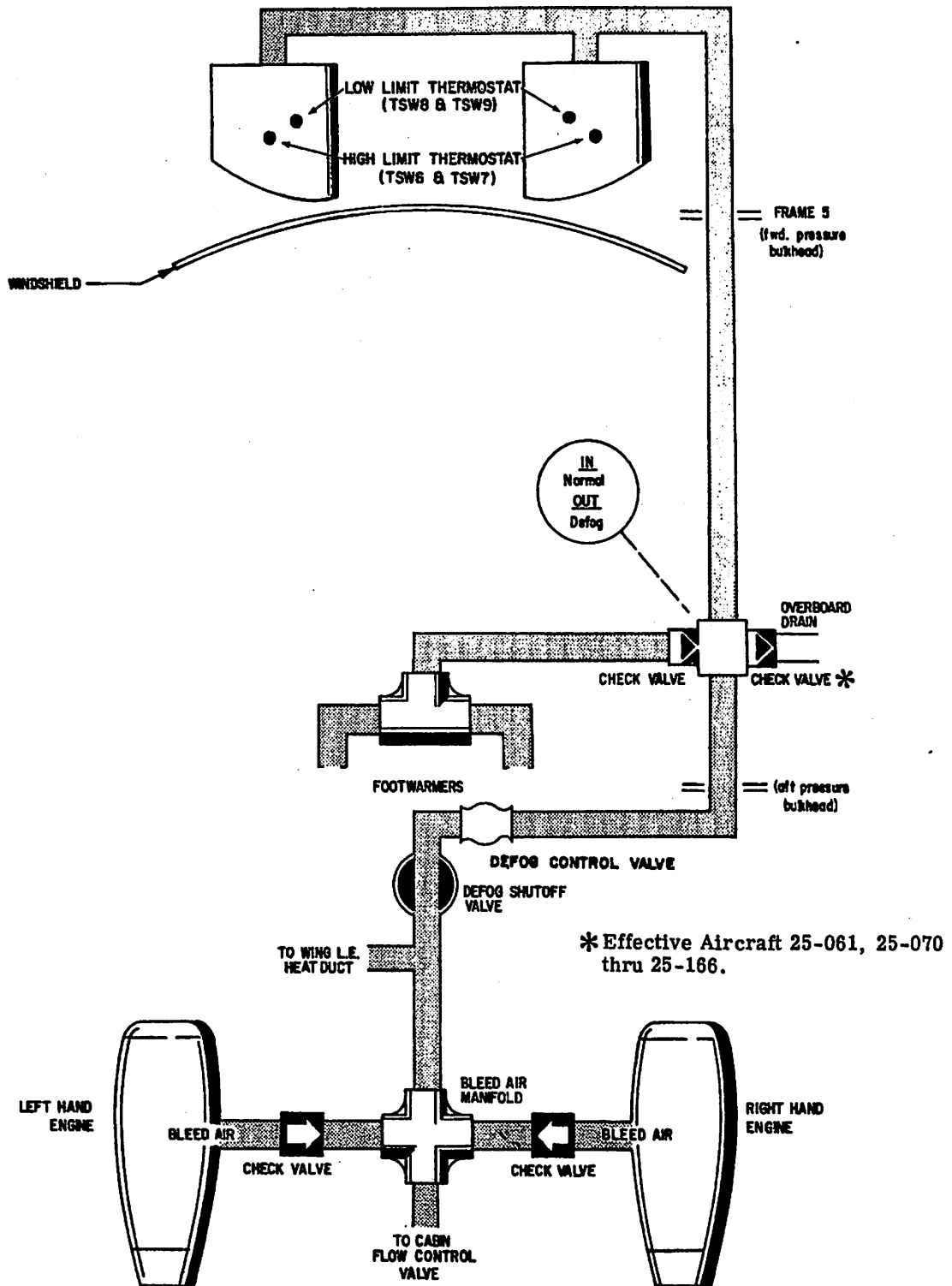
EFFECTIVITY: ALL

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Windshield Heat Control System Schematic
Figure 3

EFFECTIVITY: ALL

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WINDSHIELD - TROUBLE SHOOTING

1. General

- A. To aid in trouble shooting the windshield bleed air system, the windshield heat electrical control system schematic (figure 2), should be used with the following trouble shooting procedure.

2. Tools and Equipment

NOTE: Equivalent substitutes may be used in lieu of the following:

NAME	PART NUMBER	MANUFACTURER	USE
Voltmeter	3430A	Hewlett Packard	To check electrical circuit.
Voltmeter	260	Simpson	To check electrical circuit.

3. Trouble Shooting

- A. See figure 101 for trouble shooting procedure.

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WINDSHIELD HEAT SWITCHES SET TO "ON" AND "AUTO". NO BLEED AIR IS AVAILABLE AT EXTERNAL DEFOG OUTLETS.

NOTE: IF "WSHLD OV HT" LIGHT IS LIGHTED WITH THE ABOVE TROUBLE, DISCONNECT ELECTRICAL PLUG AND CHECK EACH INDIVIDUAL THERMOSWITCH CIRCUIT. REPLACE DEFECTIVE THERMOSWITCH.

CHECK THE FOLLOWING -

1. "WSHLD HEAT" CIRCUIT BREAKER DEPRESSED.
2. MANUAL DEFOG CONTROL VALVE IN PROPER POSITION.

NOTE: NO BLEED AIR FLOW TO THE EXTERNAL DEFOG OUTLETS, INDICATES THAT EITHER OR BOTH THE DEFOG CONTROL VALVE AND SHUTOFF VALVE HAVE CLOSED. PERFORM THE FOLLOWING ELECTRICAL CHECKS.

1. DISCONNECT ELECTRICAL PLUG FROM SHUTOFF VALVE AND CHECK FOR 28 VDC ACROSS PINS A AND C. NO 28 VDC ACROSS PINS B AND C.
2. DISCONNECT ELECTRICAL PLUG FROM DEFOG CONTROL VALVE AND CHECK FOR 28 VDC ACROSS PINS A AND B. IF

NOT OK,

1. 28 VDC PRESENT ACROSS PINS B AND C. THIS INDICATES THAT DEFOG CONTROL RELAY CONTACTS ARE STUCK IN THE ENERGIZED POSITION, APPLYING POWER TO THE COLD SIDE OF THE SHUTOFF VALVE, CLOSING THE VALVE.
2. 28 VDC NOT PRESENT. THIS INDICATES THAT THE DEFOG CONTROL VALVE CONTROL RELAY IS ENERGIZED.

NOTE: IF ANY THERMOSWITCH STICKS OR A GROUND IS APPLIED TO THE DEFOG CONTROL RELAY OR THE DEFOG CONTROL VALVE CONTROL RELAY, THE "WSHLD OV HT" WARNING LIGHT WILL BE LIGHTED.

OK, (EITHER VALVE HAS FAILED INTERNALLY) REPLACE DEFECTIVE VALVE)

Windshield Defog System Troubleshooting
Figure 101

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DEFOG CONTROL VALVE - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: Aircraft 25-061, 25-070 thru 25-208 are equipped with Whittaker pressure regulators or Sterer shutoff valves. Aircraft 25-209 and Subsequent are equipped with Sterer shutoff valves.

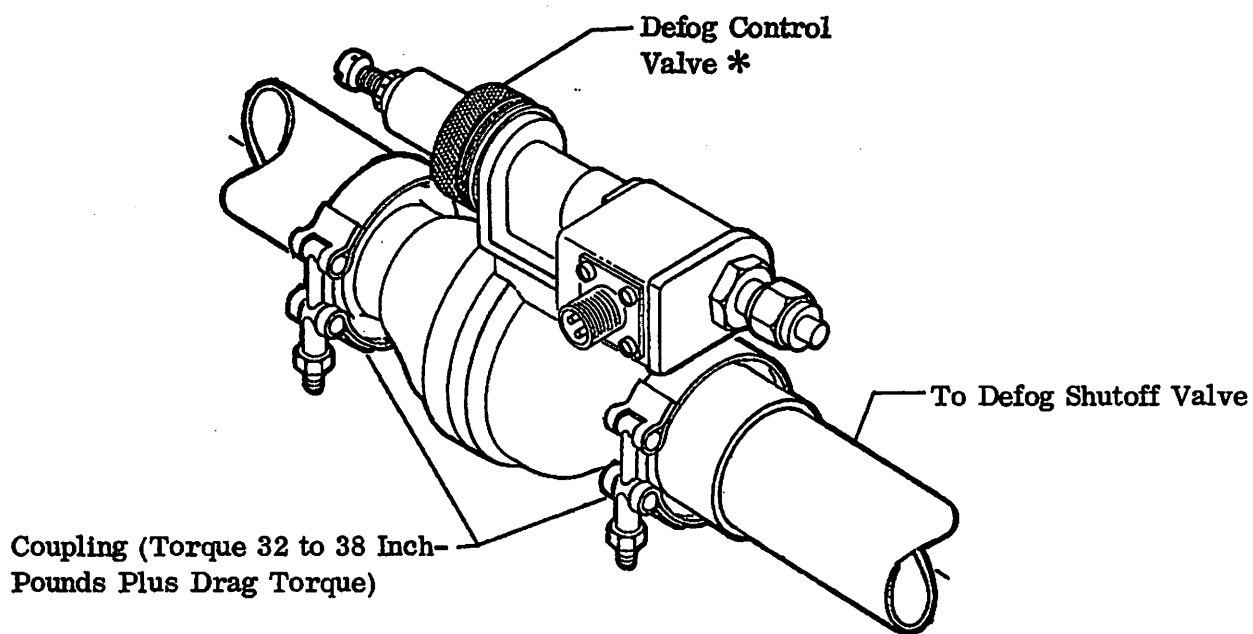
Removal and installation procedures for both Whittaker and Sterer Control Valves are identical.

- A. Remove Control Valve (See figure 201)**
 - (1) Lower tailcone access door and remove electrical power from aircraft.
 - (2) Remove insulation jackets and disconnect electrical connector from control valve solenoid.
 - (3) Loosen coupling and remove control valve from aircraft.
- B. Install Control Valve (See figure 201)**
 - (1) Install control valve and secure with coupling. Torque couplings to 32 to 38 inch-pounds plus drag torque.

NOTE: Drag torque is the amount of torque required to overcome the friction of any self-locking nut. This nut friction (drag torque) must be added to the torque callout to assure proper torquing. The tailcone bleed air ducting incorporates stainless steel clamps, nuts and bolts, which require higher nut-friction than the standard self-locking nuts. In some instances nut friction may exceed the required torque values.

- (2) Connect electrical connector to control valve solenoid and install insulation jackets.
- (3) Restore electrical power to aircraft and secure tailcone access door.

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* Aircraft 25-061, 25-070 thru 25-208 are equipped with Whittaker pressure regulators or Sterer shutoff valves. Aircraft 25-209 and Subsequent are equipped with Sterer shutoff valves.

Defog Control Valve Installation
Figure 201

EFFECTIVITY: ALL

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SHUTOFF VALVE - MAINTENANCE PRACTICES

1. General

- A. The defog shutoff valve is a motor-driven butterfly-type valve used to regulate the amount of engine bleed air routed to the external defog outlets.
- B. In a manual mode of operation, the valve is controlled by the Windshield Heat ON-OFF switch. In an automatic mode of operation, the valve is controlled by the Hi and Low limit thermostats thru the defog control relay and the Windshield Heat AUTO-MAN switch.

2. Removal/Installation

- A. Remove Valve (See figure 201)
 - (1) Lower tailcone access door and disconnect aircraft batteries.
 - (2) Remove duct insulation as required to gain access to defog shutoff valve.
 - (3) Disconnect electrical plug from shutoff valve.
 - (4) Loosen clamp sufficiently to remove defog duct from defog shutoff valve.
 - (5) Loosen remaining clamp and remove defog shutoff valve from aircraft.
- B. Install Valve (See figure 201)

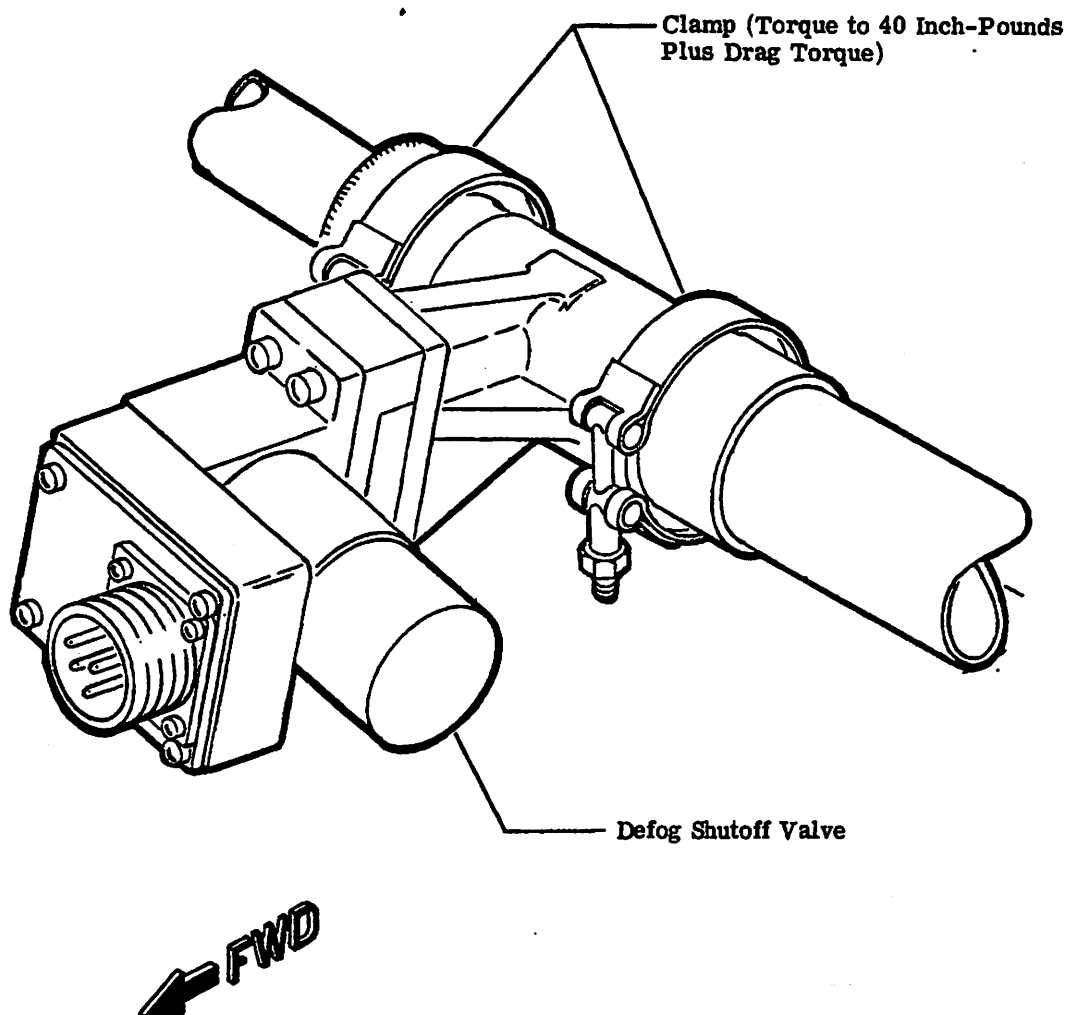
CAUTION: WHEN INSTALLING DEFOG SHUTOFF VALVE, EXTREME CARE SHOULD BE TAKEN TO ASSURE THE VALVE MATES PROPERLY WITH DISTRIBUTION DUCTS. FAILURE TO OBSERVE THIS COULD CAUSE DAMAGE TO THE SHUTOFF VALVE WHEN COUPLINGS ARE TIGHTENED.

- (1) Install defog shutoff valve on bleed air ducts and secure with clamps. Torque clamps to 40 inch-pounds plus drag torque.

NOTE: Drag torque is the amount of torque required to overcome the friction of any self-locking nut. This nut friction (drag torque) must be added to the torque callout to assure proper torquing. The tailcone bleed air ducting incorporates stainless steel clamps, nuts and bolts, which require higher nut-friction than the standard self-locking nuts. In some instances nut friction may exceed the required torque values.

- (2) Connect electrical plug to shutoff valve.
- (3) Install duct insulation.
- (4) Connect aircraft batteries and secure tailcone access door.

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Shutoff Valve Installation
Figure 201

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EXTERNAL DEFOG NOZZLE - MAINTENANCE PRACTICES

1. General

- A. An external defog nozzle is installed on each side of the windshield center post. The nozzles direct engine bleed air across both windshield halves.
- B. The pilot's defog nozzle incorporates provisions for directing alcohol on the LH half of the windshield. Refer to 30-50-00 for further information on the alcohol system.
- C. Each external defog nozzle is protected by an erosion coating. Refer to 20-60-00 for further information on the erosion coating.

2. Removal/Installation

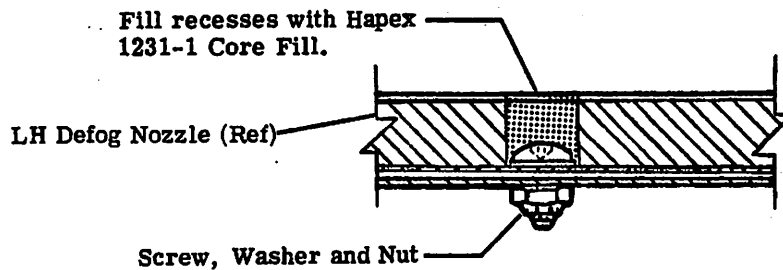
- A. Remove Defog Nozzle (Typical) (See figure 201)
 - (1) Remove nose compartment access doors.
 - (2) Disconnect wiring from high and low limit thermostats at splices.
 - (3) Disconnect alcohol system plumbing from defog nozzle (pilot side only). Loosen clamps securing hose to nozzle.
 - (4) Defog nozzle attaching screws are recessed into the nozzle and the recesses are filled with aerodynamic putty. Locate screws and remove putty using a knife or other suitable tool.
 - (5) Lower pilot's and copilot's instrument panels. Remove upper three screws securing defog nozzle and windshield and retrieve washers and nuts.
 - (6) Remove remaining screws and defog nozzle from aircraft.
- B. Install Defog Nozzle (Typical) (See figure 201)
 - (1) Install defog outlet and secure with attaching parts. Torque screws to 5 inch-pounds plus drag torque. Raise pilot's and copilot's instrument panels.

NOTE: Drag torque is the amount of torque required to overcome the friction of any self-locking nut. This nut friction (drag torque) must be added to the torque callout to assure proper torquing. The tailcone bleed air ducting incorporates stainless steel clamps, nuts and bolts, which require higher nut-friction than the standard self-locking nuts. In some instances nut friction may exceed the required torque values.

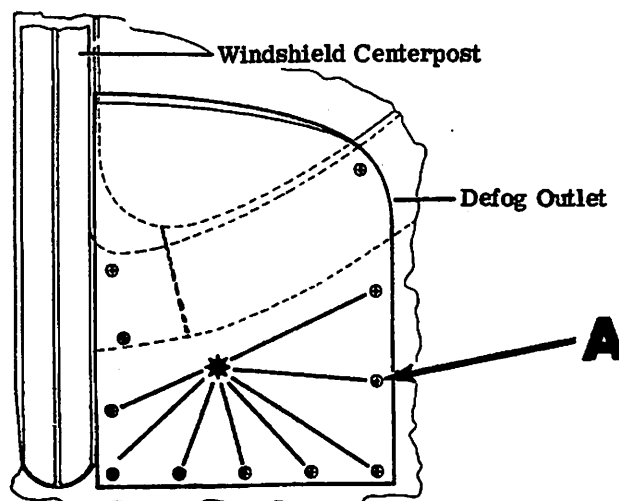
- (2) Connect hose to defog nozzle and secure with clamp.
- (3) Connect alcohol system plumbing to defog nozzle (pilot side only). Connect electrical wiring to high and low limit thermostats at splices.

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- (4) Fill recesses in defog nozzle with Hapex 1231-1 core fill resin, manufactured by Hastings Plastics Co., 1704 Colorado Ave., Santa Monica, Calif.
- (5) Repaint affected area.
- (6) Install nose compartment access doors.



Detail A



* Torque to 5 inch-pounds plus drag torque. Torque remaining defog outlet screws to 30 inch-pounds.

Note

Drag torque is the amount of torque required to overcome the friction of any self-locking nut. This nut friction (drag torque) must be added to the torque callout to assure proper torque.

External Defog Nozzle Installation
Figure 201

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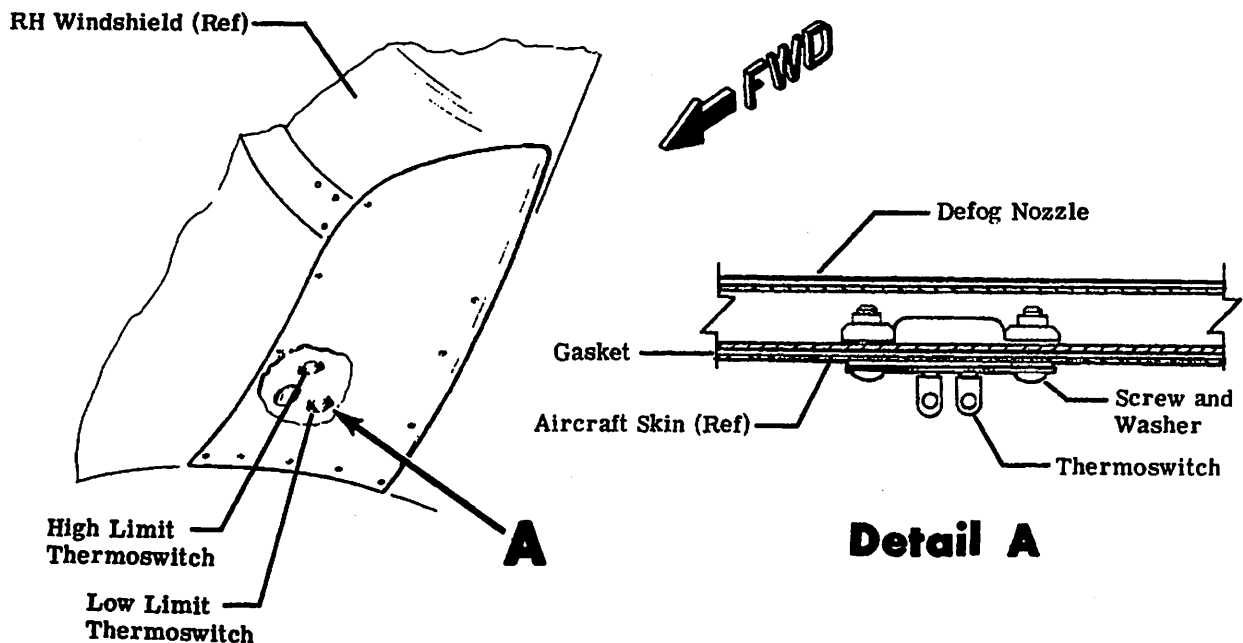
THERMOSWITCH - MAINTENANCE PRACTICES

1. General

- A. The thermostats are installed in the external defog nozzle outlets.
Two are installed in each outlet (one high and one low limit).
- B. Access to the thermostats is through the nose compartment doors.

2. Removal/Installation

- A. Remove Thermostat (Typical) (See figure 201)
 - (1) Remove nose compartment access doors.
 - (2) Disconnect thermostat wiring at splice. Tag wiring.
 - (3) Remove screws, washers and thermostat from aircraft.
- B. Install Thermostat (Typical) (See figure 201)
 - (1) Install thermostat and secure with screws and washers.
 - (2) Remove tags from wiring and connect to thermostats.
 - (3) Install nose compartment access doors.



Thermostat Installation
Figure 201

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RADOME AND WINDSHIELD ANTI-ICE - DESCRIPTION AND OPERATION

1. General

- A. The alcohol anti-ice system is used for windshield anti-ice if the normal windshield defog system malfunctions. The alcohol anti-ice system is also used for radome anti-ice.

2. Description

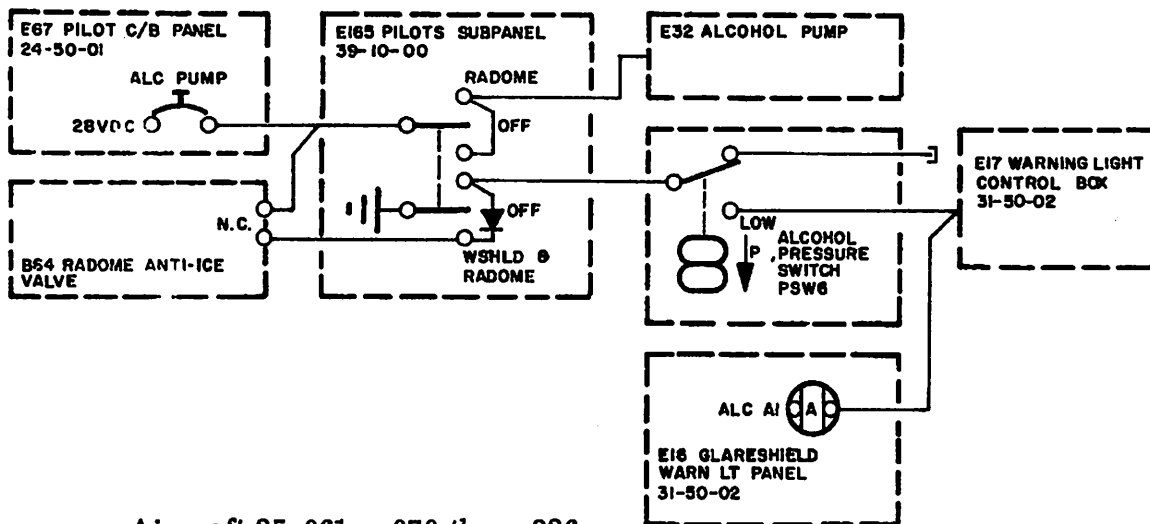
- A. The system consists of a system switch, a caution light, a 2.2 gallon tank containing methyl alcohol, a motor-driven pump, an anti-ice shutoff valve, a filter and a low pressure switch. On Aircraft 25-227 and Subsequent, the system consists of a 1.75 gallon pressurized tank containing methyl alcohol, a shutoff and pressure regulator valve, a relief valve, a float switch, a three-way valve and a filter.

3. Operation (See figures 1 and 2)

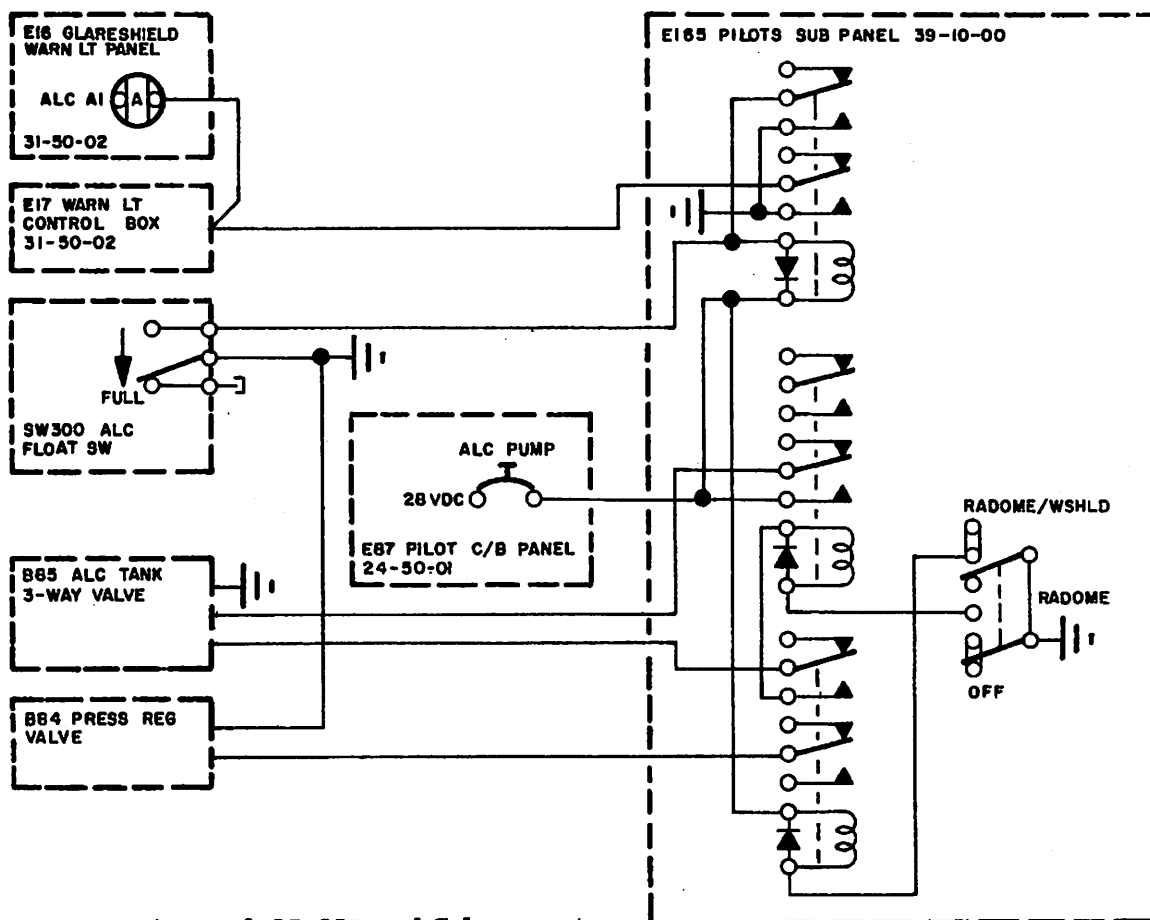
- A. On Aircraft 25-061, -070 thru -226, when the alcohol system switch is set to RADOME, the motor-driven pump is energized and supplies methyl alcohol to the radome. The shutoff valve (normally closed) prevents any methyl alcohol from reaching the pilot's defog nozzle. If the supply of methyl alcohol is depleted, the low pressure switch will actuate and light the ALC AI caution light. Should the normal defog system malfunction, the alcohol system switch is set to WSHLD/RADOME. This will allow alcohol flow to the radome and energize (open) the shutoff valve, allowing methyl alcohol flow through the orificed assembly to the pilot's defog nozzle. The methyl alcohol is sprayed on the windshield through holes drilled in a small tube which forms an integral part of the defog nozzle. A pressure relief valve opens and vents any pressure in excess of 1.1 psi into the pressure return line.
- B. On Aircraft 25-227 and Subsequent, when the alcohol system switch is set to RAD, two circuits are completed thru a control relay; (1) a power circuit is completed to open the shutoff and pressure regulator valve and (2) a power circuit is completed to position the 3-way control valve for alcohol flow to the radome. The alcohol tank, pressurized to 2.3 ($\pm$.1) psi through the pressure regulator, supplies alcohol to the radome through the filter and 3-way valve. Operation of the system is the same when the system switch is set to WSHLD & RADOME, except the 3-way control valve is positioned to supply alcohol pressure to both the windshield and the radome. When the system switch is set to OFF the pressure regulator valve is closed. The alcohol tank pressure is bled to ambient thru the relief valve. The relief valve is set at 2.6 psi to prevent overpressurization should the pressure regulator fail. A float switch installed in the lower portion of the tank illuminates the ALC AI caution light when the alcohol tank is empty. The ALC AI caution light circuit incorporates a holding circuit, to prevent the light from flickering. When the Battery Switches are set to OFF, the holding circuit is opened.

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Aircraft 25-061, -070 thru -226



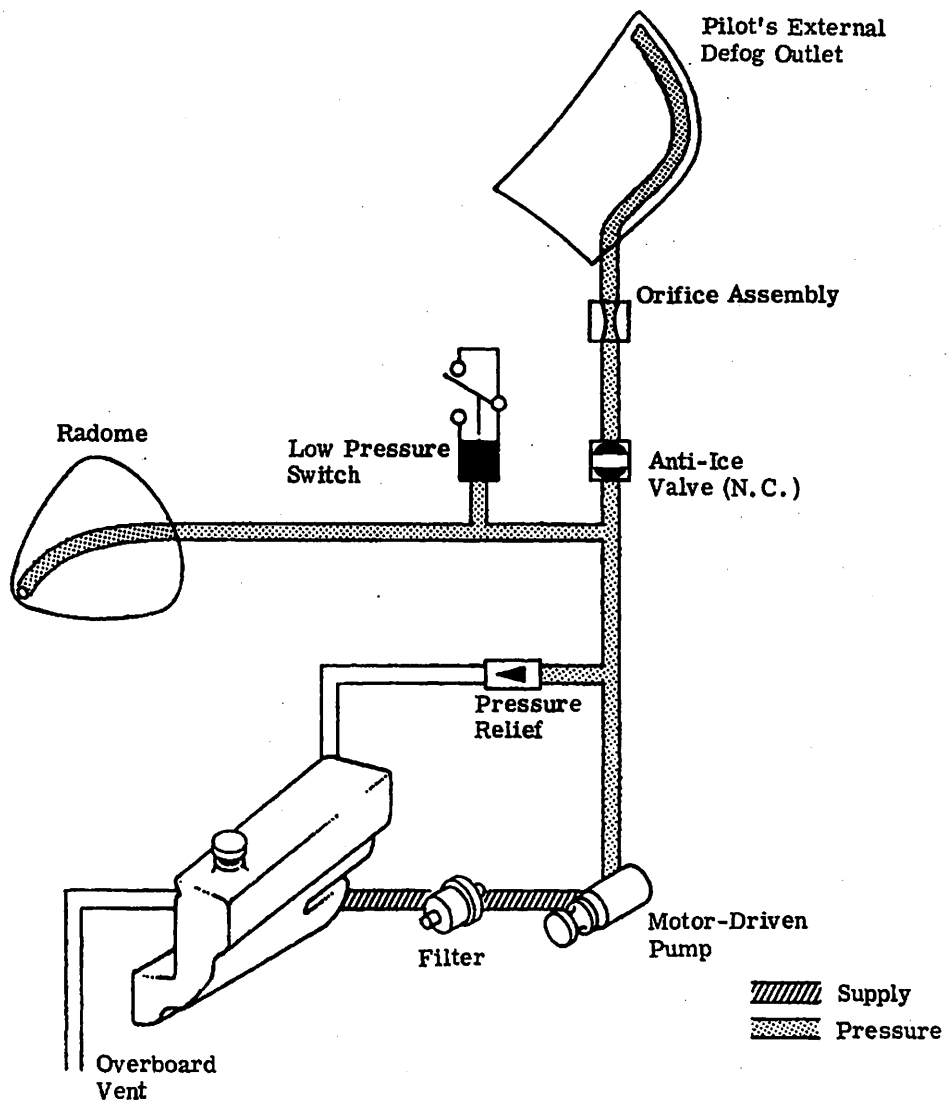
Aircraft 25-227 and Subsequent

Alcohol Anti-Ice System Electrical Control Schematic
Figure 1

EFFECTIVITY: ALL

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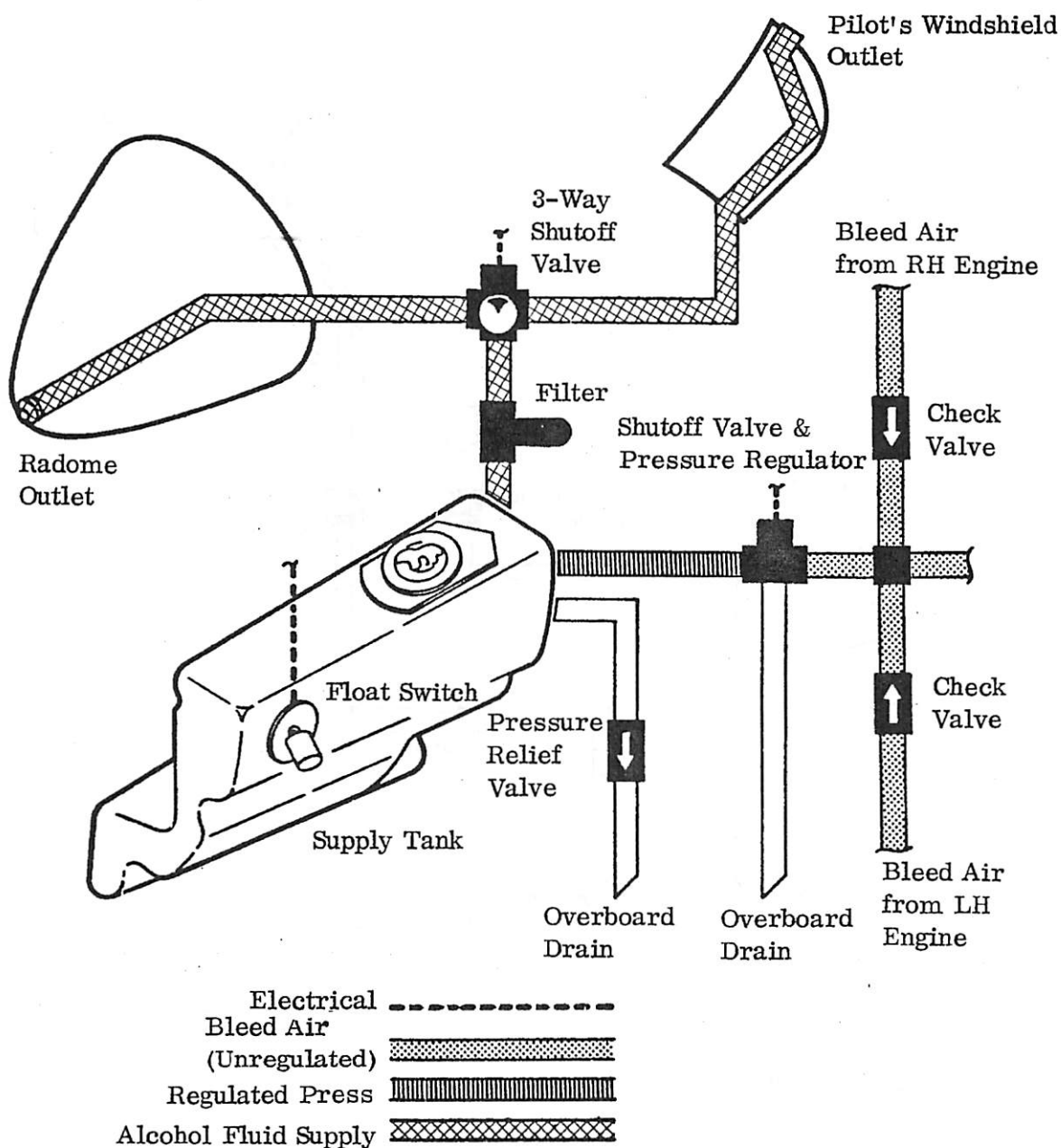
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Alcohol Anti-Ice System Schematic
Figure 2 (Sheet 1 of 2)

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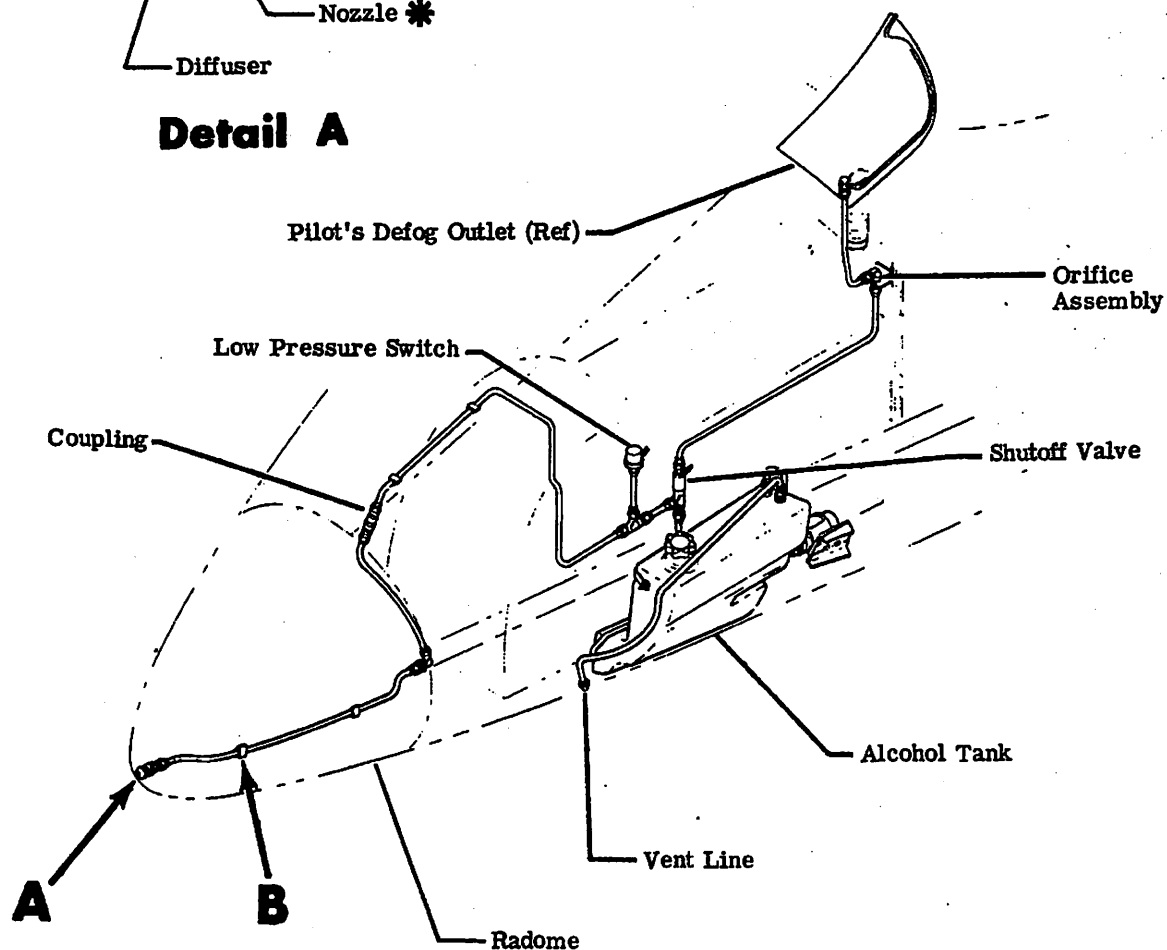
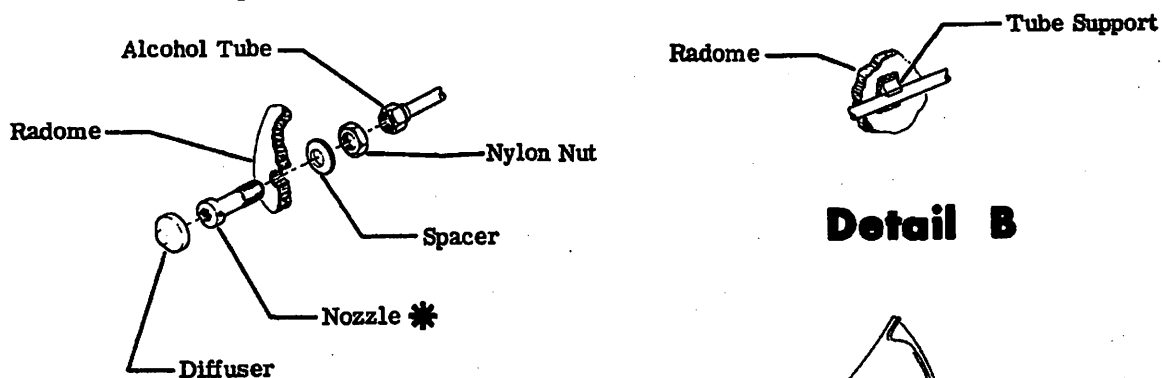
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Alcohol Anti-Ice System Schematic
Figure 2 (Sheet 2 of 2)

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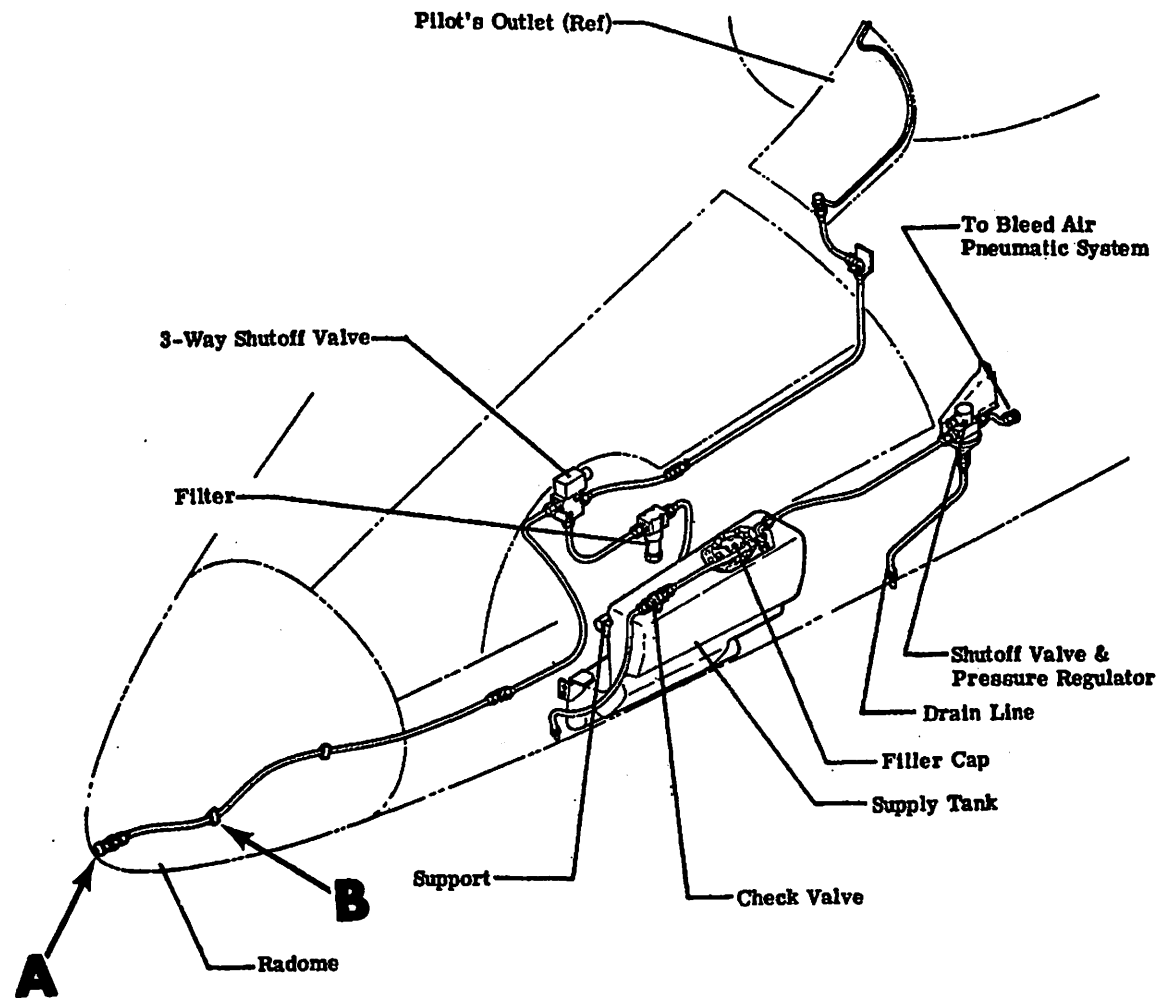
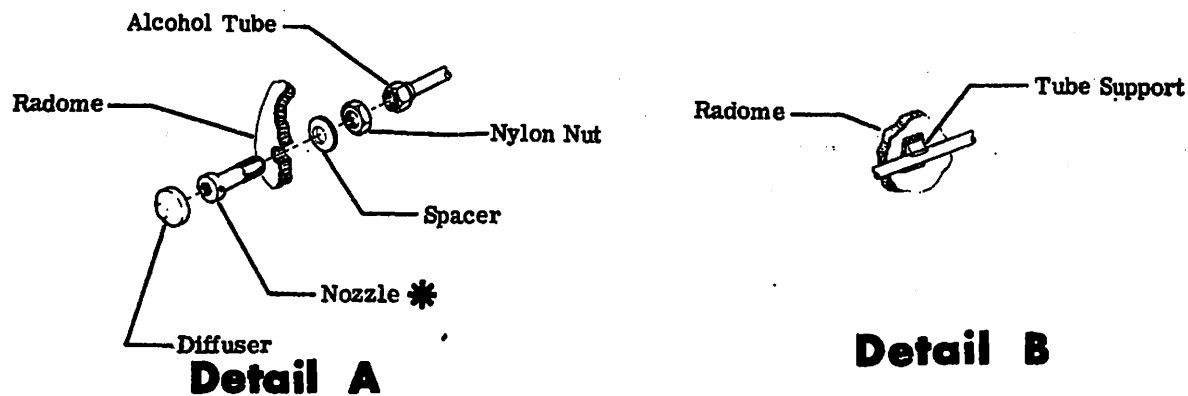
* Position nozzle assembly so that one of the three alcohol spray holes point straight down.



Alcohol Anti-Ice System Installation
Figure 3 (Sheet 1 of 2)

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* Position nozzle assembly so that one of the three alcohol spray holes point straight down.



Alcohol Anti-Ice System Installation
Figure 3 (Sheet 2 of 2)

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RADOME AND WINDSHIELD ANTI-ICE - MAINTENANCE PRACTICES

1. Inspection/Check

A. Perform Operational Check of Alcohol Anti-Ice System (Aircraft 25-227 and Subsequent)

- (1) Lower tailcone access door to gain access to alcohol anti-ice bleed air tubing.
- (2) Disconnect tubing from tee and connect an air pressure source to the tube assembly. (Refer to Chapter 36.)
- (3) Set pressure source to 50 ($\pm$ 5) psi and pressurize the tubing leading to the shutoff and pressure regulator valve.
- (4) Set Battery Switches to ON and Windshield and Radome Switch to WSHLD & RADOME.
 - (a) Check system fittings for leaks.
 - (b) Remove diffuser from radome nozzle. Alcohol shall flow from all three holes in nozzle. To unplug holes, use a maximum 0.0115 inch diameter wire.
 - (c) Alcohol shall flow from all five holes in windshield nozzle. To unplug holes use a No. 80 drill bit.
 - (d) Repeat steps (4a) thru (4c) four times.
- (5) Set Windshield and Radome Switch to RADOME. Alcohol shall flow from radome but not from windshield nozzle.
- (6) Set Windshield and Radome Switch to OFF. Alcohol flow shall stop.
- (7) Drain alcohol tank. ALC AI warning light on glareshield warning light panel will illuminate.
 - (a) If warning light does not illuminate, depress Glareshield Test Switch. If light bulb is good, check tank float switch for continuity.
- (8) Depressurize alcohol anti-ice tubing and disconnect pressure source.
- (9) Connect tubing to tee and close tailcone access door.
- (10) Wash off any alcohol remaining on aircraft.
- (11) Set Battery Switches to OFF.

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ALCOHOL FILTER - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: On Aircraft 25-061, 25-070 thru 25-226, the alcohol filter is installed in the alcohol pump supply line. On Aircraft 25-227 and Subsequent, the filter is installed in the supply line between the supply tank and the 3-way shutoff valve.

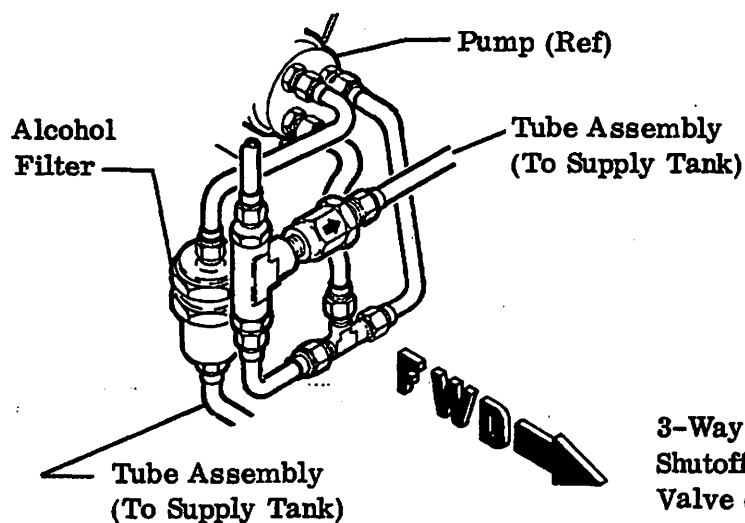
Disassemble and clean alcohol filter assembly in accordance with the time interval specified in Chapter 5.

- A. Remove Filter (Aircraft not modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft)) (See figure 201)
 - (1) Remove left nose compartment access door.
 - (2) Drain alcohol tank. (Refer to 30-50-00.)
 - (3) Disconnect tube assemblies at filter and remove filter from aircraft. Cap all exposed fittings.
- B. Install Filter (Aircraft not modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft)) (See figure 201)
 - (1) Remove caps from tube assemblies
 - (2) Connect tube assemblies to filter, noting direction arrow on filter. Arrow points in direction of flow. On Aircraft 25-061, 25-070 thru 25-226, flow is toward the pump. On Aircraft 25-227 and Subsequent, flow is toward the 3-way shutoff valve. Stamped on the filter are the words IN (in red) and OUT (in blue).
 - (3) Service the alcohol tank. (Refer to Chapter 12.)
 - (4) Install nose compartment access door.
- C. Remove Filter (Aircraft modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft)) (See figure 201)
 - (1) Remove left nose compartment access door.
 - (2) Drain alcohol tank. (Refer to 30-50-00.)
 - (3) Remove attaching parts and access cover from LH side of nose wheel box to gain access to alcohol filter.
 - (4) Remove safety wire from filter and disconnect tube assemblies at filter.
 - (5) Remove attaching parts and filter from aircraft. Cap all exposed fittings.
- D. Install Filter (Aircraft modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft)) (See figure 201)
 - (1) Install filter and secure with attaching parts.
 - (2) Safety wire filter and connect tube assemblies at filter.
 - (3) Install previously removed access cover and secure with attaching parts.
 - (4) Service alcohol tank. (Refer to Chapter 12.)
 - (5) Install nose compartment access door.

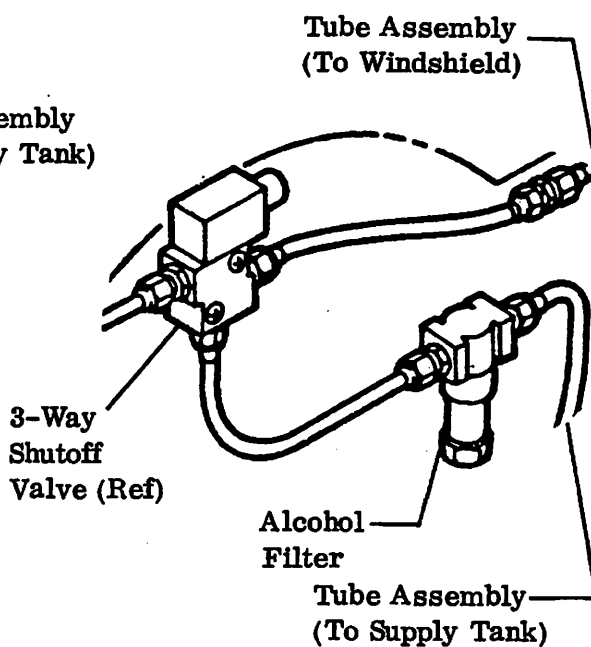
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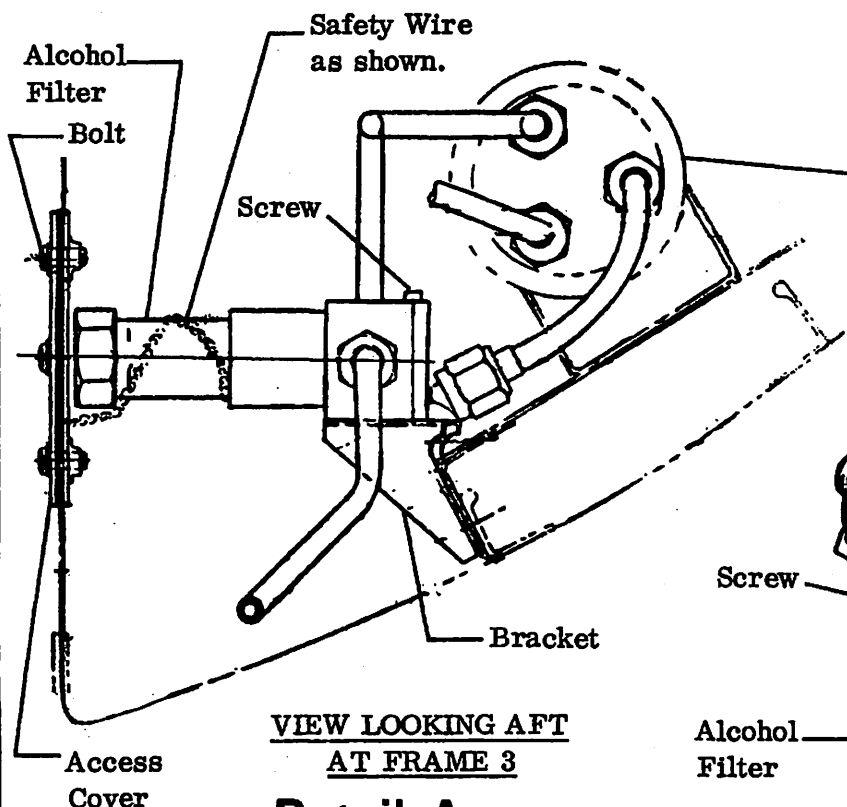
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25-061, 25-070 thru 25-226



25-227 and Subsequent

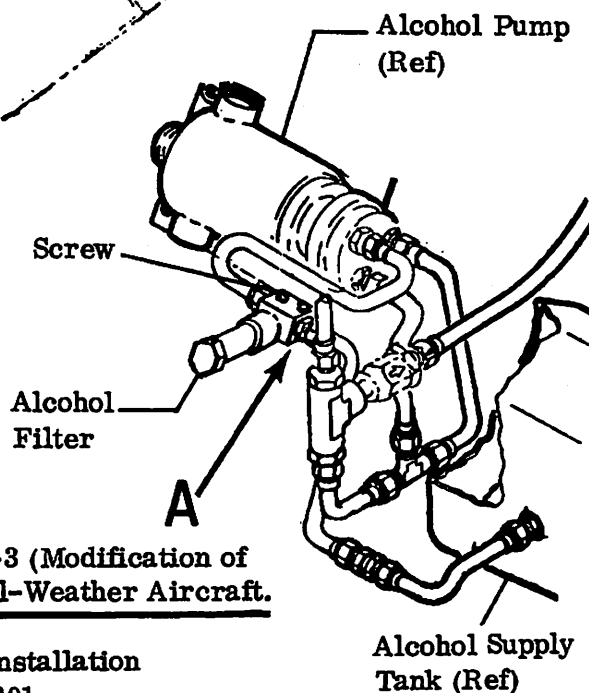


VIEW LOOKING AFT
AT FRAME 3

Detail A

Aircraft modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft.

Alcohol Filter Installation
Figure 201



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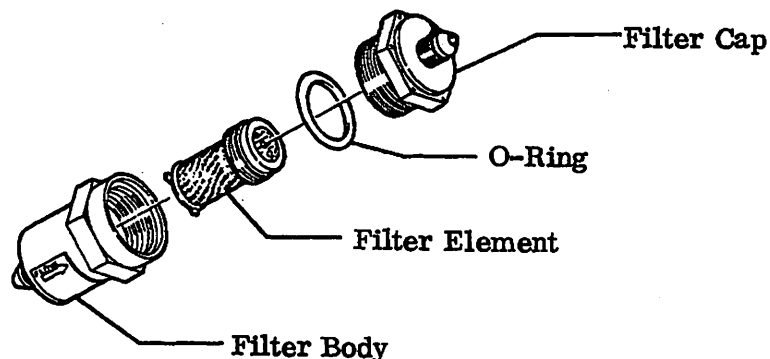
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2. Cleaning/Painting

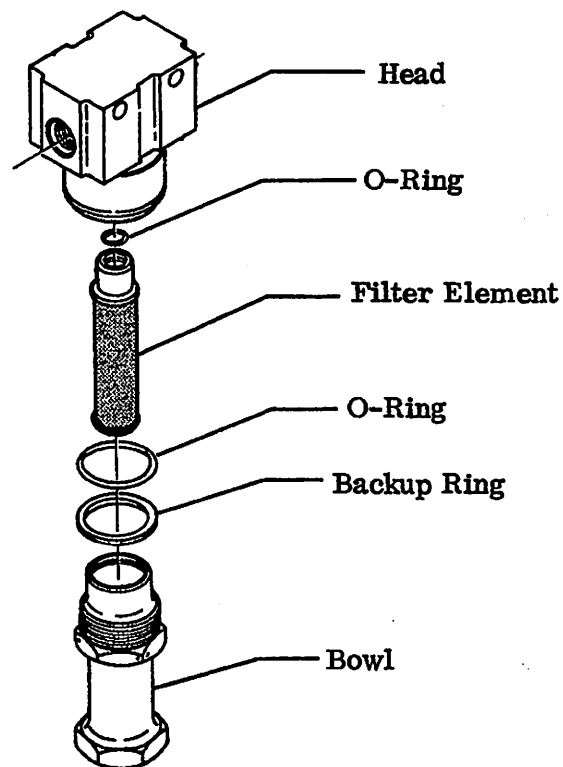
- A. To avoid system clogging, the alcohol filter assembly must be removed and cleaned in accordance with current inspection intervals in Chapter 5.
- B. Clean Alcohol Filter Assembly (25-070 thru 25-226 not modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft))
(See figure 201)
- (1) Remove alcohol filter assembly.
 - (2) Unscrew filter cap from filter body and remove filter element and O-ring.
 - (3) Wash filter element, O-ring, cap, and body in methyl alcohol and dry with dry, compressed air.
 - (4) Inspect O-ring and, if necessary, replace.
 - (5) Insert filter element into body, place O-ring on cap, and install cap.
 - (6) Tighten cap hand tight and safety wire.
 - (7) Install filter assembly.
- C. Clean Alcohol Filter Assembly (25-227 and Subsequent and prior Aircraft modified per AMK 78-3 (Modification of Alcohol Anti-Ice System for All-Weather Aircraft)) (See figure 202)
- (1) Remove alcohol filter assembly.
 - (2) Unscrew bowl from head and remove filter element, O-rings, and backup ring.
 - (3) Wash head, filter element, backup ring, O-rings, and bowl in methyl alcohol and dry with dry, compressed air.
 - (4) Inspect backup ring and O-rings and, if necessary, replace.
 - (5) Assemble O-rings and backup ring.
 - (6) Position filter element into head and tighten bowl hand tight. Safety wire.
 - (7) Install filter assembly.



25-070 thru 25-226 Not Modified per
AMK 78-3 (Modification of Alcohol Anti-Ice
System for All-Weather Aircraft)
Filter Assembly

Alcohol Filter Cleaning
Figure 202 (Sheet 1 of 2)

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Filter Assembly

Alcohol Filter Cleaning
Figure 202 (Sheet 2 of 2)

11-4313

EFFECTIVITY: 25-227 and Subsequent and Prior Aircraft
Modified per AMK 78-3 (Modification of
Alcohol Anti-Ice System for All-Weather Aircraft)

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ALCOHOL PUMP - MAINTENANCE PRACTICES

1. General

A. The alcohol pump is a cam-rotor-type, controlled by a three-position switch on the Anti-Ice and Avionics panel. The pump delivers a constant flow rate.

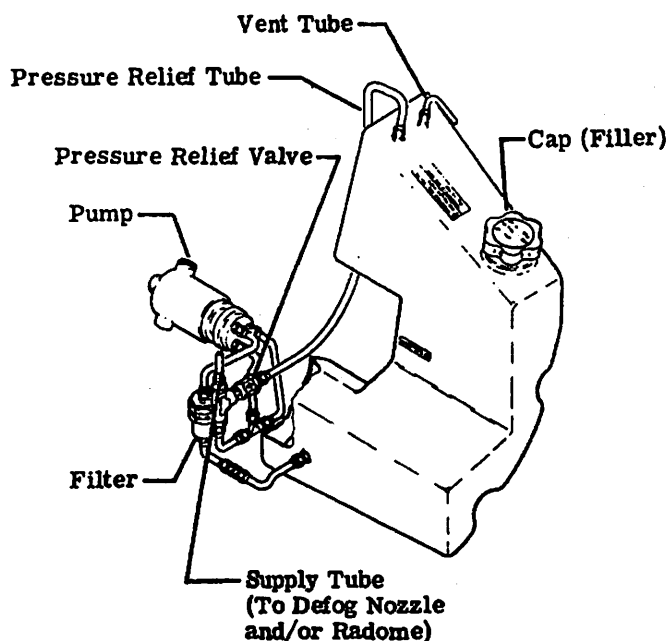
2. Removal/Installation

A. Remove Pump (See figure 201)

- (1) Remove left nose compartment access door.
- (2) Remove avionics equipment as required to gain access to pump.
- (3) Drain alcohol tank.
- (4) Disconnect electrical plug from pump.
- (5) Disconnect tube assemblies from pump. Cap tube assemblies.
- (6) Remove screws, washers and mounting plate with pump attached from aircraft.

B. Install Pump (See figure 201)

- (1) Install plate with pump attached and secure with screws and washers.
- (2) Connect electrical plug to pump.
- (3) Remove caps from tube assemblies and connect tube assemblies to pump.
- (4) Service alcohol tank. Refer to 12-00-00.
- (5) Install avionics equipment previously removed.
- (6) Install left nose compartment access door.



**Alcohol Pump Installation
Figure 201**

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ALCOHOL SHUTOFF VALVE - MAINTENANCE PRACTICES

1. General

- A.** The alcohol shutoff valve (solenoid type) is installed in the alcohol anti-ice line leading to the pilot's defog nozzle. The valve is normally closed with no electrical power applied.

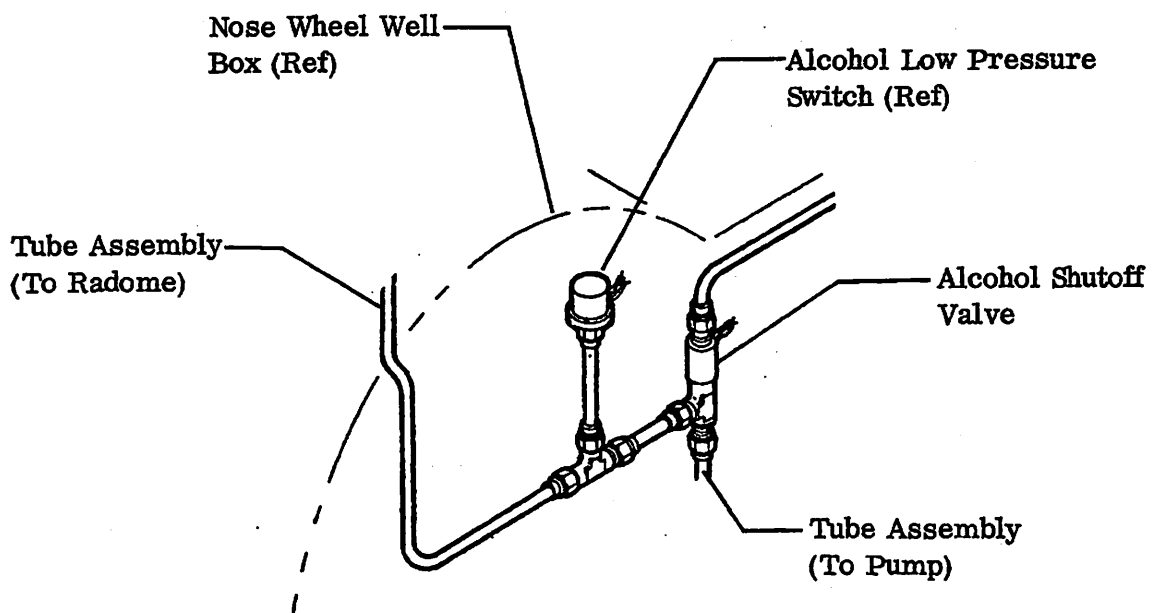
2. Removal/Installation

A. Remove Shutoff Valve (See figure 201)

- (1) Remove left nose compartment access door.
- (2) Disconnect wiring from shutoff valve.
- (3) Disconnect tube assemblies from shutoff valve and remove valve from aircraft. Cap tube assemblies.

B. Install Shutoff Valve (See figure 201)

- (1) Remove caps from tube assemblies. Connect tube assemblies to shutoff valve.
- (2) Connect wiring to shutoff valve.
- (3) Install nose compartment access door.



**Alcohol Shutoff Valve Installation
Figure 201**

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ALCOHOL PRESSURE SWITCH - MAINTENANCE PRACTICES

1. General

- A.** The pressure switch is located in the radome alcohol line. The switch is actuated if the alcohol supply is depleted or the pump malfunctions. Actuation of the switch completes a ground circuit to light the ALC AI caution light.

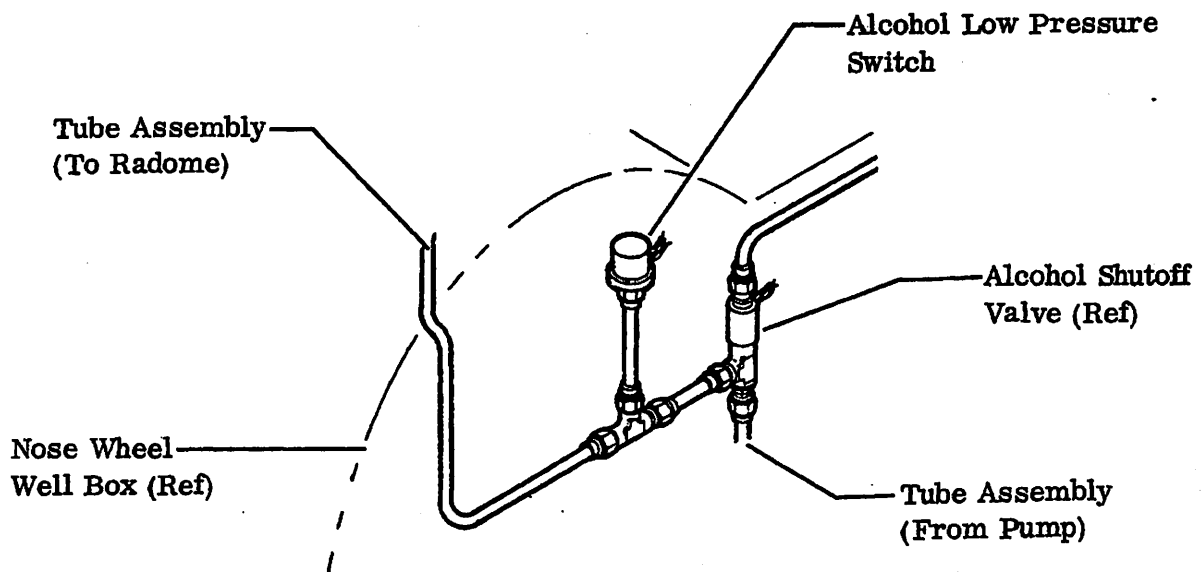
2. Removal/Installation

A. Remove Pressure Switch (See figure 201)

- (1) Remove left nose compartment access door.
- (2) Disconnect electrical wiring at splices.
- (3) Disconnect tube assembly from pressure switch.
- (4) Remove screw, washer, clamps and pressure switch from aircraft.

B. Install Pressure Switch (See figure 201)

- (1) Install pressure switch, clamps and secure with screw and washer.
- (2) Connect tube assembly to pressure switch.
- (3) Connect electrical wiring.
- (4) Install nose compartment access door.



**Alcohol Pressure Switch Installation
Figure 201**

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ALCOHOL PRESSURE RELIEF VALVE - MAINTENANCE PRACTICES

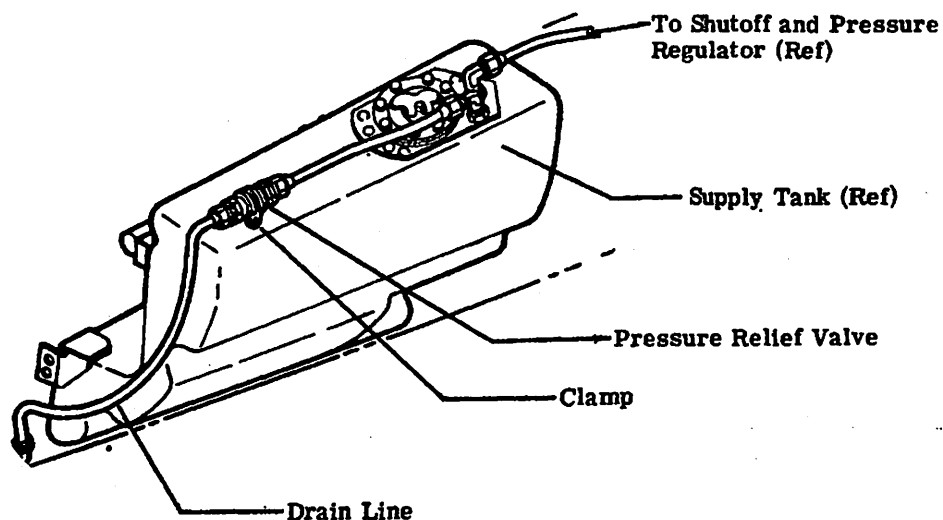
1. General

- A. An alcohol pressure relief valve is installed in the system pressure line. The relief valve opens and vents alcohol pressure in excess of 1.1 psi into the supply tank return line. On Aircraft 25-227 and Subsequent, the relief valve is installed in the supply tank overboard drain line. The valve is set to relieve any tank pressure in excess of 2.6 psi. The relief valve also incorporates a bleed off feature which allows the alcohol tank to depressurize when the alcohol system is not being utilized.

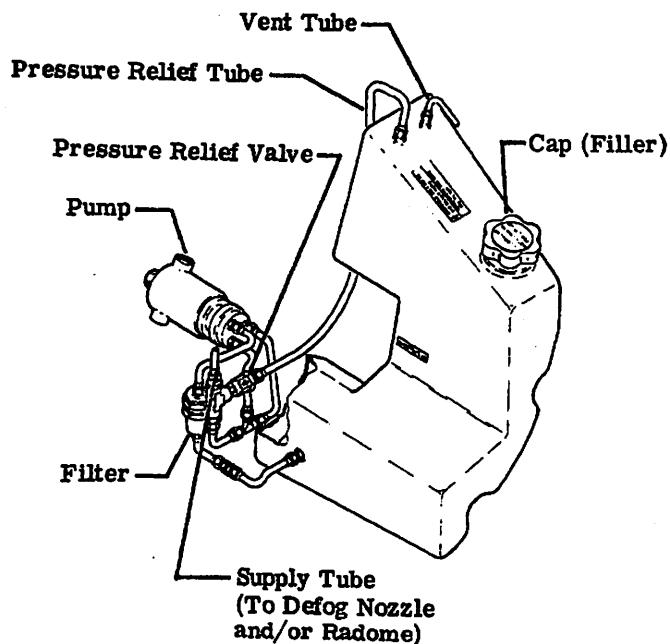
2. Removal/Installation

- A. Remove Relief Valve (See figure 201)
- (1) Remove left nose compartment access door.
 - (2) Drain alcohol tank (if applicable).
 - (3) Disconnect tube assembly from relief valve. Cap tube assembly. On Aircraft 25-227 and Subsequent, disconnect overboard drain tubes from relief valve. Remove attaching parts, clamps and relief valve from aircraft.
 - (4) Remove relief valve from tee.
- B. Install Relief Valve (See figure 201)
- (1) Install relief valve in tee noting direction arrow. Arrow points toward supply tank. On Aircraft 25-227 and Subsequent, install relief valve with clamps and attaching parts, noting the direction arrow. Arrow points toward overboard drain.
 - (2) Remove cap from tube assembly(s) and connect tube assembly to relief valve.
 - (3) Service alcohol tank. Refer to 12-00-00.
 - (4) Install nose compartment access door.

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Aircraft 25-227 and Subsequent



ROTATED 90° COUNTERCLOCKWISE

Aircraft 25-061, -070 thru -226

Alcohol Pressure Relief Valve Installation
Figure 201

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maintenance manual

ALCOHOL SUPPLY TANK - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: The alcohol tank is installed in the nose compartment on the LH side. The tank has a capacity of 2.2 gallons. On Aircraft 25-227 and Subsequent, the alcohol tank is a pressurized tank, incorporating a float switch, with a capacity of 1.75 gallons.

A. Remove Tank (See figure 201)

- (1) Remove left nose compartment access door.
- (2) Drain tank.
- (3) Remove avionics equipment as required to gain access to supply tank.
- (4) Disconnect supply, return, and vent tubes from tank and cap tube assemblies. On Aircraft 25-227 and Subsequent, disconnect electrical wiring from float switch. Disconnect supply, vent, and pressurization tubes from tank. Cap all exposed tubes and fittings.
- (5) Loosen strap assembly sufficiently to allow removal of tank. On Aircraft 25-227 and Subsequent, remove attaching parts and upper and lower supports. Remove alcohol tank from aircraft.

B. Install Tank (See figure 201)

- (1) Apply Loctite Nut Lock to strap assembly bolt threads.
- (2) Install tank and position strap assemblies with pads between strap and tank.
- (3) Tighten nuts on strap assembly bolts. On Aircraft 25-227 and Subsequent, install upper and lower supports and secure with attaching parts.
- (4) Remove caps from vent, supply, and return lines and connect lines to tank. On Aircraft 25-227 and Subsequent, remove caps from supply, vent, and pressure tubes and fittings and connect lines to tank. Connect electrical wiring to float switch.
- (5) Install previously removed avionics equipment.
- (6) Service supply tank. (Refer to Chapter 12.)
- (7) Install LH nose compartment access door.

2. Draining Alcohol Tank

A. Drain Tank (Aircraft 25-061, 25-070 thru 25-226) (See figure 201)

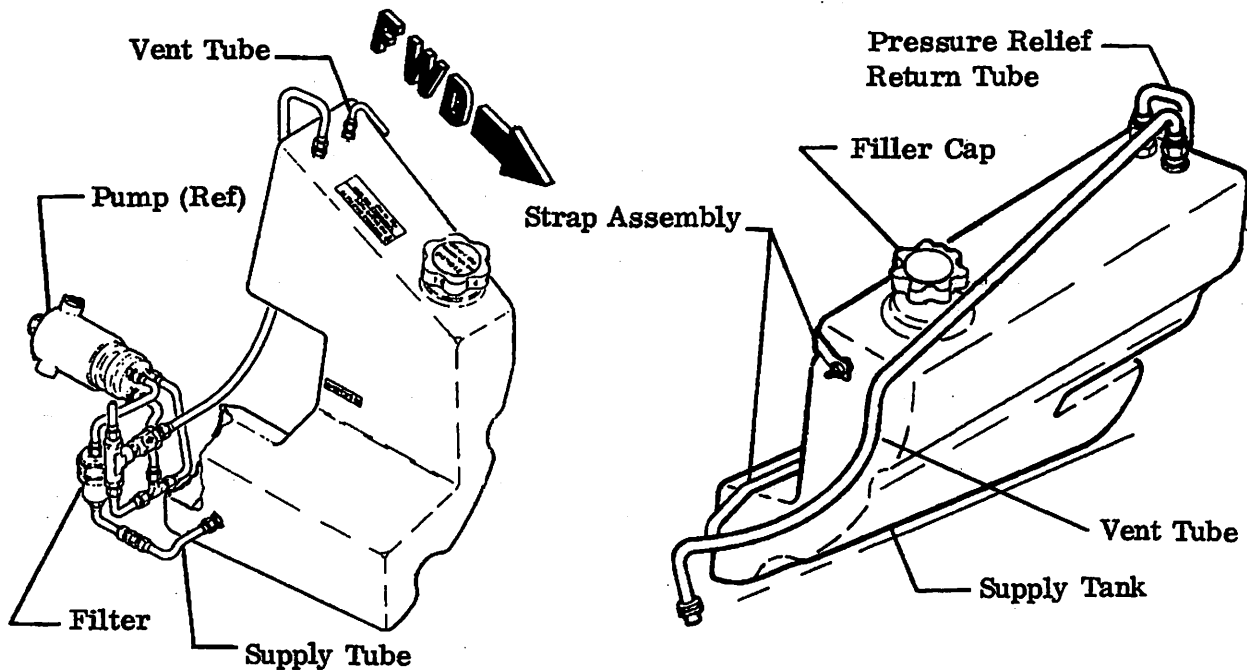
- (1) Remove LH nose compartment access door.
- (2) Disconnect defog nozzle and radome supply tube from tee.
- (3) Connect hose to tee and route hose overboard into a suitable container.

NOTE: Appropriate caution should be taken to soak up any spillage when supply tube is disconnected.

- (4) Set Windshield and Radome Switch to Radome and allow pump to empty tank.

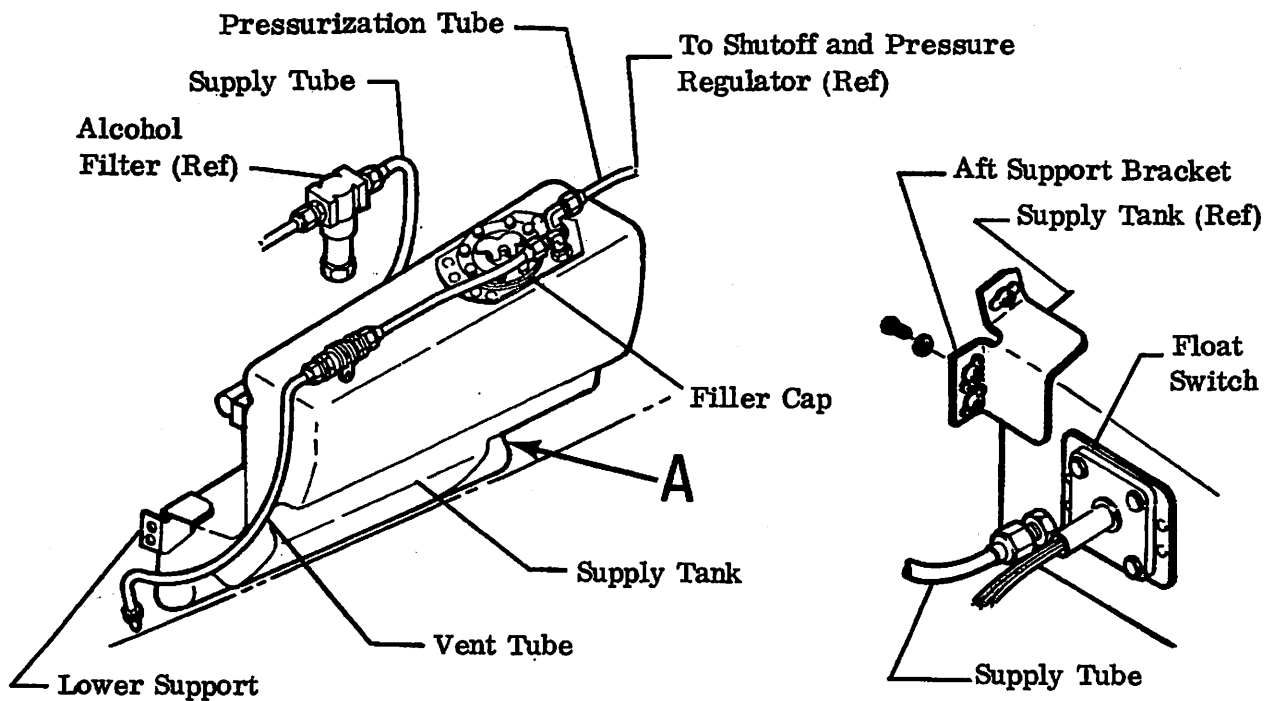
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maintenance manual



ROTATED 90° COUNTERCLOCKWISE

Aircraft 25-061, -070 thru -226



Aircraft 25-227 and Subsequent

Alcohol Supply Tank Installation
Figure 201

Detail A

13-251N
13-94C

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NOTE: Due to the location of the alcohol tank supply port, some residual alcohol will remain in the tank.

- B. **Drain Tank (Aircraft 25-227 and Subsequent) (See figure 201)**
- (1) Remove LH nose compartment access door.
 - (2) Open and remove supply tank filler cap.
 - (3) Siphon alcohol from tank into a suitable container.

NOTE: Some residual alcohol will remain in the tank.

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ALCOHOL ANTI-ICE SHUTOFF AND PRESSURE REGULATOR VALVE -
MAINTENANCE PRACTICES

1. General

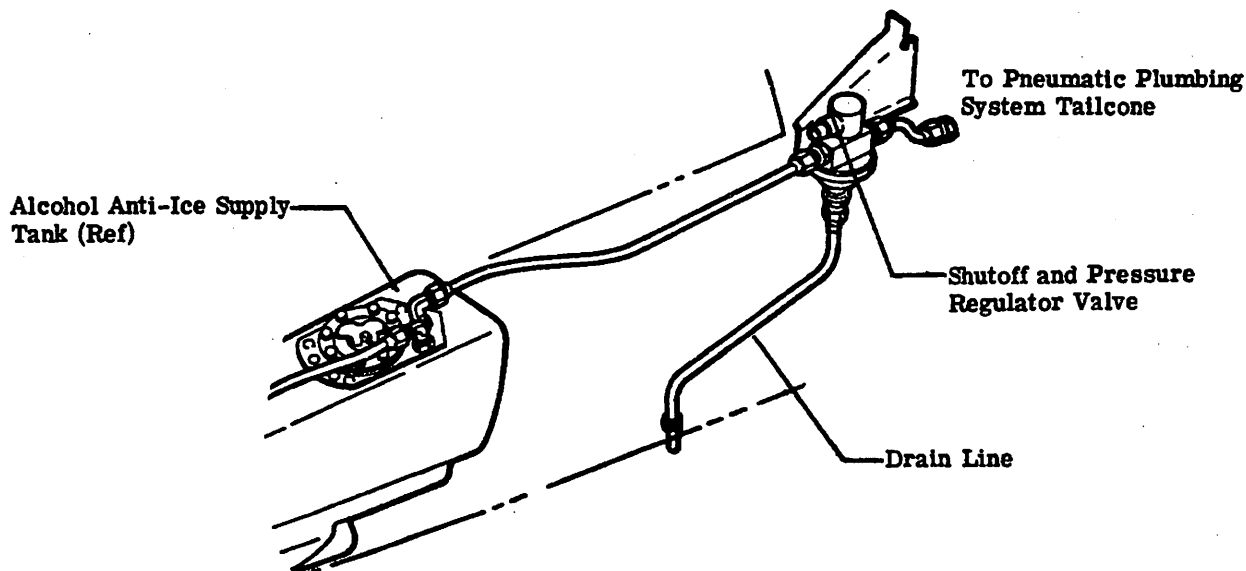
- A. The shutoff and pressure regulator valve is installed on the LH side of the nose compartment. The regulator is connected to engine bleed air ducts in the tailcone.

2. Operation

- A. The shutoff and pressure regulator valve, when energized open, regulates alcohol tank pressure at 2.3 (± 1) psi (normal). The regulator also incorporates a backup feature which vents overboard any pressure in excess of 4.0 psi. It also incorporates a vacuum relief feature. Should the ambient pressure exceed 1.0 psi above the alcohol tank pressure, the ambient pressure will flow thru the pressure regulators overboard drain and into the tank.

3. Removal/Installation

- A. Remove Shutoff and Pressure Regulator (See figure 202)
- (1) Remove left nose compartment access door.
 - (2) Disconnect pressure tubes and drain tube from regulator. Cap all tubes and exposed fittings.
 - (3) Remove safety wire, attaching parts and regulator from aircraft.
- B. Install Shutoff and Pressure Regulator Valve (See figure 202)
- (1) Install regulator and secure with attaching parts. Safety wire bolts.
 - (2) Remove caps and connect lines to regulator.
 - (3) Install left nose compartment access door.



Shutoff and Pressure Regulator Valve Installation
Figure 202

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3-WAY SHUTOFF VALVE - MAINTENANCE PRACTICES

1. General

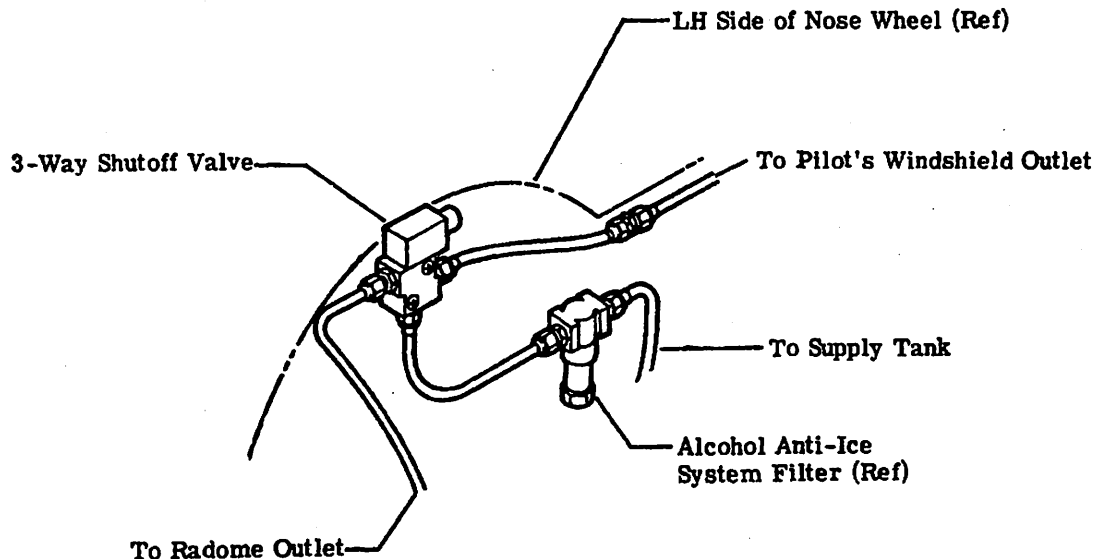
- A. The 3-way valve is installed on the LH side of the nose wheel well box in the nose compartment.

2. Operation

- A. The valve is solenoid operated and is controlled by the alcohol system WSHLD & RADOME - RAD- OFF Switch on the pilot's switch panel.

3. Removal/Installation

- A. Remove 3-Way Shutoff Valve (See figure 201)
- (1) Remove left nose compartment access doors.
 - (2) Disconnect electrical plug from valve.
 - (3) Disconnect tubes from valve. Cap all exposed tubes and fittings.
 - (4) Loosen and remove screws, valve and washers from aircraft.
- B. Install 3-Way Shutoff Valve (See figure 201)
- (1) Install valve on nose wheel box and secure with screws and washers. Assure that washers are installed between valve and side of nose wheel box.
 - (2) Remove caps from tubes and fittings and connect tubes to valve.
 - (3) Connect electrical plug to valve
 - (4) Install left nose compartment access door.



**3-Way Shutoff Valve Installation
Figure 201**

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ALCOHOL ANTI-ICE FLOAT SWITCH - MAINTENANCE PRACTICES

1. General

- A. The float switch is installed in the lower aft end of the alcohol tank.

2. Operation

- A. The float switch completes a ground circuit to energize the ALC AI caution light control relay. When the relay is energized a ground circuit is completed to illuminate the ALC AI caution light. The caution light control relay incorporates a holding circuit which keeps the relay energized once ground is applied. This prevents the caution light from flickering due to the bobbing motion of the float switch. When the Battery Switches are set to OFF and the alcohol tank serviced the holding circuit is opened.

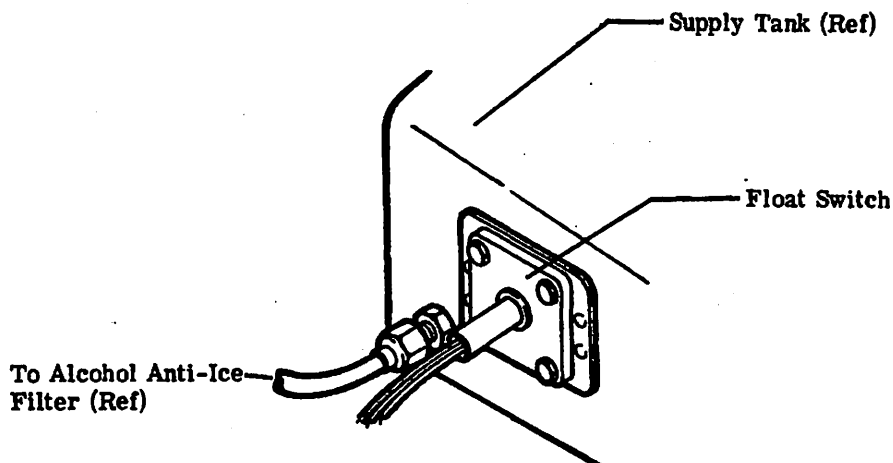
3. Removal/Installation

- A. Remove Float Switch (See figure 201)

- (1) Remove alcohol supply tank from aircraft. Refer to 30-50-06.
- (2) Loosen and remove attaching parts and float switch from tank.

- B. Install Float Switch (See figure 201)

- (1) Check condition of float switch O-ring. Replace if O-ring shows signs of distortion or deterioration.
- (2) Install float switch in tank with the word TOP in the up position.
- (3) Secure float switch to tank with attaching parts.



Alcohol Anti-Ice Float Switch
Figure 201

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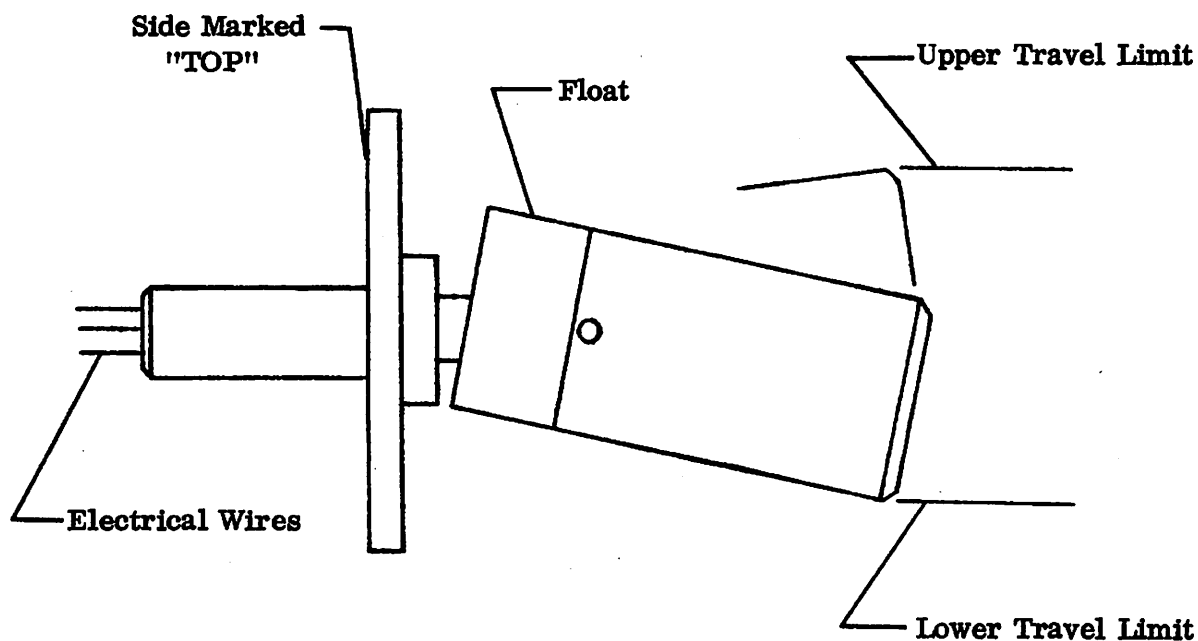
4. Adjustment/Test

A. Perform Alcohol Anti-Ice Float Switch Functional Test (See figure 202)

- (1) Remove float switch, with electrical wiring intact, from alcohol tank.
- (2) Set Battery Switches to ON.
- (3) Move float from lower travel limit to upper travel limit to ensure float travels freely.
- (4) With float switch at lower travel limit, ALC AI warning light shall illuminate.

NOTE: A holding circuit, once energized, will keep the ALC AI warning light illuminated. Set Battery Switches to OFF to deenergize the holding circuit.

- (5) ALC AI warning light shall not illuminate at upper travel limit.
- (6) Set Battery Switches to OFF.
- (7) Install float switch.



Alcohol Anti-Ice Float Switch Functional Test Setup
Figure 202

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DETECTION - DESCRIPTION AND OPERATION

1. General

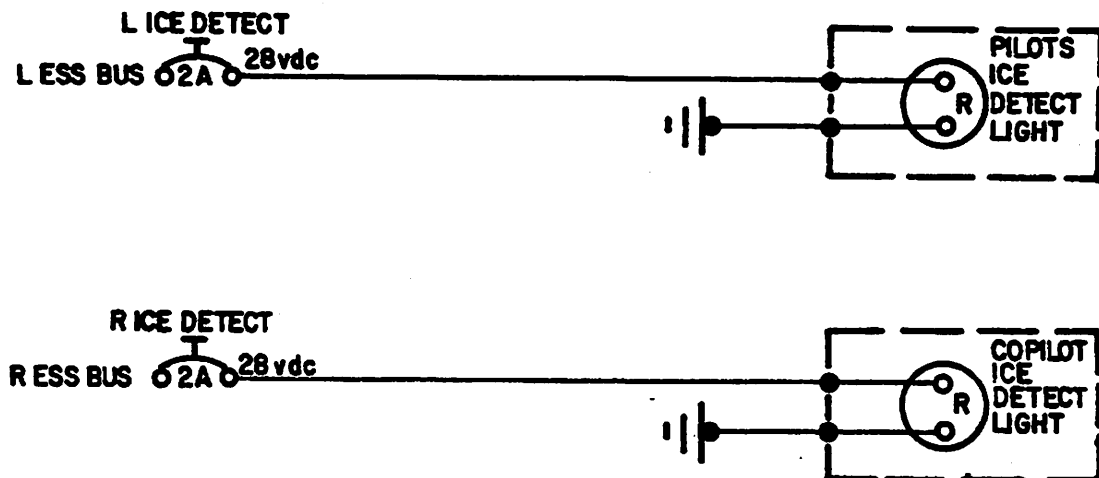
- A. Ice detection is accomplished by monitoring the wing heat indicator, the stabilizer heat indicator, and the windshield ice detect lights.
- B. Refer to 30-10-01 and 30-10-02 for further information on the temperature indicators.

2. Description

- A. The windshield ice detector system consists of two red light assemblies mounted on the glareshield and two circuit breakers. The light on the pilot's side is located in an area which is cleared by the windshield defog system. The light on the copilot's side is located in an area outside the windshield defog airflow system. These locations are such that if the windshield defog system is in operation, there will still be an indication available on the copilot's side.

3. Operation (See figure 1.)

- A. Setting the Battery Switches to ON applies 28 vdc through individual circuit breakers to its respective light. When particles of ice or moisture form, light refraction takes place resulting in the appearance of two red areas, approximately 1-1/2 inches in diameter, on the windshield.



Ice Detect Light Electrical Control Schematic
Figure 1

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ICE DETECT LIGHT - MAINTENANCE PRACTICES

1. General

A. The ice detect lights are installed on the forward glareshield adjacent to each windshield half.

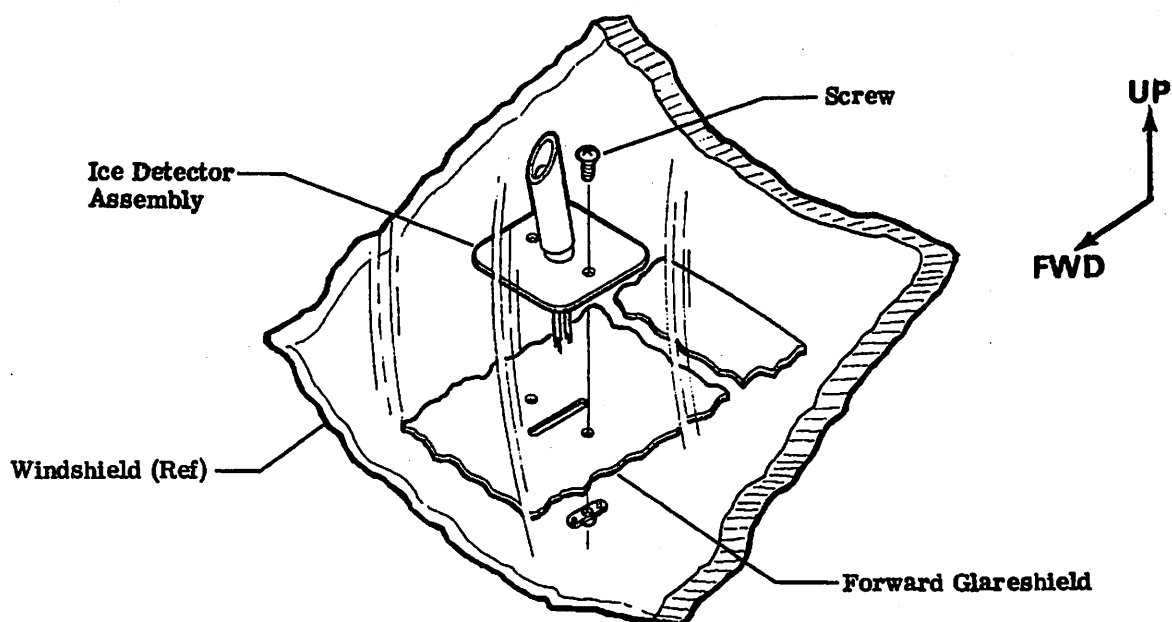
2. Removal/Installation

A. Remove Light Bulb (See figure 201)

- (1) Slip neoprene tube off ice detector assembly.
- (2) Snap off red lens to expose bulb.
- (3) Extract bulb using a piece of tacky putty.

B. Install Light Bulb (See figure 201)

- (1) Install new bulb (62 Shelly) assuring wire leads enter holes in socket.
- (2) Replace neoprene tube with bevel facing forward.
- (3) Check operation by turning on either battery switch and holding an object between light and windshield.



**Ice Detect Light Installation
Figure 201**